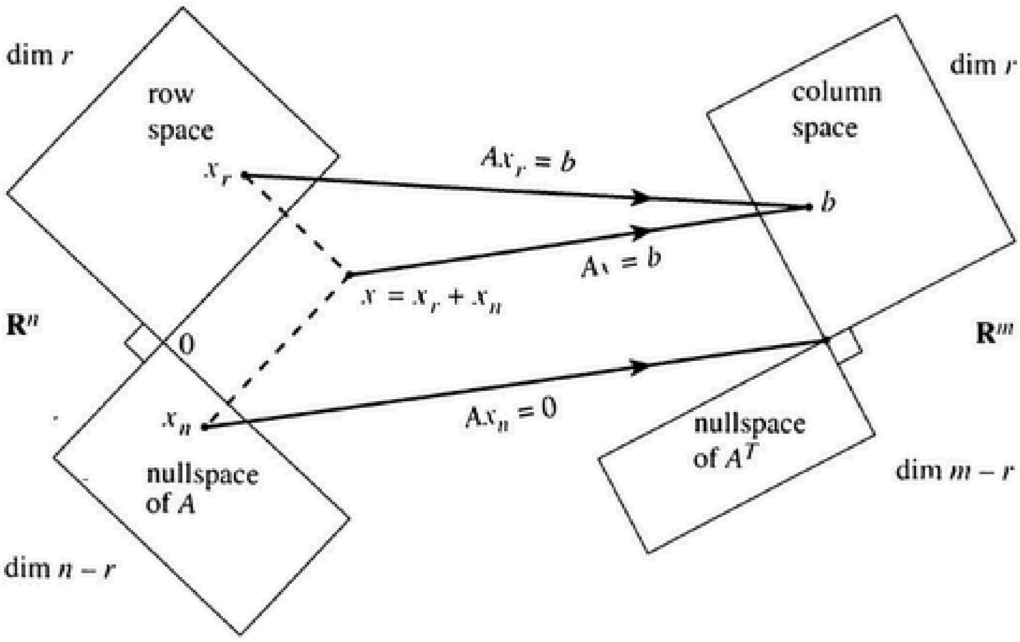
LINEAR ALGEBRA



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Contents

CHAPTER 1

Numbers and Vectors

Main objectives:

- Introduction to number systems;
- Introduction to vectors, related operators, basis for $\mathbb{R}^n$, and subspaces of $\mathbb{R}^n$.

1.1. Number Systems

Let us begin by the definition of frequently used terms in the coming lectures. We note that there are other set of numbers (such as *irrational* and *complex* numbers) that are overlooked in the following.

Frequently used numbers in mathematics:

(i) The set of numbers $\mathbb{N} = \{1, 2, \dots\}$ is called **natural** numbers;
(ii) The set of numbers $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ is called **integer** numbers;
(iii) The set of numbers $\mathbb{Q} = \{a/b \mid a, b \in \mathbb{Z} \text{ and } b \neq 0\}$ is called **rational** numbers;
(iv) The set of all numbers that are continuous quantities which can represent a distance along a line is called **real** numbers and denoted by $\mathbb{R}$. In other words, any real number can be determined by a possibly infinite decimal representation.

We next describe fundamental axioms for working with real number systems.

Addition axioms on $\mathbb{R}$:

(a1) $a + b = b + a$ for $a, b \in \mathbb{R}$;
(a2) $(a + b) + c = a + (b + c)$ for $a, b, c \in \mathbb{R}$;
(a3) $0 + a = a + 0 = a$ for $a \in \mathbb{R}$;
(a4) $a + (-a) = (-a) + a = 0$ for $a \in \mathbb{R}$;.

Multiplication axioms on $\mathbb{R}$:

(m1) $a * b = b * a$ for $a, b \in \mathbb{R}$;
(m2) $(a * b) * c = a * (b * c)$ for $a, b, c \in \mathbb{R}$;
(m3) $1 * a = a * 1 = a$ for $a \in \mathbb{R}$;
(m4) $a * a^{-1} = a^{-1} * a = 1$ for $0 \neq a \in \mathbb{R}$;.

Addition-multiplication axiom on $\mathbb{R}$:

(am1) $a * (b + c) = a * b + a * c$ for $a, b, c \in \mathbb{R}$.

Example 1. *Let us consider the following example combining the above-mentioned axioms.*

- $(1 + 2) + 3 = 1 + (2 + 3) = 6$*;*
- $5 * (1/5) = (1/5) * 5 = 1$*;*
- $3 * (4 + 5) = 3 * 4 + 3 * 5 = 12 + 15 = 27$*;*
- $(1/9) * (4 + 5) = (1/9) * 4 + (1/9) * 5 = 4/9 + 5/9 = 1$*;*

1.2. Vectors

In this subsection, we will discuss vectors in $\mathbb{R}^n$, some of their properties, and related necessary operators. A vector in $\mathbb{R}^n$ is a one-dimensional array that whose elements are all from $\mathbb{R}$, i.e.,

$$a \in \mathbb{R}^n \quad \Leftrightarrow \quad a = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}, \quad a_i \in \mathbb{R} \quad i = 1, \ldots, n.$$

Zeros and ones vectors. A vector $a \in \mathbb{R}^n$ is called bf zeros vector if all of its elements are zero, which is denoted by 0_n, i.e.,

$$\mathbf{0}_n = a = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

A vector $b \in \mathbb{R}^n$ is called **ones vector** if all of its elements are one, which is denoted by $\mathbf{1}_n$, i.e.,

$$\mathbf{1}_n = b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}.$$

Example 2. *The zeros and ones vector in $\mathbb{R}^3$ are given in the following*

$$\mathbf{0}_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{1}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

Equal vectors. Two vectors a and b are **equal** if all of their elements are equal, which is $a_i = b_i$ for $i = 1, \ldots, n$, i.e.,

$$A = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \quad \Leftrightarrow \quad a_i = b_i, \quad i = 1, \ldots, n.$$

In the following, we study the main operations of vectors that will be frequently used in the remainder of the lecture.

Scalar multiplication: For the constant $\alpha \in \mathbb{R}$ and the vector $a \in \mathbb{R}^n$, the scalar multiplication of α with a is given by

$$\alpha a = \alpha \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} \alpha a_1 \\ \alpha a_2 \\ \vdots \\ \alpha a_n \end{bmatrix}.$$

Two main properties of scalar multiplication:

(a) **Associative property.** For the constants $\alpha, \beta \in \mathbb{R}$ and the vector $a \in \mathbb{R}^n$, it holds that

$$(\alpha\beta)a = \alpha(\beta a);$$

(b) **Distributive property.** For the constants $\alpha, \beta \in \mathbb{R}$ and the vector $a \in \mathbb{R}^n$, it hold that

$$(\alpha + \beta)a = \alpha a + \beta a.$$

Example 3. *Let us consider the scalars $\alpha, \beta \in \mathbb{R}$ and the vector $a \in \mathbb{R}^3$ given by*

$$\alpha = 2, \quad , \beta = -3, \quad a = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.$$

Then, we have

$$(\alpha\beta)a = \alpha(\beta a) = \alpha \begin{bmatrix} -3 \\ -6 \\ -9 \end{bmatrix} = \begin{bmatrix} -6 \\ -12 \\ -18 \end{bmatrix},$$

and

$$(\alpha + \beta)a = \alpha a + \beta a = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} + \begin{bmatrix} -3 \\ -6 \\ -9 \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \\ -3 \end{bmatrix}.$$

Vectors addition. For two vectors $a, b \in \mathbb{R}^n$, the **addition** of two vectors is given by

$$a + b = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} + \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} a_1 + b_1 \\ a_2 + b_2 \\ \vdots \\ a_n + b_n \end{bmatrix}.$$

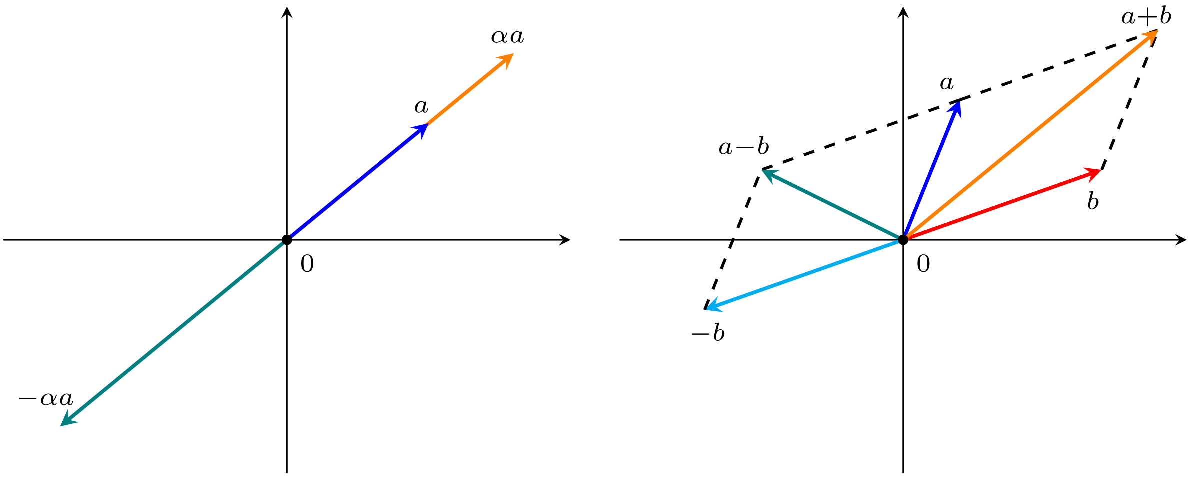
Two main properties of vectors addition:

(a) **Commutative property.** For the vector $a, b \in \mathbb{R}^n$, it holds that

$$a + b = b + a.$$

(b) **Associative property.** For the vector $a, b, c \in \mathbb{R}^n$, it hold that

$$(a + b) + c = a + (b + c).$$



(A) scalar multiplication. (B) vector addition and subtraction.

FIGURE 1.1. Subfigure (A) stands for scalar multiplication of scalar α and $-\alpha$ with vector a, and Subfigure (B) illustrates addition and subtraction of two vectors a and b.

We are in a position to define linear combinations of two vectors.

Linear combination. For two vectors $a, b \in \mathbb{R}^n$ and scalars $\alpha, \beta \in \mathbb{R}$, the **linear combination** of these vectors given by

$$\alpha a + \beta b = \alpha \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} + \beta \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = \begin{bmatrix} \alpha a_1 + \beta b_1 \\ \alpha a_2 + \beta b_2 \\ \vdots \\ \alpha a_n + \beta b_n \end{bmatrix}.$$

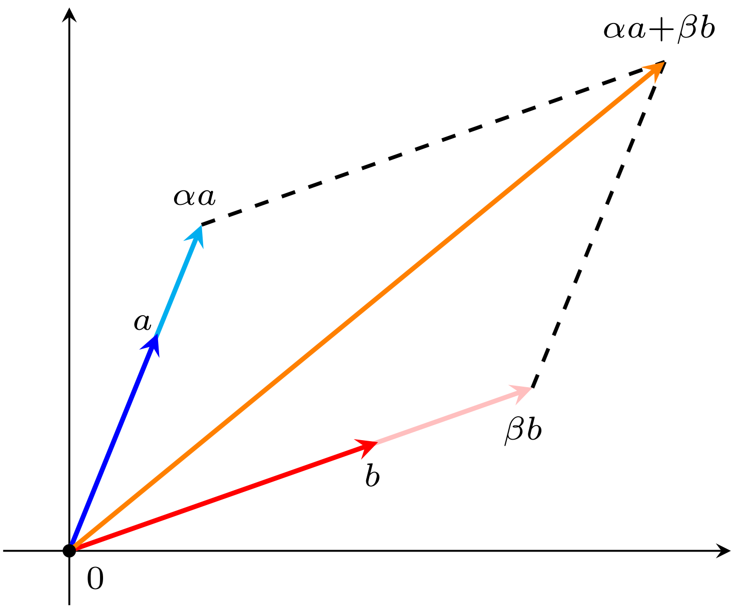


FIGURE 1.2. Linear combination of two vectors $a, b \in \mathbb{R}^2$.

Example 4. *We consider a linear combination of vectors*

$$a = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad b = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \quad \alpha = 5, \quad \beta = 2.$$

Then, we have

$$\alpha a + \beta b = 5 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + 2 \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \\ 15 \end{bmatrix} + \begin{bmatrix} 8 \\ 10 \\ 12 \end{bmatrix} = \begin{bmatrix} 13 \\ 20 \\ 27 \end{bmatrix}.$$

Length of a vector. The length of a vector $a \in \mathbb{R}^n$ (or **Euclidean norm of** a) is given by

$$\|a\| = \sqrt{a_1^2 + a_2^2 + \ldots + a_n^2}.$$

Two main properties:

(a) **Length of a scaled vector.** For a scalar $\alpha \in \mathbb{R}$, we have

$$\|\alpha a\| = \sqrt{\alpha^2 a_1^2 + \alpha^2 a_2^2 + \ldots + \alpha^2 a_n^2} = \sqrt{\alpha^2 (a_1^2 + a_2^2 + \ldots + a_n^2)} = |\alpha| \|a\|;$$

(b) **Triangular inequality.** For vectors $a, b \in \mathbb{R}^n$, we have

$$\|a + b\| \leq \|a\| + \|b\|.$$

Example 5. *We consider a linear combination of vectors*

$$a = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad b = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}.$$

Then, we have

$$\|a\| = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{14}, \quad \|b\| = \sqrt{4^2 + 5^2 + 6^2} = \sqrt{77}$$

and

$$\| - 5a\| = 5\sqrt{14}, \quad \|2b\| = 2\sqrt{77}$$

and

$$\|a + b\| = \sqrt{5^2 + 7^2 + 9^2} = \sqrt{155}, \quad \sqrt{155} = \|a + b\| \leq \|a\| + \|b\| = \sqrt{14} + \sqrt{77},$$

which is the triangle inequality.

Unit vectors. A vector $a \in \mathbb{R}^n$ whose length equals one is called **unit vector**, i.e.,

$$\|a\| = 1.$$

Example 6. *Let us consider the vectors*

$$a = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad b = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad c = \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{bmatrix}, \quad d = \begin{bmatrix} \cos(\theta) \\ \sin(\theta) \end{bmatrix}, \quad e = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

It is clear that

$$\begin{aligned} \|a\| &= \sqrt{1 + 0 + 0} = 1, \\ \|b\| &= \sqrt{0 + 0 + 1 + 0} = 1, \\ \|c\| &= \sqrt{\tfrac{1}{4} + \tfrac{1}{4} + \tfrac{1}{4} + \tfrac{1}{4}} = 1, \\ \|d\| &= \sqrt{\cos^2(\theta) + \sin^2(\theta)} = 1, \\ \|e\| &= \sqrt{1 + 4} = \sqrt{5}, \end{aligned}$$

which means that the vectors a, b, c, d are unit while d is not unit.

Example 7. *Let us consider a vector $0 \neq a \in \mathbb{R}^n$, and let us form the vector $a/\|a\|$, i.e.,*

$$\frac{a}{\|a\|} = \begin{bmatrix} \frac{a_1}{\|a\|} \\ \frac{a_2}{\|a\|} \\ \vdots \\ \frac{a_n}{\|a\|} \end{bmatrix}.$$

We now compute the length of the resulting vector, i.e.,

$$\begin{aligned} \|\tfrac{a}{\|a\|}\| &= \sqrt{\frac{a_1^2}{\|a\|^2} + \frac{a_2^2}{\|a\|^2} + \ldots + \frac{a_n^2}{\|a\|^2}} \\ &= \sqrt{\frac{a_1^2+a_2^2+\ldots+a_n^2}{\|a\|^2}} = \frac{\|a\|}{\|a\|} = 1, \end{aligned}$$

which implies that $a/\|a\|$ is a unit vector for $a \neq 0$.

Inner product of two vectors. For given vectors $a, b \in \mathbb{R}^n$, the **inner product** of them is given by

$$\langle a, b\rangle = a_1b_1 + a_2b_2 + \ldots + a_nb_n = \|a\|\|b\|\cos(\theta),$$

where θ is the angle between two vectors a and b.
Four main properties of the inner product:

(a) **Commutative property.** For vectors $a, b \in \mathbb{R}^n$, we have
$$\langle a, b\rangle = \langle b, a\rangle;$$
(b) **Associative property.** For vectors $a, b \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$, we have
$$\langle \alpha a, b\rangle = \alpha \langle b, a\rangle;$$
(c) **Distributive property.** For vectors $a, b, c \in \mathbb{R}^n$, we have
$$\langle a + b, c\rangle = \langle a, c\rangle + \langle b, c\rangle.$$
(d) **Cauchy-Schwartz inequality.** For vectors $a, b \in \mathbb{R}^n$, we have
$$\langle a, b\rangle = \|a\|\|b\|\cos(\theta) \leq \|a\|\|b\|, \quad a, b \in \mathbb{R}^n,$$
which is named the Cauchy-Schwarz inequality.

Example 8. *Let us consider the vectors*

$$a = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad b = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \quad c = \begin{bmatrix} -4 \\ 1 \\ -2 \end{bmatrix}.$$

Then, we have

$$\langle a, b\rangle = \langle b, a\rangle = 1 * 4 + 2 * 5 + 3 * 6 = 32, \quad \langle a, b\rangle = 2 * 32 = 64,$$

and

$$\langle a + b, c\rangle = \langle a, c\rangle + \langle b, c\rangle = (-4 + 2 - 6) + (-16 + 5 - 12) = -8 - 23 = -31.$$

We know from Example 8 that $\|a\| = \sqrt{14}$, $\|b\| = \sqrt{77}$, and $\langle a, b\rangle = 32$. Then, it is clear that

$$32 = \langle a, b\rangle \leq \|a\|\|b\| = \sqrt{14}\sqrt{77},$$

which are certified by the Cauchy-Schwartz inequality.

It follows from the definition of the inner product that

$$\cos(\theta) = \frac{\langle a,b\rangle}{\|a\|\|b\|}, \tag{1.1}$$

which implies that $\cos(\theta) = 0$ if $\langle a, b\rangle = 0$, i.e., $\theta = \pi/2$. Hence, two vectors a and b are **prependicular** if their inner product is zero (i.e., $\langle a, b\rangle = 0$).

Example 9. *We consider a linear combination of vectors*

$$a = \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix}, \quad b = \begin{bmatrix} 4 \\ 5 \\ 7 \end{bmatrix}.$$

Then, it is clear that

$$\langle a, b \rangle = 4 + 10 - 14 = 0,$$

implying that a and b are prependicular.

Example 10. *Show the triangular inequality using the Cauchy-Schwarz inequality. We note that for a vector $a \in \mathbb{R}^n$, we have*

$$\langle a, a \rangle = \|a\| \|a\| \cos(0) = \|a\|^2.$$

Together with the Cauchy-Schwarz inequality, this implies

$$\begin{aligned} \|a + b\|^2 &= \langle a + b, a + b \rangle = \langle a, a \rangle + \langle a, b \rangle + \langle b, a \rangle + \langle b, b \rangle \\ &= \|a\|^2 + \|b\|^2 + 2 \langle a, b \rangle \leq \|a\|^2 + \|b\|^2 + 2\|a\|\|b\| = (\|a\| + \|b\|)^2, \end{aligned}$$

which finally leads to the triangle inequality

$$\|a + b\| \leq \|a\| + \|b\|,$$

for any $a, b \in \mathbb{R}^n$.

1.3. Linear dependence and independence

In this subsection, we will discuss several important topics, such as linear independence of vectors in $\mathbb{R}^n$, which will be frequently used in the coming sections.

Linear independence. Let $a_1, a_2, \ldots, a_n$ be vectors in $\mathbb{R}^n$. Then, the vectors $a_1, a_2, \ldots, a_n$ are **linearly dependent** if there exist scalars $\alpha_1, \alpha_2, \ldots, \alpha_n$ (not all zero) such that

$$\alpha_1 a_1 + \alpha_2 a_2 + \ldots + \alpha_n a_n = 0. \tag{1.1}$$

If such scalars $\alpha_1, \alpha_2, \ldots, \alpha_n$ do not exist, then the vectors $a_1, a_2, \ldots, a_n$ are **linearly independent**. On the other words, the vectors $a_1, a_2, \ldots, a_n$ are linearly independent if the equation (2.1) leads to

$$\alpha_1 = \alpha_2 = \ldots = \alpha_n = 0.$$

Example 11. *Let us consider the vectors*

$$a_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad a_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \quad a_3 = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}.$$

Then, it is clear that for $\alpha_1 = 1$, $\alpha_2 = 0$, and $\alpha_3 = -1/2$ we have

$$\alpha_1 a_1 + \alpha_2 a_2 + \alpha_3 a_3 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} - 1/2 \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix},$$

which means that a_1, a_2, and a_3 are dependent. Now, let us consider the following vectors

$$b_1 = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}, \quad b_2 = \begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix}, \quad b_3 = \begin{bmatrix} 0 \\ 0 \\ 9 \end{bmatrix}.$$

Then, writing the equation (2.1) leads to

$$\alpha_1 b_1 + \alpha_2 b_2 + \alpha_3 b_3 = \alpha_1 \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} + \alpha_2 \begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix} + \alpha_3 \begin{bmatrix} 0 \\ 0 \\ 9 \end{bmatrix} = \begin{bmatrix} 2\alpha_1 \\ 5\alpha_2 \\ 9\alpha_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

This clearly leads to $\alpha_1 = \alpha_2 = \alpha_3 = 0$, which means that b_1, b_2, and b_3 are independent.

1.4. Spanning vectors and bases for $\mathbb{R}^n$

In this subsection, we will discuss spanning vectors and basis for $\mathbb{R}^n$, which play central role in linear algebra.

Spanning vectors. The vectors $a_1, a_2, \ldots, a_n$ **span** $\mathbb{R}^n$ if every vector a in $\mathbb{R}^n$ can be written as a linear combination of these vectors, i.e., there exist scalars $\alpha_1, \alpha_2, \ldots, \alpha_n$ such that

$$a = \alpha_1 a_1 + \alpha_2 a_2 + \ldots + \alpha_n a_n.$$

Example 12. *Let us consider the following vectors*

$$b_1 = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}, \quad b_2 = \begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix}, \quad b_3 = \begin{bmatrix} 0 \\ 0 \\ 9 \end{bmatrix},$$

and let

$$a = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

be an arbitrary vector in $\mathbb{R}^3$*. Then, we have*

$$\tfrac{a_1}{2} b_1 + \tfrac{a_2}{5} b_2 + \tfrac{a_3}{9} b_3 = \tfrac{a_1}{2} \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} + \tfrac{a_2}{5} \begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix} + \tfrac{a_3}{9} \begin{bmatrix} 0 \\ 0 \\ 9 \end{bmatrix} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = a.$$

This clearly implies that setting $\alpha_1 = a_1/2$*,* $\alpha_2 = a_2/5$*, and* $\alpha_3 = a_3/9$ *we can generate an arbitrary vector* a *of* $\mathbb{R}^3$*, which means that* b_1*,* b_2*, and* b_3 *are spanning* $\mathbb{R}^3$*.*

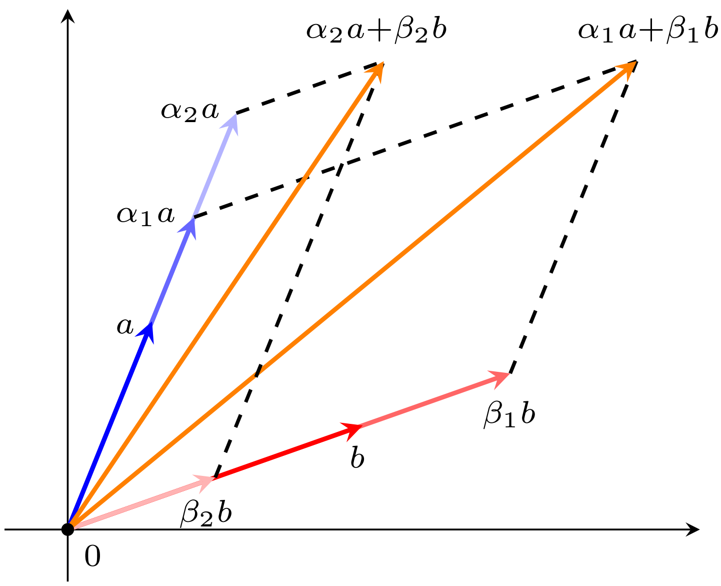


FIGURE 1.3. Spaning space of two vectors $a, b \in \mathbb{R}^2$.

Bases of $\mathbb{R}^n$ The vectors $a_1, a_2, \ldots, a_n$ are called a **basis** for $\mathbb{R}^n$ if they are spanning $\mathbb{R}^n$ and are independent too.

Example 13. *Let us consider the following vectors*

$$b_1 = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}, \quad b_2 = \begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix}, \quad b_3 = \begin{bmatrix} 0 \\ 0 \\ 9 \end{bmatrix},$$

and let

$$a = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

be an arbitrary vector in $\mathbb{R}^3$. It follows from Example 11 that the vectors b_1, b_2, and b_3 are spanning $\mathbb{R}^3$. Moreover, Example 11 implies that they are independent too. Therefore, the set of vectors b_1, b_2, and b_3 are a basis for $\mathbb{R}^3$.

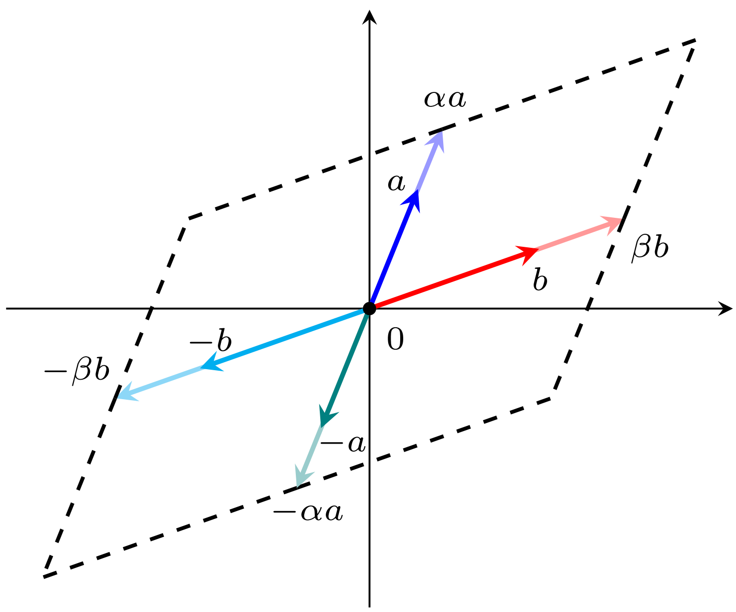


FIGURE 1.4. Two nonzero vectors $a, b \in \mathbb{R}^2$ with different directions are a basis of $\mathbb{R}^2$.

Elementary vectors of $\mathbb{R}^n$. The vectors

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \cdots, \quad e_n = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 0 \\ 1 \end{bmatrix},$$

are called **elementary vectors** of $\mathbb{R}^n$.

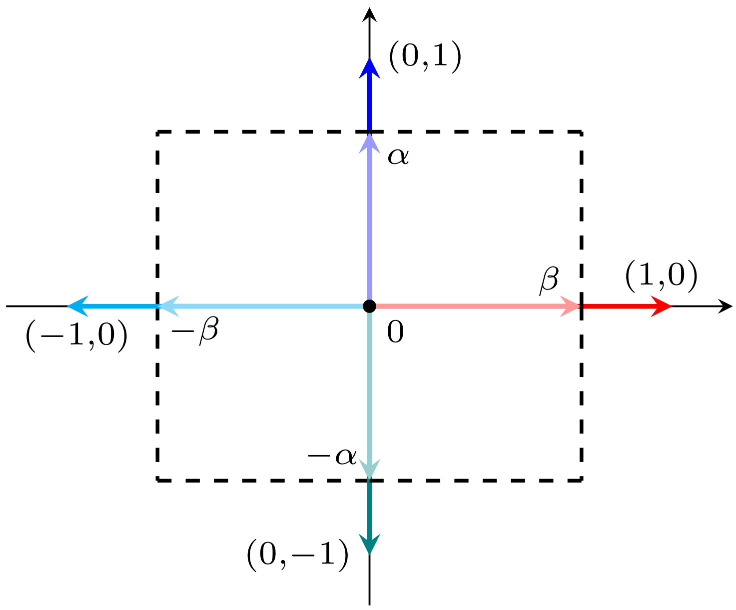


FIGURE 1.5. Elementary basis for $\mathbb{R}^2$.

Remark 14 (elementary vectors are a basis on $\mathbb{R}^n$)**.** *Let us show that elementary vectors as a basis for $\mathbb{R}^n$. For an arbitrary vector $a \in \mathbb{R}^n$, we have*

$$a_1e_1 + a_2e_2 + \ldots + a_ne_n = \begin{bmatrix} a_1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ a_2 \\ 0 \\ \vdots \\ 0 \end{bmatrix} + \ldots + \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 0 \\ a_n \end{bmatrix} = a,$$

which ensures that the elementary vectors span $\mathbb{R}^n$. Moreover, the equation

$$\alpha_1e_1 + \alpha_2e_2 + \ldots + \alpha_ne_n = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix},$$

leads to the trivial solution $\alpha_1 = \alpha_2 = \ldots = \alpha_n = 0$. Hence, the elementary vectors are independent. Together with spanning $\mathbb{R}^n$, this implies that the elementary vectors are a basis for $\mathbb{R}^n$.

1.5. Linear subspace of R^n

In this subsection, we will describe an important concept in linear algebra called **subspace**, which will be appear frequently in the coming sections.

Linear subspace. Let S be a nonempty subset of $\mathbb{R}^n$. Then, S is said to be a **subspace** if the following hold:

(a) **Containing zero.** $0 \in S$;
(b) **Closedness w.r.t. addition.** $s_1, s_2 \in S$ leads to $s_1 + s_2 \in S$;
(c) **Closedness w.r.t. scalar multiplication.** $s_1 \in S$ and $\alpha \in \mathbb{R}$ implies $\alpha s_1 \in S$.

Example 15 (subspaces of $\mathbb{R}^2$ and $\mathbb{R}^3$)**.** *The subspaces of $\mathbb{R}^2$ are*

- $\{0\}$*;*
- *Lines through origin;*
- $\mathbb{R}^2$ *itself.*

The subspaces of $\mathbb{R}^3$ are

- $\{0\}$*;*

- *Lines through the origin;*
- *Planes through the origin;*
- $\mathbb{R}^3$ *itself.*

Example 16. *Let us consider the set*

$$S = \left\{ [t, t^3]^T \mid t \in \mathbb{R} \right\},$$

where we verify if this set is a subspace. It is clear that

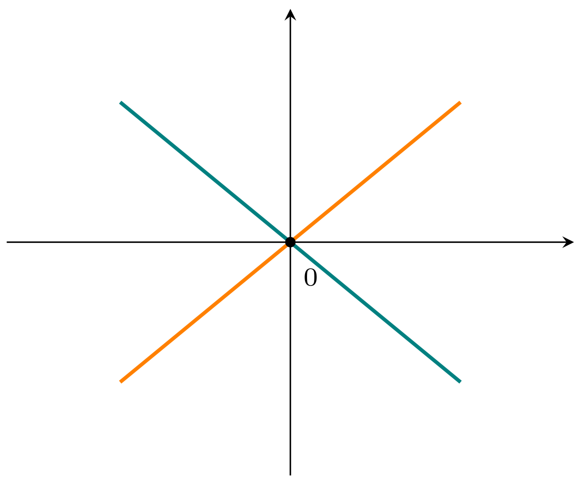
(a) Setting $t = 0$ *implies* $[0, 0]^T \in S$*;*

(a) For $t_1 \neq t_2 \neq 0$*, we obtain*

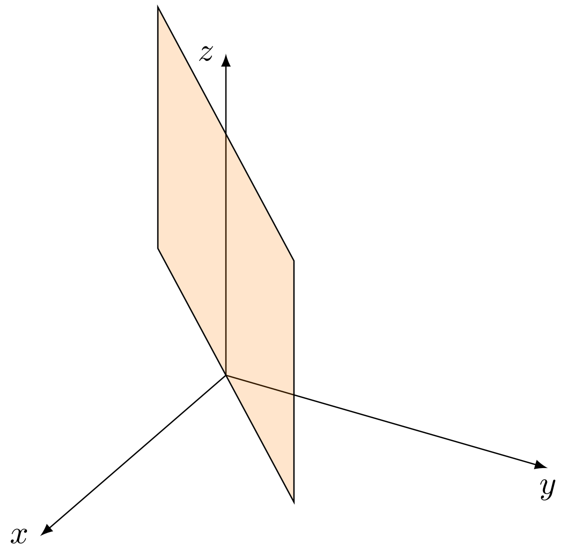
$$\begin{bmatrix} t_1 \\ t_1^3 \end{bmatrix} + \begin{bmatrix} t_2 \\ t_2^3 \end{bmatrix} = \begin{bmatrix} t_1 + t_2 \\ t_1^3 + t_2^3 \end{bmatrix}.$$

Since $(t_1 + t_2)^3 \neq t_1^3 + t_2^3$*, then* $\begin{bmatrix} t_1 \\ t_1^3 \end{bmatrix} + \begin{bmatrix} t_2 \\ t_2^3 \end{bmatrix} \notin S$*,*

which clearly implies that S *is not a subspace.*



(A) subspaces of $\mathbb{R}^2$.



(B) a subspace of $\mathbb{R}^3$.

FIGURE 1.6. Subfigure (A) stands for two sunspaces of $\mathbb{R}^2$, and Subfigure (B) illustrates a subspace of $\mathbb{R}^2$.

Span and subspace. Let $x_1, x_2, \ldots, x_p \in \mathbb{R}^n$ and $S = \text{Span}\{x_1, x_2, \ldots, x_p\}$. Then, from the definition of a spanning set, we obtain

(a) $0 = 0x_1 + 0x_2 + \ldots + 0x_p \in S$;

(b) For $s_1, s_2 \in S$, we have

$$\begin{aligned} s_1 + s_2 &= (\alpha_1 x_1 + \alpha_2 x_2 + \ldots + \alpha_p x_p) + (\beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_p x_p) \\ &= (\alpha_1 + \beta_1)x_1 + (\alpha_2 + \beta_2)x_2 + \ldots + (\alpha_p + \beta_p)x_p \in S; \end{aligned}$$

(c) For $s_1 \in S$ and $\gamma \in \mathbb{R}$, we have

$$\gamma s_1 = \gamma(\alpha_1 x_1 + \alpha_2 x_2 + \ldots + \alpha_p x_p) = \gamma\alpha_1 x_1 + \gamma\alpha_2 x_2 + \ldots + \gamma\alpha_p x_p \in S.$$

This clearly implies that $S = \text{Span}\{x_1, x_2, \ldots, x_p\}$ is subspace of $\mathbb{R}^n$.

CHAPTER 2

Matrices

In this chapter, we will introduce the class of matrices and its important subclasses, some of their properties, and the necessary operations on them.

Our main objectives are to:

(a) present the class of matrices and verify some of their properties;
(b) describe the necessary operations on matrices;
(c) introduce several important subclasses of matrices.

2.1. Matrices

A **matrix** in $\mathbb{R}^{m\times n}$ is a two-dimensional array that has m rows and n columns in which all of its elements are from $\mathbb{R}$, i.e.,

$$A \in \mathbb{R}^{m\times n} \quad \Leftrightarrow \quad A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}_{m\times n}, \quad a_{ij} \in \mathbb{R} \quad i = 1, \dots, n, \; j = 1, \dots, n.$$

We note that if $m \neq n$, A is a **rectangular** matrix; otherwise $(m = n)$, it is a **square** matrix. This matrix can be represented by rows ($r_i^T \in \mathbb{R}^n$, $i = 1, \dots, m$) or its columns ($c_i \in \mathbb{R}^m$, $i = 1, \dots, n$) as follows

$$A = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_m \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} = \begin{bmatrix} c_1 & c_2 & \cdots & c_n \end{bmatrix}.$$

Equal matrices. Two matrices A and B are **equal** if all of their elements are equal, which is $a_{ij} = b_{ij}$ for $i = 1, \dots, m$ and $j = 1, \dots, n$, i.e.,

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} = \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m1} & b_{m2} & \cdots & b_{mn} \end{bmatrix} = B \Leftrightarrow a_{ij} = b_{ij}, i = 1, \dots, m, j = 1, \dots, n.$$

Example 17. *Let us consider the* 3×3 *matrices*

$$A = \begin{bmatrix} 3 & 9 & 7 \\ 1 & 5 & 2 \\ 4 & 6 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & a & 7 \\ b & 5 & 2 \\ 4 & c & d \end{bmatrix}.$$

Then, the rows of A *are*

$$r_1 = \begin{bmatrix} 3 & 9 & 7 \end{bmatrix}, \quad r_2 = \begin{bmatrix} 1 & 5 & 2 \end{bmatrix}, \quad r_3 = \begin{bmatrix} 4 & 6 & 1 \end{bmatrix},$$

and the columns of A *are*

$$c_1 = \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix}, \quad c_2 = \begin{bmatrix} 9 \\ 5 \\ 6 \end{bmatrix}, \quad c_3 = \begin{bmatrix} 7 \\ 2 \\ 1 \end{bmatrix}.$$

In addition $A = B$ *if* $a = 9,\; b = 1, c = 6,$ *and* $d = 1$.

2.2. Important matrix classes

We next introduce several important classes of matrices that will be coming up frequently in our lectures and also in many real-world applications.

Symmetric matrix. A $n \times n$ matrix A is called **symmetric** if $a_{ij} = a_{ji}$ for $1 \leq i, j \leq n$, i.e.,

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{12} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{nn} \end{bmatrix}.$$

Zero matrix. A $m \times n$ matrix A is called **zero** or **null** if $a_{ij} = 0$ for $1 \leq i \leq m,\ 1 \leq j \leq n$, i.e.,

$$O_{m\times n} = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix}.$$

Diagonal matrix. A $n \times n$ matrix A is called **diagonal** if $a_{ij} = 0$ for $1 \leq i, j \leq n$ and $i \neq j$, i.e.,

$$A = \begin{bmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{bmatrix}.$$

Identity matrix. A $n \times n$ matrix A is called **identity** if it is diagonal with $a_{ii} = 1$ for $1 \leq i \leq n$, i.e.,

$$A = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}.$$

Scalar matrix. A $n \times n$ matrix A is called **scalar** if it is a multiplication of an identity matrix, $a_{ii} = c$ for $1 \leq i \leq n$ and $C \neq 0$, i.e.,

$$A = \begin{bmatrix} c & 0 & \cdots & 0 \\ 0 & c & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & c \end{bmatrix}.$$

Upper triangular matrix. A $n \times n$ matrix A is called **upper triangular** if $a_{ij} = 0$ for $j < i$, i.e.,

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{bmatrix}.$$

Lower triangular matrix. A $n \times n$ matrix A is called **upper triangular** if $a_{ij} = 0$ for $i < j$, i.e.,

$$A = \begin{bmatrix} a_{11} & 0 & \cdots & 0 \\ a_{21} & a_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}.$$

Diagonally dominant matrix. Let A be an $n \times n$ matrix, i.e.,

$$A = \begin{bmatrix} \mathbf{a_{11}} & a_{12} & \cdots & a_{1n} \\ a_{12} & \mathbf{a_{22}} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & \mathbf{a_{nn}} \end{bmatrix}.$$

Then,

(a) the matrix A is called **diagonally dominant** if for every row of the matrix, the magnitude of the diagonal entry in a row is larger than or equal to the sum of the magnitudes of all the other (non-diagonal) entries in that row, i.e.,

$$|a_{ii}| \geq \sum_{j \neq i} |a_{ij}|, \tag{2.1}$$

for all $i = 1, 2, \ldots, n$;

(b) the matrix A is called **strictly diagonally dominant** if for every row of the matrix, the magnitude of the diagonal entry in a row is larger than the sum of the magnitudes of all the other (non-diagonal) entries in that row, i.e.,

$$|a_{ii}| > \sum_{j \neq i} |a_{ij}|, \tag{2.2}$$

for all $i = 1, 2, \ldots, n$.

We next provide some examples of the above-mentioned matrices.

Example 18. *Let us give some 3×3 examples of the above matrices.*

(1) A is symmetric, but B is not symmetric:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 1 \\ 3 & 1 & 8 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & 4 & 1 \\ 5 & 7 & 2 \\ 6 & 3 & 4 \end{bmatrix}.$$

(2) A is a diagonal matrix, and B is an identity matrix:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 8 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

(3) A is zero matrix, and B is a scalar matrix:

$$A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}.$$

(4) A is an upper triangular matrix, and B is a lower triangular matrix:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 5 & 1 \\ 0 & 0 & 8 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & 0 & 0 \\ 5 & 7 & 0 \\ 6 & 3 & 4 \end{bmatrix}.$$

(5) A is a diagonally dominant matrix, and B is not a diagonally dominant matrix:

$$A = \begin{bmatrix} -5 & 2 & -2 \\ 1 & 7 & -6 \\ 3 & 0 & 8 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & 0 & 0 \\ 5 & 7 & -3 \\ 1 & 3 & 4 \end{bmatrix}.$$

(6) A is a strictly diagonally dominant matrix, and B is not a strictly diagonally dominant matrix:

$$A = \begin{bmatrix} 4 & 0 & -2 \\ 1 & 7 & 0 \\ -2 & 5 & -9 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & 0 & 0 \\ 5 & 7 & 0 \\ 1 & -3 & -4 \end{bmatrix}.$$

2.3. Matrix operations

In this subsection, we study the main operations on matrices that will be needed in the remainder of this document.

Matrix transpose. The **transpose** of a $m \times n$ matrix A is the $n \times m$ matrix A^T such that $(A^T)_{ij} = a_{ji}$ for $i = 1, \ldots, m$ and $j = 1, \ldots, n$, i.e.,

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}_{m\times n} \quad \Rightarrow \quad A^T = \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{bmatrix}_{n\times m}.$$

This means that if A is an $m \times n$ matrix, then A^T is an $n \times m$ matrix. Basically, by switching the rows of a matrix with its columns, one can easily get its transpose matrix.

Example 19. *For a given $A \in \mathbb{R}^{n\times n}$, the matrix*

$$A + A^T = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} + \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{n1} \\ a_{12} & a_{22} & \cdots & a_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{nn} \end{bmatrix} = \begin{bmatrix} 2a_{11} & a_{12}+a_{21} & \cdots & a_{1n}+a_{n1} \\ a_{21}+a_{12} & 2a_{22} & \cdots & a_{2n}+a_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1}+a_{1n} & a_{n2}+a_{2n} & \cdots & 2a_{nn} \end{bmatrix},$$

which is a symmetric matrix.

Trace of a matrix. Let A be an $n \times n$ square matrix. The **trace** of A is given by $\operatorname{tr}(A) = a_{11} + a_{22} + \ldots + a_{nn}$, i.e.,

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}_{n\times n} \quad \Rightarrow \operatorname{tr}(A) = a_{11} + a_{22} + \ldots + a_{nn}.$$

Example 20. *Let us consider the matrices*

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 1 \\ 3 & 6 & 8 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & 4 & 1 & 5 \\ 5 & 7 & 2 & 3 \\ 6 & 3 & 4 & 9 \end{bmatrix}.$$

Then, we have

$$A^T = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 6 \\ 3 & 1 & 8 \end{bmatrix}, \quad B^T = \begin{bmatrix} 4 & 5 & 6 \\ 4 & 7 & 3 \\ 1 & 2 & 4 \\ 5 & 3 & 9 \end{bmatrix}, \quad \operatorname{tr}(A) = 1 + 5 + 8 = 14.$$

Scalar multiplication. For $A \in \mathbb{R}^{m \times n}$ and $\alpha \in \mathbb{R}$, the **scalar multiplication** of α to A is given by

$$\alpha A = \alpha \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} = \begin{bmatrix} \alpha a_{11} & \alpha a_{12} & \cdots & \alpha a_{1n} \\ \alpha a_{21} & \alpha a_{22} & \cdots & \alpha a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha a_{m1} & \alpha a_{m2} & \cdots & \alpha a_{mn} \end{bmatrix}.$$

Matrices addition. For $A, B \in \mathbb{R}^{m \times n}$, the **matrix addition** of A and B is given by

$$\begin{aligned} A + B &= \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m1} & b_{m2} & \cdots & b_{mn} \end{bmatrix} \\ &= \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \cdots & a_{1n} + b_{1n} \\ a_{21} + b_{21} & a_{22} + b_{22} & \cdots & a_{2n} + b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} + b_{m1} & a_{m2} + b_{m2} & \cdots & a_{mn} + b_{mn} \end{bmatrix}. \end{aligned}$$

Note that for matrix addition $A + B$ the number of rows and columns of A and B should be the same.

Linear combination of matrices. For $A, B \in \mathbb{R}^{m \times n}$ and $\alpha, \beta \in \mathbb{R}$, the **linear combination** of A and B is given by

$$\begin{aligned} \alpha A + \beta B &= \begin{bmatrix} \alpha a_{11} & \alpha a_{12} & \cdots & \alpha a_{1n} \\ \alpha a_{21} & \alpha a_{22} & \cdots & \alpha a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha a_{m1} & \alpha a_{m2} & \cdots & \alpha a_{mn} \end{bmatrix} + \begin{bmatrix} \beta b_{11} & \beta b_{12} & \cdots & \beta b_{1n} \\ \beta b_{21} & \beta b_{22} & \cdots & \beta b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \beta b_{m1} & \beta b_{m2} & \cdots & \beta b_{mn} \end{bmatrix} \\ &= \begin{bmatrix} \alpha a_{11} + \beta b_{11} & \alpha a_{12} + \beta b_{12} & \cdots & \alpha a_{1n} + \beta b_{1n} \\ \alpha a_{21} + \beta b_{21} & \alpha a_{22} + \beta b_{22} & \cdots & \alpha a_{2n} + \beta b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha a_{m1} + \beta b_{m1} & \alpha a_{m2} + \beta b_{m2} & \cdots & \alpha a_{mn} + \beta b_{mn} \end{bmatrix}. \end{aligned}$$

Example 21. *Let us consider the matrices*

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 1 \\ 3 & 6 & 8 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & 4 & 1 \\ 5 & 7 & 2 \\ 6 & 3 & 4 \end{bmatrix}, \quad \alpha = 2, \quad \beta = 3.$$

Then, we have

$$\alpha A+\beta B = 2\begin{bmatrix}1&2&3\\2&5&1\\3&6&8\end{bmatrix}+3\begin{bmatrix}4&4&1\\5&7&2\\6&3&4\end{bmatrix}=\begin{bmatrix}2&4&6\\4&10&2\\6&12&16\end{bmatrix}+\begin{bmatrix}12&12&3\\15&21&6\\18&9&12\end{bmatrix}=\begin{bmatrix}14&16&9\\19&31&8\\24&21&28\end{bmatrix}.$$

Matrix-vector multiplication. For $A\in\mathbb{R}^{m\times n}$ and $x\in\mathbb{R}^n$, the **matrix-vector multiplication** of Ax is given by

$$Ax=\begin{bmatrix}a_{11}&a_{12}&\cdots&a_{1n}\\a_{21}&a_{22}&\cdots&a_{2n}\\\vdots&\vdots&\ddots&\vdots\\a_{m1}&a_{m2}&\cdots&a_{mn}\end{bmatrix}\begin{bmatrix}x_1\\x_2\\\vdots\\x_n\end{bmatrix}=\begin{bmatrix}\langle r_1,x\rangle\\\langle r_2,x\rangle\\\vdots\\\langle r_m,x\rangle\end{bmatrix}=\begin{bmatrix}a_{11}x_1+a_{12}x_2+\cdots+a_{1n}x_n\\a_{21}x_1+a_{22}x_2+\cdots+a_{2n}x_n\\\vdots\\a_{m1}x_1+a_{m2}x_2+\cdots+a_{mn}x_n\end{bmatrix}.$$

Note that in matrix-vector multiplication of A and x (i.e., Ax), we need the number of columns of A should be the same as the number of rows of x.

Example 22. *We consider a linear combination of matrices*

$$A=\begin{bmatrix}1&2&3\\2&5&1\\3&6&8\end{bmatrix},\quad B=\begin{bmatrix}4&4&1\\5&7&2\\6&3&4\end{bmatrix},\quad a=\begin{bmatrix}3\\2\\1\end{bmatrix},\quad b=\begin{bmatrix}2\\5\\4\end{bmatrix}.$$

Then, we have

$$Aa+Bb=\begin{bmatrix}1&2&3\\2&5&1\\3&6&8\end{bmatrix}\begin{bmatrix}3\\2\\1\end{bmatrix}+\begin{bmatrix}4&4&1\\5&7&2\\6&3&4\end{bmatrix}\begin{bmatrix}2\\5\\4\end{bmatrix}=\begin{bmatrix}10\\17\\29\end{bmatrix}+\begin{bmatrix}32\\53\\43\end{bmatrix}=\begin{bmatrix}42\\70\\72\end{bmatrix}.$$

Matrix-matrix multiplication. Let us consider the matrices $A\in\mathbb{R}^{m\times n}$ and $B\in\mathbb{R}^{n\times p}$, i.e.,

$$A=\begin{bmatrix}a_{11}&a_{12}&\cdots&a_{1n}\\a_{21}&a_{22}&\cdots&a_{2n}\\\vdots&\vdots&\ddots&\vdots\\a_{m1}&a_{m2}&\cdots&a_{mn}\end{bmatrix},\quad B=\begin{bmatrix}b_{11}&b_{12}&\cdots&b_{1p}\\b_{21}&b_{22}&\cdots&b_{2p}\\\vdots&\vdots&\ddots&\vdots\\b_{n1}&b_{n2}&\cdots&b_{np}\end{bmatrix}.$$

Then, the matrix-matrix multiplication AB is an $m\times p$ matrix given by

$$AB=\begin{bmatrix}a_{11}&a_{12}&\cdots&a_{1n}\\a_{21}&a_{22}&\cdots&a_{2n}\\\vdots&\vdots&\ddots&\vdots\\a_{m1}&a_{m2}&\cdots&a_{mn}\end{bmatrix}\begin{bmatrix}b_{11}&b_{12}&\cdots&b_{1p}\\b_{21}&b_{22}&\cdots&b_{2p}\\\vdots&\vdots&\ddots&\vdots\\b_{n1}&b_{n2}&\cdots&b_{np}\end{bmatrix}=\begin{bmatrix}\langle r_1^A,c_1^B\rangle&\langle r_1^A,c_2^B\rangle&\cdots&\langle r_1^A,c_p^B\rangle\\\langle r_2^A,c_1^B\rangle&\langle r_2^A,c_2^B\rangle&\cdots&\langle r_2^A,c_p^B\rangle\\\vdots&\vdots&\ddots&\vdots\\\langle r_m^A,c_1^B\rangle&\langle r_m^A,c_2^B\rangle&\cdots&\langle r_m^A,c_p^B\rangle\end{bmatrix}.$$

Note that in matrix-matrix multiplication of A and B (i.e., AB) we need that the number of columns of A should be the same as the number of rows of B.

$$\begin{bmatrix}a_{11}&\cdots&a_{1k}&\cdots&a_{1n}\\\vdots&&\vdots&&\vdots\\a_{i1}&\cdots&a_{ik}&\cdots&a_{in}\\\vdots&&\vdots&&\vdots\\a_{m1}&\cdots&a_{mk}&\cdots&a_{mn}\end{bmatrix}\cdot\begin{bmatrix}b_{11}&\cdots&b_{1j}&\cdots&b_{1p}\\\vdots&&\vdots&&\vdots\\b_{k1}&\cdots&b_{kj}&\cdots&b_{kp}\\\vdots&&\vdots&&\vdots\\b_{n1}&\cdots&b_{nj}&\cdots&b_{np}\end{bmatrix}=\begin{bmatrix}c_{11}&\cdots&c_{1j}&\cdots&c_{1p}\\\vdots&&\vdots&&\vdots\\c_{i1}&\cdots&\boxed{c_{ij}}&\cdots&c_{ip}\\\vdots&&\vdots&&\vdots\\c_{m1}&\cdots&c_{mj}&\cdots&c_{mp}\end{bmatrix}$$

FIGURE 2.1. Matrix-matrix product of two matrices $A,B\in\mathbb{R}^{m\times n}$.

Example 23. *We consider the following matrices*

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 1 \\ 3 & 6 & 8 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & 4 & 1 \\ 5 & 7 & 2 \\ 6 & 3 & 4 \end{bmatrix}.$$

Then, it holds that

$$AB = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 1 \\ 3 & 6 & 8 \end{bmatrix} \begin{bmatrix} 4 & 4 & 1 \\ 5 & 7 & 2 \\ 6 & 3 & 4 \end{bmatrix} = \begin{bmatrix} 32 & 27 & 17 \\ 39 & 46 & 16 \\ 90 & 72 & 47 \end{bmatrix}.$$

The main properties of matrix multiplication are given in the following:

(a) **Associative law is true.** For $A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{n\times p}$, and $C \in \mathbb{R}^{p\times q}$, it holds that
$$(AB)C = A(BC);$$

(b) **Left distributive law is true.** For $A \in \mathbb{R}^{m\times n}$ and $B, C \in \mathbb{R}^{n\times p}$, it holds that
$$A(B + C) = AB + AC;$$

(c) **Right distributive law is true.** For $A, B \in \mathbb{R}^{m\times n}$ and $C \in \mathbb{R}^{n\times p}$, it holds that
$$(A + B)C = AC + BC;$$

(d) **Commutative law is false.** For $A, B \in \mathbb{R}^{m\times m}$, it holds that
$$AB \neq BA.$$

Example 24. *Let us consider the* 3×3 *matrices*

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 1 \\ 3 & 6 & 8 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & 4 & 1 \\ 5 & 7 & 2 \\ 6 & 3 & 4 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Then, we have

$$AB = \begin{bmatrix} 32 & 27 & 17 \\ 39 & 46 & 16 \\ 90 & 72 & 47 \end{bmatrix}, \ (AB)C = \begin{bmatrix} 59 & 32 & 49 \\ 85 & 39 & 55 \\ 162 & 90 & 137 \end{bmatrix}, \ BC = \begin{bmatrix} 8 & 4 & 5 \\ 12 & 5 & 7 \\ 9 & 6 & 10 \end{bmatrix}, \ A(BC) = \begin{bmatrix} 59 & 32 & 49 \\ 85 & 39 & 55 \\ 162 & 90 & 137 \end{bmatrix}$$

and

$$\begin{bmatrix} 32 & 27 & 17 \\ 39 & 46 & 16 \\ 90 & 72 & 47 \end{bmatrix} = AB \neq BA = \begin{bmatrix} 15 & 34 & 24 \\ 25 & 57 & 38 \\ 24 & 51 & 53 \end{bmatrix}.$$

Matrix polynomial. Let A be an $n \times n$ matrix. Then,

$$A^0 = I_n, \quad A^q = AA\ldots A, \quad A^pA^q = A^{p+q}, \quad (A^p)^q = A^{pq}$$

and

$$P(A) = a_0A^0 + a_1A + a_2A^2 + \ldots + a_qA^q.$$

Example 25. *Let us consider the* 3×3 *matrix*

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 1 \\ 3 & 6 & 8 \end{bmatrix}.$$

Then, we have

$$A^2 = AA = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 1 \\ 3 & 6 & 8 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 1 \\ 3 & 6 & 8 \end{bmatrix} = \begin{bmatrix} 14 & 40 & 29 \\ 15 & 65 & 19 \\ 39 & 84 & 79 \end{bmatrix},$$

and

$$A^3 = A^2 A = \begin{bmatrix} 14 & 40 & 29 \\ 15 & 65 & 19 \\ 39 & 84 & 79 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 1 \\ 3 & 6 & 8 \end{bmatrix} = \begin{bmatrix} 181 & 412 & 314 \\ 202 & 469 & 253 \\ 444 & 972 & 833 \end{bmatrix},$$

and

$$5I + A^2 + A^3 = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix} + \begin{bmatrix} 14 & 40 & 29 \\ 15 & 65 & 19 \\ 39 & 84 & 79 \end{bmatrix} + \begin{bmatrix} 181 & 412 & 314 \\ 202 & 469 & 253 \\ 444 & 972 & 833 \end{bmatrix} = \begin{bmatrix} 200 & 452 & 343 \\ 217 & 539 & 272 \\ 483 & 1056 & 917 \end{bmatrix}.$$

Transpose properties. Let A and B be two $m \times n$ matrices and $\alpha, \beta \in \mathbb{R}$. Then,
(a) $(A^T)^T = A$;
(b) $(\alpha A + \beta B)^T = \alpha A^T + \beta B^T$;
(c) $(AB)^T = B^T A^T$.

Example 26. *Let us consider the* 3×3 *matrices*

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 1 \\ 3 & 6 & 8 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & 4 & 1 \\ 5 & 7 & 2 \\ 6 & 3 & 4 \end{bmatrix}, \quad \alpha = 2, \quad \beta = 5.$$

Then, we have

$$(\alpha A + \beta B)^T = \alpha A^T + \beta B^T = 2 \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 6 \\ 3 & 1 & 8 \end{bmatrix} + 5 \begin{bmatrix} 4 & 5 & 6 \\ 4 & 7 & 3 \\ 1 & 2 & 4 \end{bmatrix} = \begin{bmatrix} 22 & 29 & 36 \\ 24 & 45 & 27 \\ 11 & 12 & 36 \end{bmatrix}$$

and

$$(AB)^T = B^T A^T = \begin{bmatrix} 4 & 5 & 6 \\ 4 & 7 & 3 \\ 1 & 2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 6 \\ 3 & 1 & 8 \end{bmatrix} = \begin{bmatrix} 32 & 39 & 90 \\ 27 & 46 & 78 \\ 17 & 16 & 47 \end{bmatrix}.$$

Trace properties. Let A and B be two $m \times n$ matrices and $\alpha, \beta \in \mathbb{R}$. Then,
(a) $\mathrm{tr}(A^T) = \mathrm{tr}(A)$;
(b) $\mathrm{tr}(\alpha A + \beta B) = \alpha \mathrm{tr}(A) + \beta \mathrm{tr}(B)$;
(c) $\mathrm{tr}(A^T B) = \mathrm{tr}((A^T B)^T) = \mathrm{tr}(B^T A) = \sum_{i=1}^m \sum_{j=1}^n a_{ij} b_{ij}$.

Example 27. *Let us consider the* 3×3 *matrices*

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 1 \\ 3 & 6 & 8 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & 4 & 1 \\ 5 & 7 & 2 \\ 6 & 3 & 4 \end{bmatrix}, \quad \alpha = 2, \quad \beta = 5.$$

Then, we have

$$\mathrm{tr}(\alpha A + \beta B) = \alpha \mathrm{tr}(A) + \beta \mathrm{tr}(B) = 2 * 14 + 5 * 15 = 103$$

and

$$\mathrm{tr}(A^T B) = \mathrm{tr}((A^T B)^T) = \mathrm{tr}(B^T A) = 4 + 10 + 18 + 8 + 35 + 18 + 3 + 2 + 32 = 130,$$

as certified by the trace properties.

2.4. Elementary matrices and a basis for $\mathbb{R}^{m \times n}$

We conclude this section by providing a basis for the space of matrices. To do so, we define **elementary matrices** and show they are independent matrices and span $\mathbb{R}^{m \times n}$.

Linear independence matrices. Let $A_1, A_2, \ldots, A_p \in \mathbb{R}^{m\times n}$. Then, the vectors $A_1, A_2, \ldots, A_p$ are **linearly dependent** if there exist scalars $\alpha_1, \alpha_2, \ldots, \alpha_p$ (not all zero) such that

$$\alpha_1 A_1 + \alpha_2 A_2 + \ldots + \alpha_p A_p = 0. \tag{2.1}$$

If such scalars $\alpha_1, \alpha_2, \ldots, \alpha_p$ do not exist, then the vectors $A_1, A_2, \ldots, A_p$ are **linearly independent**. On the other words, the vectors $A_1, A_2, \ldots, A_p$ are linearly independent if the equation (2.1) leads to

$$\alpha_1 = \alpha_2 = \ldots = \alpha_p = 0.$$

Spanning matrices. The matrices $A_1, A_2, \ldots, A_p$ **span** $\mathbb{R}^{m\times n}$ if every matrix $A \in \mathbb{R}^{m\times n}$ can be written as a linear combination of these matrices, i.e., there exist scalars $\alpha_1, \alpha_2, \ldots, \alpha_p$ such that

$$A = \alpha_1 A_1 + \alpha_2 A_2 + \ldots + \alpha_p A_p.$$

Bases of $\mathbb{R}^{m\times n}$. The matrices $A_1, A_2, \ldots, A_p$ are called a **basis** for $\mathbb{R}^{m\times n}$ if they are spanning $\mathbb{R}^{m\times n}$ and are independent too.

Elementary matrices of $\mathbb{R}^{m\times n}$. The matrices

$$E_{11} = \begin{bmatrix} 1 & 0 & \ldots & 0 \\ 0 & 0 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & 0 \end{bmatrix}, \quad E_{12} = \begin{bmatrix} 0 & 1 & \ldots & 0 \\ 0 & 0 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & 0 \end{bmatrix}, \quad \cdots, \quad E_{1n} = \begin{bmatrix} 0 & 0 & \ldots & 1 \\ 0 & 0 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & 0 \end{bmatrix}, \quad \cdots,$$

$$E_{m1} = \begin{bmatrix} 0 & 0 & \ldots & 0 \\ 0 & 0 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & \ldots & 0 \end{bmatrix}, \quad E_{m2} = \begin{bmatrix} 0 & 0 & \ldots & 0 \\ 0 & 0 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 1 & \ldots & 0 \end{bmatrix}, \quad \cdots, \quad E_{mn} = \begin{bmatrix} 0 & 0 & \ldots & 0 \\ 0 & 0 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & 1 \end{bmatrix}.$$

are called **elementary matrices** of $\mathbb{R}^{m\times n}$.

We now show that the set $E = \{E_{11}, E_{12}, \ldots, E_{mn}\}$ is a basis for spaces of matrices. To do so, let us first show that these matrices are linearly independent, i.e.,

$$\begin{aligned} &\alpha_1 E_{11} + \alpha_2 E_{12} + \ldots + \alpha_{mn} E_{mn} \\ &= \alpha_1 \begin{bmatrix} 1 & 0 & \ldots & 0 \\ 0 & 0 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & 0 \end{bmatrix} + \alpha_2 \begin{bmatrix} 0 & 1 & \ldots & 0 \\ 0 & 0 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & 0 \end{bmatrix} + \ldots + \alpha_n \begin{bmatrix} 0 & 0 & \ldots & 1 \\ 0 & 0 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & 0 \end{bmatrix} \\ &+ \ldots + \alpha_{(m-1)n+1} \begin{bmatrix} 0 & 0 & \ldots & 0 \\ 0 & 0 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & \ldots & 0 \end{bmatrix} + \alpha_{(m-1)n+2} \begin{bmatrix} 0 & 0 & \ldots & 0 \\ 0 & 0 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 1 & \ldots & 0 \end{bmatrix} + \ldots + \alpha_{mn} \begin{bmatrix} 0 & 0 & \ldots & 0 \\ 0 & 0 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & 1 \end{bmatrix} \\ &= \begin{bmatrix} \alpha_1 & \alpha_2 & \ldots & \alpha_n \\ \alpha_{n+1} & \alpha_{n+2} & \ldots & \alpha_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{(m-1)n+1} & \alpha_{(m-1)n+2} & \ldots & \alpha_{(m-1)n+n} \end{bmatrix} = \begin{bmatrix} 0 & 0 & \ldots & 0 \\ 0 & 0 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & 0 \end{bmatrix}, \end{aligned}$$

which clearly implies that $\alpha_1 = \alpha_2 = \ldots = \alpha_{mn} = 0$, i.e., the matrices $E_{11}, E_{12}, \ldots, E_{mn}$ are linearly independent.

On the other hand, for any matrix $A \in \mathbb{R}^{m\times n}$, i.e.,

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix},$$

we have

$$a_{11}E_{11} + a_{12}E_{12} + \ldots + a_{mn}E_{mn} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} = A,$$

which means that $E_{11}, E_{12}, \ldots, E_{mn}$ span $\mathbb{R}^{m\times n}$. Together with the linear independence of the matrices $E_{11}, E_{12}, \ldots, E_{mn}$, this implies that $E_{11}, E_{12}, \ldots, E_{mn}$ are a basis for $\mathbb{R}^{m\times n}$.

CHAPTER 3

Linear Systems of Equations

In this chapter, we describe linear equations and linear systems of equations, and we will present a systematic way to find a possible solution to such systems.

Our main objectives are:

- to describe what a linear system of equations;
- to present how to find a solution (if any) of a linear system of equations.

3.1. Linear Equation

We begin by defining a *linear equation.* Let us consider some unknown variables $x, y, z \in \mathbb{R}$. Then, we may consider the linear combination of these variables as

$$2x - 3y = 5, \quad \frac{1}{2}x + y = 9, \quad 4x - 5y + 7z = 12,$$

where there are no products and/or equations of unknown variables x, y and z (i.e., the degree of the related polynomial is one). Such linear combinations are referred to as **linear equation** that we define next in a more general form.

Linear equation. Let us consider a vectors $a, x \in \mathbb{R}^n$ and a scalar $b \in \mathbb{R}$, i.e.,

$$a = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \in \mathbb{R}^n, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n, \quad , b \in \mathbb{R}.$$

Then, the equation

$$\langle a, x \rangle = a_1x_1 + a_2x_2 + \ldots + a_nx_n = b \tag{3.1}$$

is referred as a **linear equation**, where $a_1, a_2 \ldots, a_n$ are its n *coefficients* and $x_1, \ldots, x_n$ are its n *unknown variables.* A set of numbers $s_1, s_2 \ldots, s_n$ is a *solution* of the equation (3.1) if we set $x_1 = s_1, \ldots, x_n = s_n$, then the equation (3.1) is satisfied.

Example 28. *Let us consider the vectors* $a = \begin{bmatrix} -2 & 3 \end{bmatrix}^T \in \mathbb{R}^2$ *and* $x = \begin{bmatrix} x_1 & x_2 \end{bmatrix}^T \in \mathbb{R}^2$, *and let* $\langle a, x \rangle = -2x_1 + 3x_2 = 1$. *Then, it is clear that the numbers*

$$s_1 = 1, s_2 = 1, \quad s_1 = \tfrac{1}{2}, s_2 = \tfrac{2}{3}, \quad s_1 = -\tfrac{3}{2}, s_2 = -\tfrac{2}{3}, \ldots$$

are solutions of this equation. By simple algebra manipulation of this equation, we have

$$x_2 = \tfrac{1}{3} + \tfrac{2}{3}x_1 \quad \forall x_1 \in \mathbb{R}.$$

This means that all points on the line $x_2 = \frac{1}{3} + \frac{2}{3}x_1$ *is a solution of this equation, as illustrated in Figure 3.1.*

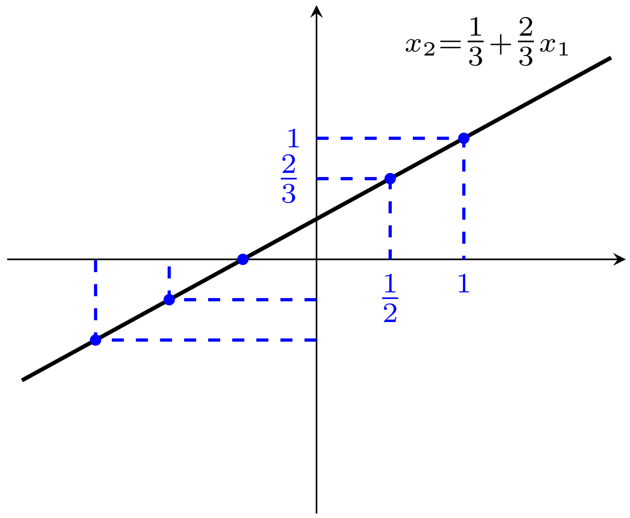


FIGURE 3.1. Solutions of the equation $-2x_1 + 3x_2 = 1$ are the line $x_2 = \frac{1}{3} + \frac{2}{3}x_1$.

Solutions of a linear equation. If $b = 0$, then $x_1 = x_2 = \ldots = x_n = 0$ is always a solution for this linear equation. In general, if $a_n \neq 0$, using simple algebraic manipulation, one will find solutions of the equation (3.1) by

$$x_n = \tfrac{b}{a_n} - (\tfrac{a_1}{a_n}x_1 + \ldots + \tfrac{a_{n-1}}{a_n}x_{n-1}), \quad \forall x_1, \ldots, x_{n-1} \in \mathbb{R}. \tag{3.2}$$

which is a n-dimensional the *hyperplane* that is a line if $n = 1$ (see Figure 3.1). We note that if the number of equations is more than one, then we may not have such a closed-form solution.

3.2. Linear systems of equations

In this subsection, we describe systems of linear equations (systems usually refer to more than one equation) and discuss solutions of such linear systems of equations. Let us begin with a simple example.

Example 29 (Finding solutions of linear systems graphically)**.** *Let us consider the linear system of equations*

$$\begin{cases} -2x_1 + 3x_2 = 1, \\ -4x_1 + 6x_2 = 2, \end{cases} \tag{3.1}$$

where the second equation is twice the first equation. Therefore, we reduce to the equation $-2x_1 + 3x_2 = 1$*, which has infinitely many solutions* $x_2 = -\frac{1}{3} + \frac{2}{3}x_1$ *for any* $x_1 \in \mathbb{R}$*.*

We now consider the linear system

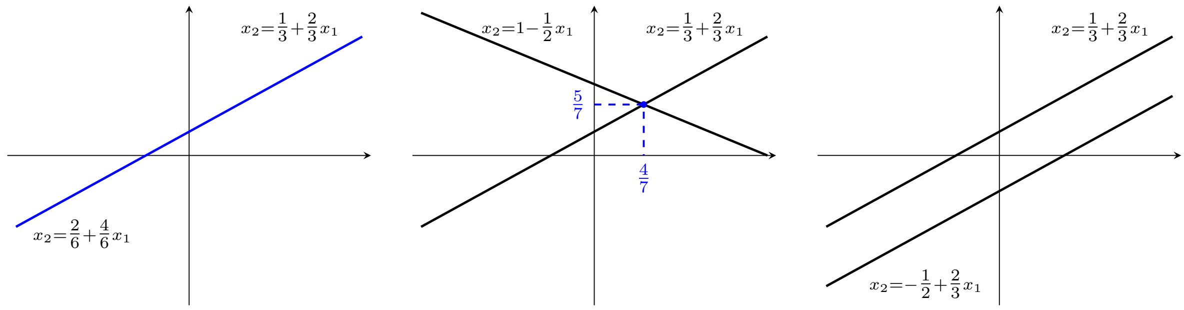
$$\begin{cases} -2x_1 + 3x_2 = 1, \\ x_1 + 2x_2 = 2. \end{cases} \tag{3.2}$$

Each of these equations are a line that we can draw and search for points that intersect. It is clear from Subfigure (a) of Figure 3.2 that there is a unique solution $(4/7, 5/7)$ *for this system.*

We also consider the linear systems of equations

$$\begin{cases} -2x_1 + 3x_2 = 1, \\ x_1 + 2x_2 = 3/2, \end{cases} \tag{3.3}$$

As shown in Subfigure (b) of Figure 3.2, the two lines are parallel and therefore there is no intersection, i.e., there is no solution for the latter system.



(A) Infini. many solutions

(B) Unique solution

(C) No solution

FIGURE 3.2. Subfigure (a) indicates that there are infinitely many solutions for the system (3.1), Subfigure (b) shows that there is the solution $(4/7, 5/7)$ for the system (3.2), and Subfigure (c) illustrates that there is no solution for the system (3.3).

Linear system of equations. Let $r_1, r_2, \ldots, r_m \in \mathbb{R}^n$, $x \in \mathbb{R}^n$, and $b \in \mathbb{R}^m$, i.e.,

$$r_1 = \begin{bmatrix} a_{11} \\ a_{12} \\ \vdots \\ a_{1n} \end{bmatrix} \in \mathbb{R}^n, \quad r_2 = \begin{bmatrix} a_{21} \\ a_{22} \\ \vdots \\ a_{2n} \end{bmatrix} \in \mathbb{R}^n, \quad \ldots, \quad r_m = \begin{bmatrix} a_{m1} \\ a_{m2} \\ \vdots \\ a_{mn} \end{bmatrix} \in \mathbb{R}^n, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n.$$

Then, the corresponding linear system of equations can be represented in the general form

$$\left\{ \begin{array}{lll} \langle r_1, x \rangle = a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n & = & b_1, \\ \langle r_2, x \rangle = a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n & = & b_2, \\ \vdots & \vdots & \vdots \\ \langle r_m, x \rangle = a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n & = & b_m, \end{array} \right. \tag{3.4}$$

where

$$\begin{gathered} a_{11}, a_{12}, \cdots, a_{1n}, \\ a_{21}, a_{22}, \cdots, a_{2n}, \\ \vdots \\ a_{m1}, a_{m2}, \cdots, a_{mn} \end{gathered}$$

are nm *coefficients*,

$$\begin{gathered} x_1, \\ x_2, \\ \vdots \\ x_n \end{gathered}$$

are n *unknown variables*,

$$\begin{gathered} b_1, \\ b_2, \\ \vdots \\ b_m \end{gathered}$$

are m *right-hand side constants*.

In the first case of Example 29, we have

$$r_1 = \begin{bmatrix} -2 \\ 3 \end{bmatrix}, \quad r_2 = \begin{bmatrix} -4 \\ 6 \end{bmatrix},$$

the coefficients are $a_{11} = -2, a_{12} = 3, a_{21} = -4, a_{22} = -6$, unknown variables are x_1 and x_2, and right-hand side constants are $b_1 = 1$ and $b_2 = 2$.

Matrix form of a linear system of equations. The linear system of equations (3.4) can be written in the matrix form

$$Ax = b,$$

where

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}, \tag{3.5}$$

where A is the *coefficient matrix* (a rectangular array of numbers).

(a) If $m < n$ (i.e., the number of equations is less than the number of unknown variables), it is called **underdetermined**;
(b) If $m > n$ (i.e., the number of equations is more than the number of unknown variables), it is called **overdetermined**;
(c) If $m = n$ (i.e., the number of equations is the same as the number of unknown variables), it is called **square**;
(d) If $b = 0$, this system is called **homogeneous**. In this case, $x = 0$ is always a solution, which is called **trivial solution**.

Example 30. *Let us consider the linear systems of equations*

$$\begin{cases} x_1 - 3x_2 + 9x_3 = 5, \\ -2x_1 + x_2 - 4x_3 = 7, \end{cases}, \quad \begin{cases} 5x_1 - x_2 = 1, \\ 2x_1 - 6x_2 = 3, \\ -x_1 + 5x_2 = -4, \end{cases}, \quad \begin{cases} 5x_1 - x_2 + x_3 = 2, \\ 2x_1 - 6x_2 + 2x_3 = -5, \\ -x_1 + 5x_2 + 8x_3 = 9. \end{cases}$$

In the first system, we have

$$A = \begin{bmatrix} 1 & -3 & 9 \\ -2 & 1 & -4 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad b = \begin{bmatrix} 5 \\ 7 \end{bmatrix},$$

which is an underdetermined system of equations with $m = 2$ and $n = 3$. In the second system, we have

$$A = \begin{bmatrix} 5 & -1 \\ 2 & -6 \\ -1 & 5 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 3 \\ -4 \end{bmatrix},$$

which is an overdetermined system of equations with $m = 3$ and $n = 2$. In the third system, we have

$$A = \begin{bmatrix} 5 & -1 & 1 \\ 2 & -6 & 2 \\ -1 & 5 & 8 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad b = \begin{bmatrix} 2 \\ -5 \\ 9 \end{bmatrix},$$

which is an square system of equations with $m = n = 3$.

Row picture of a system of equations. Let us consider a matrix $A \in \mathbb{R}^{m \times n}$ with columns $r_1, r_2, \ldots, r_m \in \mathbb{R}^n$, $x \in \mathbb{R}^n$, and $b \in \mathbb{R}$, i.e.,

$$A = \begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mn} \end{bmatrix}, \quad r_1 = \begin{bmatrix} a_{11} \\ a_{12} \\ \vdots \\ a_{1n} \end{bmatrix}, \quad r_2 = \begin{bmatrix} a_{21} \\ a_{22} \\ \vdots \\ a_{2n} \end{bmatrix}, \quad \ldots, \quad r_m = \begin{bmatrix} a_{m1} \\ a_{m2} \\ \vdots \\ a_{mn} \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}.$$

Then, the linear system $Ax = b$ can be rewritten as

$$Ax = \begin{bmatrix} \langle r_1, x \rangle \\ \langle r_2, x \rangle \\ \vdots \\ \langle r_m, x \rangle \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix},$$

where each of $\langle r_i, x \rangle = b_i$ $(i = 1, \ldots, m)$ is a hyperplane. Solutions of this system of equations are points where these hyperplanes intersect. Therefore, as suggested from Example 29, any system of equations with m equations and n variables satisfies one of the following conditions:

(a) There are infinitely many solutions;
(b) There is a unique solution;
(c) there is no solution.

If there is a solution for the system of equations (i.e., Cases (i) and (ii)), it is called **consistent**; however, if there is no solution (Case (iii)), it is called **inconsistent**.

Column picture of a system of equations. Let us consider a matrix $A \in \mathbb{R}^{m \times n}$ with columns $c_1, c_2, \ldots, c_n \in \mathbb{R}^m$, $x \in \mathbb{R}^n$, and $b \in \mathbb{R}$, i.e.,

$$A = \begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mn} \end{bmatrix}, \quad c_1 = \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix}, \quad c_2 = \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix}, \quad \ldots, \quad c_n = \begin{bmatrix} a_{n1} \\ a_{n2} \\ \vdots \\ a_{mn} \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}.$$

Then, the linear system $Ax = b$ can be rewritten as

$$Ax = x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} + \ldots + x_n \begin{bmatrix} a_{n1} \\ a_{n2} \\ \vdots \\ a_{mn} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}.$$

Hence, we have

$$Ax = x_1 c_1 + x_2 c_2 + \ldots + x_n c_n = b,$$

which implies that b is constructed by a linear combination of columns of the matrix A.

Solving diagonal system of equations. Let us solve a linear system of equations with a diagonal coefficient matrix A, i.e.,

$$Ax = \begin{bmatrix} a_{11} & 0 & \ldots & 0 \\ 0 & a_{22} & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix},$$

with $a_{ii} \neq 0$ for $i = 1, 2, \ldots, n$. This consequently leads to

$$x_1 = \tfrac{b_1}{a_{11}}, \quad x_2 = \tfrac{b_2}{a_{22}}, \quad \ldots, x_n = \tfrac{b_n}{a_{nn}}.$$

Solving lower triangular system of equations. Let us solve a linear system of equations with a lower triangular coefficient matrix A, i.e.,

$$Ax = \begin{bmatrix} a_{11} & 0 & \dots & 0 \\ a_{21} & a_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix},$$

with $a_{ii} \neq 0$ for $i = 1, 2, \dots, n$. For this system of equations, we start from the first equation and compute x_1, substitute x_1 into the second equation, and compute x_2. We continue this procedure until computing x_n.

Solving upper triangular system of equations. Let us solve a linear system of equations with an upper triangular coefficient matrix A, i.e.,

$$Ax = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ 0 & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix},$$

with $a_{ii} \neq 0$ for $i = 1, 2, \dots, n$. For this system of equations, we start from the last equation and compute x_n, substitute x_n into the $(n-1)$th equation and compute x_{n-1}. We continue this procedure until computing x_1.

Example 31. *Let us consider the following systems of equations*

$$\begin{cases} 2x_1 = 3, \\ 3x_2 = 6, \\ 7x_3 = 12, \end{cases}, \quad \begin{cases} 4x_1 = 12, \\ 3x_1 + 6x_2 = 6, \\ 6x_1 - 2x_2 + 5x_3 = 14, \end{cases}, \quad \begin{cases} 3x_1 + x_2 - x_3 = 4, \\ x_2 + 4x_3 = 6, \\ -2x_3 = 4. \end{cases}$$

The first system has a diagonal coefficient matrix, and the solution is given by

$$x_1 = \tfrac{3}{2}, \quad x_2 = \tfrac{6}{3} = 2, \quad x_3 = \tfrac{12}{7}.$$

The second system has a lower triangular coefficient matrix. Hence, we begin from the first equation, leading to $x_1 = 3$. Substituting this into the second equation leads to $9 + 6x_2 = 6$, implying $x_2 = -1/2$. Substituting $x_1 = 3$ and $x_2 = -1/2$ into the third equation, we come to $19 + 5x_3 = 14$ leading to $x_3 = 1$, i.e.,

$$x_1 = 3, \quad x_2 = -\tfrac{1}{2}, \quad x_3 = 1.$$

The third system has an upper triangular coefficient matrix. Hence, we begin from the last equation leading to $x_3 = -2$. Substituting this into the second equation leads to $-2 + 4x_3 = 6$, implying $x_2 = 2$. Substituting $x_3 = -2$ and $x_2 = 2$ into the first equation, we come to $3x_1 + 4 = 4$ leading to $x_3 = 0$, i.e.,

$$x_1 = 0, \quad x_2 = 2, \quad x_3 = -2.$$

In the next section, we will present a systematic way to solve general systems of equations of the form(3.4).

3.3. Gauss-Jordan elimination

In this subsection, we describe a systematic way of solving a system of equations, which is called Gauss-Jordan elimination. Let us begin with the following simple example.

Example 32. *Let us consider the following linear system of equations*

$$\begin{cases} -2x_1 + 3x_2 = 1, \\ x_1 + 2x_2 = 2. \end{cases}$$

We multiply the first equation by $-1/2$ and add -1 times the first equation to the second one, i.e.,

$$\begin{cases} x_1 - 3/2x_2 = -1/2, \\ 7/2x_2 = 5/2. \end{cases}$$

First, we multiply the second equation by $2/7$ *and then multiply it by* $3/2$ *and add it to the first equation, i.e.,*

$$\begin{cases} x_1 = 8/14, \\ x_2 = 5/7, \end{cases}$$

which implies that $x_1 = 4/7$ *and* $x_2 = 5/7$ *is the unique solution of this system, as verified in Subfigure (b) of Figure 3.2.*

The above example provides us with a systematic idea for solving general systems of equations; however, before presenting this scheme, we need the following definitions.

Augmented matrix and elementary row operations Let $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$.

- **Augmented matrix.**

$$A = \left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right] \tag{3.1}$$

- **Scaling row by a nonzero constant** c**.**

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right] \xrightarrow{r_1 \to cr_1} \left[\begin{array}{cccc|c} ca_{11} & ca_{12} & \cdots & ca_{1n} & cb_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right]$$

- **Sum of two rows.**

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right] \xrightarrow{r_1 \to r_1 + r_2} \left[\begin{array}{cccc|c} a_{11}+a_{21} & a_{12}+a_{22} & \cdots & a_{1n}+a_{2n} & b_1+b_2 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right]$$

- **Swap two rows.**

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right] \xrightarrow{r_1 \leftrightarrow r_2} \left[\begin{array}{cccc|c} a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right]$$

Example 33. *Let us reconsider the system in Example 32. The corresponding augmented matrix is given by*

$$\left[\begin{array}{cc|c} -2 & 3 & 1 \\ 1 & 2 & 2 \end{array}\right].$$

We verify a solution for this system of equations using elementary row operations. To do so, we may follow the next steps:

$$\left[\begin{array}{cc|c} -2 & 3 & 1 \\ 1 & 2 & 2 \end{array}\right] \xrightarrow{r_1 \to -1/2 r_1} \left[\begin{array}{cc|c} 1 & -3/2 & -1/2 \\ 1 & 2 & 2 \end{array}\right] \xrightarrow{r_2 \to -r_1 + r_2} \left[\begin{array}{cc|c} 1 & -3/2 & -1/2 \\ 0 & 7/2 & 5/2 \end{array}\right]$$

and

$$\left[\begin{array}{cc|c} 1 & -3/2 & -1/2 \\ 0 & 7/2 & 5/2 \end{array}\right] \xrightarrow{r_2 \to 2/7 r_2} \left[\begin{array}{cc|c} 1 & -3/2 & -1/2 \\ 0 & 1 & 5/7 \end{array}\right] \xrightarrow{r_1 \to r_1 + 3/2 r_2} \left[\begin{array}{cc|c} 1 & 0 & 4/7 \\ 0 & 1 & 5/7 \end{array}\right],$$

which leads to the unique solution $(4.7, 5/7)$.

The above example illustrates a systematic way for solving a general system of equations, which we present next as the Gauss-Jordan elimination algorithm.

Algorithm 1: Gauss-Jordan Elimination Algorithm

```
begin
    Swap the rows such that all zero rows are on the bottom;
    repeat
        Swap the rows such that the row with the largest, leftmost nonzero entry is on top;
        Scale the top row such that the top row's leading entry becomes unity;
        Add multiples of the top row to others such that the other entries of this column are zero;
    until All the leading entries are unity
    Swap the rows such that the leading entry of each nonzero row is to the right of the leading
      entry of the above row;
end
```

Example 34 (inconsistent system)**.** *Let us consider the linear system*

$$\begin{cases} x_1 + 2x_2 - 3x_3 = 5, \\ x_1 + 3x_2 + 5x_3 = 4, \\ 2x_1 + 5x_2 + 2x_3 = 8, \end{cases}$$

Applying the Gauss-Jordan elimination algorithm, we come to

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & 5 \\ 1 & 3 & 5 & 4 \\ 2 & 5 & 2 & 8 \end{array}\right] \xrightarrow{r_1 \leftrightarrow r_3} \left[\begin{array}{ccc|c} 2 & 5 & 2 & 8 \\ 1 & 3 & 5 & 4 \\ 1 & 2 & -3 & 5 \end{array}\right] \xrightarrow{1/2r_1 \to r_1} \left[\begin{array}{ccc|c} 1 & 5/2 & 1 & 4 \\ 1 & 3 & 5 & 4 \\ 1 & 2 & -3 & 5 \end{array}\right] \xrightarrow{-r_1+r_2 \to r_2,\ -r_1+r_3 \to r_3}$$

$$\left[\begin{array}{ccc|c} 1 & 5/2 & 1 & 4 \\ 0 & 1/2 & 4 & 0 \\ 0 & -1/2 & -4 & 1 \end{array}\right] \xrightarrow{2r_2 \to r_2} \left[\begin{array}{ccc|c} 1 & 5/2 & 1 & 4 \\ 0 & 1 & 8 & 0 \\ 0 & -1/2 & -4 & 1 \end{array}\right] \xrightarrow{1/2r_2+r_3 \to r_3,\ -5/2r_2+r_1 \to r_1} \left[\begin{array}{ccc|c} 1 & 0 & -19 & 4 \\ 0 & 1 & 8 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

It is clear that the latter r_3 leads to $0 = 1$, which is a contradiction. Therefore, this linear system is inconsistent.

Example 35 (infinitely many solutions)**.** *Let us consider the linear system*

$$\begin{cases} 2x_1 + x_2 - x_3 = 4, \\ x_1 + 3x_2 + 2x_3 = 5, \\ 3x_1 + 4x_2 + x_3 = 9, \end{cases}$$

Applying the Gauss-Jordan elimination algorithm, we come to

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 4 \\ 1 & 3 & 2 & 5 \\ 3 & 4 & 1 & 9 \end{array}\right] \xrightarrow{r_1 \leftrightarrow r_3} \left[\begin{array}{ccc|c} 3 & 4 & 1 & 9 \\ 1 & 3 & 2 & 5 \\ 2 & 1 & -1 & 4 \end{array}\right] \xrightarrow{1/3r_1 \to r_1} \left[\begin{array}{ccc|c} 1 & 4/3 & 1/3 & 3 \\ 1 & 3 & 2 & 5 \\ 2 & 1 & -1 & 4 \end{array}\right] \xrightarrow{-r_1+r_2 \to r_2,\ -2r_1+r_3 \to r_3}$$

$$\left[\begin{array}{ccc|c} 1 & 4/3 & 1/3 & 3 \\ 0 & 5/3 & 5/3 & 2 \\ 0 & -5/3 & -5/3 & -2 \end{array}\right] \xrightarrow{3/5r_2 \to r_2} \left[\begin{array}{ccc|c} 1 & 4/3 & 1/3 & 3 \\ 0 & 1 & 1 & 6/5 \\ 0 & -5/3 & -5/3 & -2 \end{array}\right] \xrightarrow{5/3r_2+r_3 \to r_3,\ -4/3r_2+r_1 \to r_1}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 7/5 \\ 0 & 1 & 1 & 6/5 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Since $r_3 = 0$, it is clear that the system has infinitely many solutions.

CHAPTER 4

Matrix Four Spaces

In this chapter, we describe spaces (column space, row space, nullspace, left nullspace) corresponding to a given matrix and verify their properties.

Our main objectives involve:

- introducing matrix four spaces (i.e., row space, column space, nullspace, and left nullspace);
- defining the rank of matrices;
- verifying the dimensions of row space, column space, nullspace, and left nullspace.

4.1. Column space

In this subsection, we introduce the column space of an arbitrary matrix A and investigate some of its properties.

Column space. For a given matrix $A \in \mathbb{R}^{m\times n}$, the set of all linear combinations of columns of A forms the **column space** which is denoted by $C(A)$. Since

$$Ax = x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} + \ldots + x_n \begin{bmatrix} a_{n1} \\ a_{n2} \\ \vdots \\ a_{mn} \end{bmatrix},$$

we have

$$C(A) = \{Ax \in \mathbb{R}^m \mid x \in \mathbb{R}^n\}. \tag{4.1}$$

It is clear that the linear system on equations $Ax = b$ has a solution if and only if $b \in C(A)$.

Example 36. *Let us consider the matrices*

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 1 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix}.$$

It is clear that

$$C(A) = \left\{Ax \mid x \in \mathbb{R}^2\right\} = \left\{ x_1 \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix} \mid x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right\}$$

Therefore, the linear system $Ax = b$ has a solution if and only if $b \in \operatorname{span}\left\{c_1^A, c_2^A\right\}$ (c_i^A denotes the column i of A), which is a subspace spanned by these two vectors. On the other hand, for B, we have

$$C(B) = \left\{Bx \mid x \in \mathbb{R}^2\right\} = \left\{ x_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} \mid x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right\}$$
$$= \left\{ (x_1 + 2x_2) \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \mid x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right\}.$$

Hence, $C(B)$ is spanned by the column $c_1 = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}^T$.

Column space is a subspace of $\mathbb{R}^m$. For a given matrix $A \in \mathbb{R}^{m\times n}$, it follows from the definition of the column space that

- $0 \in C(A)$ because $A0 = 0 \in C(A)$;
- If $c_1, c_2 \in C(A)$, then there exists $x_1, x_2 \in \mathbb{R}^n$ such that $c_1 = Ax_1$ and $c_2 = Ax_2$, i.e.,
$$c_1 + c_2 = Ax_1 + Ax_2 = A(x_1 + x_2) \in C(A);$$
- If $c_1 \in C(A)$ and $\gamma \in \mathbb{R}$, then
$$\gamma c_1 = \gamma A x_1 = A(\gamma x_1) \in C(A).$$

‘ Therefore, $C(A)$ is a subspace of $\mathbb{R}^m$.

Example 37. *Let us describe the column space of the matrices*
$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 4 \\ 3 & 6 \end{bmatrix}, \quad D = \begin{bmatrix} 2 & 6 & 1 \\ 0 & 0 & 5 \end{bmatrix}.$$
The columns of A are independent vectors and therefore span $\mathbb{R}^2$. For B, we have $c_2^B = 2c_1^B$ that means c_1^B and c_2^B are dependent, i.e., $C(B) = \left\{\alpha \begin{bmatrix} 2 & 3 \end{bmatrix}^T \mid \alpha \in \mathbb{R}\right\}$. For D, we have
$$\begin{aligned} C(D) &= \left\{ \alpha \begin{bmatrix} 2 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 6 \\ 0 \end{bmatrix} + \gamma \begin{bmatrix} 1 \\ 5 \end{bmatrix} \right\} \\ &= \left\{ (\alpha + 3\beta) \begin{bmatrix} 2 \\ 0 \end{bmatrix} + \gamma \begin{bmatrix} 1 \\ 5 \end{bmatrix} \right\}, \end{aligned}$$
which is $\mathbb{R}^2$.

4.2. Nullspace and rank of matrices

In this subsection, we introduce the nullspace and rank of an arbitrary matrix A and investigate some of their properties.

Nullspace. For a given matrix $A \in \mathbb{R}^{m\times n}$, the set of all solutions of the homogeneous linear system $Ax = 0$ is called **nullspace** of A, which is denoted by $N(A)$, i.e.,
$$N(A) = \{x \in \mathbb{R}^n \mid Ax = 0\}. \tag{4.1}$$
It is clear that the zero vector $x = 0$ satisfies $Ax = 0$, implying that $0 \in N(A)$.

Example 38. *Let us consider the matrix*
$$A = \begin{bmatrix} 3 & 1 \\ 6 & 2 \end{bmatrix},$$
where we aim at identifying its nullspace. Applying the Gauss-Jordan elimination leads to
$$\left[\begin{array}{cc|c} 3 & 1 & 0 \\ 6 & 2 & 0 \end{array}\right] \xrightarrow{r_1 \leftrightarrow r_2} \left[\begin{array}{cc|c} 6 & 2 & 0 \\ 3 & 1 & 0 \end{array}\right] \xrightarrow{r_1 \to 1/6 r_1} \left[\begin{array}{cc|c} 1 & 1/3 & 0 \\ 3 & 1 & 0 \end{array}\right] \xrightarrow{r_2 \to r_2 - 3r_1} \left[\begin{array}{cc|c} 1 & 1/3 & 0 \\ 0 & 0 & 0 \end{array}\right],$$
Hence, all points satisfying $x_1 + \frac{1}{3}x_2 = 0$ are solutions of this linear system. From the row picture point of view, we can see that $3x_1 + x_2 = 0$ is the same as $6x_1 + 2x_2 = 0$. Therefore, we come to
$$N(A) = \left\{x \in \mathbb{R}^2 \mid 3x_1 + x_2 = 0\right\},$$
which is a line in two-dimensional space. For example, we can see $s = \begin{bmatrix} 1 & -3 \end{bmatrix}^T$ is a **special solution** *of this equation by setting $x_2 = 3$ and the remainder of the solutions are multiples of this vector, i.e.,*
$$N(A) = \operatorname{span}\left\{ \begin{bmatrix} 1 \\ -3 \end{bmatrix} \right\}.$$
Note that one can select a different value for x_2; however, the nullspace will be the same as that presented in the last identity.

Example 39. *Let us find the nullspace of the matrices*

$$A = \begin{bmatrix} 5 & 2 \\ 1 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} A \\ 2A \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ 1 & 4 \\ 10 & 4 \\ 2 & 8 \end{bmatrix}, \quad C = \begin{bmatrix} A & 2A \end{bmatrix} = \begin{bmatrix} 5 & 2 & 10 & 4 \\ 1 & 4 & 2 & 8 \end{bmatrix}.$$

Then, for the matrix A, we have

$$\left[\begin{array}{cc|c} 5 & 2 & 0 \\ 1 & 4 & 0 \end{array}\right] \xrightarrow{r_1 \to 1/5 r_1} \left[\begin{array}{cc|c} 1 & 2/5 & 0 \\ 1 & 4 & 0 \end{array}\right] \xrightarrow{r_2 \to r_2 - r_1} \left[\begin{array}{cc|c} 1 & 2/5 & 0 \\ 0 & 18/5 & 0 \end{array}\right] \xrightarrow{r_2 \to 5/18 r_2} \left[\begin{array}{cc|c} 1 & 2/5 & 0 \\ 0 & 1 & 0 \end{array}\right]$$
$$\xrightarrow{r_1 \to r_1 - 2/5 r_2} \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \end{array}\right],$$

which implies that the trivial solution $x = \begin{bmatrix} 0 & 0 \end{bmatrix}^T$ is the only point in $N(A)$.

For the matrix B, we have

$$\left[\begin{array}{cc|c} 5 & 2 & 0 \\ 1 & 4 & 0 \\ 10 & 4 & 0 \\ 2 & 8 & 0 \end{array}\right] \xrightarrow{r_1 \leftrightarrow r_3} \left[\begin{array}{cc|c} 10 & 4 & 0 \\ 1 & 4 & 0 \\ 5 & 2 & 0 \\ 2 & 8 & 0 \end{array}\right] \xrightarrow{r_1 \to 1/10 r_1} \left[\begin{array}{cc|c} 1 & 4/10 & 0 \\ 1 & 4 & 0 \\ 5 & 2 & 0 \\ 2 & 8 & 0 \end{array}\right] \xrightarrow{r_2 \to r_2 - r_1,\ r_3 \to r_3 - 5r_1,\ r_4 \to r_4 - 2r_1}$$
$$\left[\begin{array}{cc|c} 1 & 4/10 & 0 \\ 0 & 36/10 & 0 \\ 0 & 16/10 & 0 \\ 0 & 76/10 & 0 \end{array}\right] \xrightarrow{r_2 \leftrightarrow r_4} \left[\begin{array}{cc|c} 1 & 4/10 & 0 \\ 0 & 76/10 & 0 \\ 0 & 16/10 & 0 \\ 0 & 36/10 & 0 \end{array}\right] \xrightarrow{r_1 \to 10/76 r_1} \left[\begin{array}{cc|c} 1 & 4/10 & 0 \\ 0 & 1 & 0 \\ 0 & 16/10 & 0 \\ 0 & 36/10 & 0 \end{array}\right]$$
$$\xrightarrow{r_1 \to r_1 - 4/10 r_2,\ r_3 \to r_3 - 16/10 r_2,\ r_4 \to r_4 - 36/10 r_1} \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right].$$

Hence, the nullspace of A and B is the same.

For the matrix C, we have

$$\left[\begin{array}{cccc|c} 5 & 2 & 10 & 4 & 0 \\ 1 & 4 & 2 & 8 & 0 \end{array}\right] \xrightarrow{r_1 \to 1/5 r_1} \left[\begin{array}{cccc|c} 1 & 2/5 & 2 & 4/5 & 0 \\ 1 & 4 & 2 & 8 & 0 \end{array}\right] \xrightarrow{r_2 \to r_2 - r_1} \left[\begin{array}{cccc|c} 1 & 2/5 & 2 & 4/5 & 0 \\ 0 & 18/5 & 0 & 36/5 & 0 \end{array}\right] \xrightarrow{r_2 \to 5/18 r_2}$$
$$\left[\begin{array}{cccc|c} 1 & 2/5 & 2 & 4/5 & 0 \\ 0 & 1 & 0 & 2 & 0 \end{array}\right] \xrightarrow{r_1 \to r_1 - 2/5 r_2} \left[\begin{array}{cccc|c} 1 & 0 & 2 & 0 & 0 \\ 0 & 1 & 0 & 2 & 0 \end{array}\right],$$

leading to the equations

$$x_1 = -2x_3, \quad x_2 = -2x_4.$$

Hence, we come to

$$N(C) = \left\{ x \in \mathbb{R}^4 \mid x_1 = -2x_3,\ x_2 = -2x_4 \right\}.$$

Setting $x_3 = 0, x_4 = 1$, we come to the special solution $\begin{bmatrix} -2 & 0 & 1 & 0 \end{bmatrix}^T$, and setting $x_3 = 1, x_4 = 0$ we come to the special solution $\begin{bmatrix} 0 & -2 & 0 & 1 \end{bmatrix}^T$. Therefore, the nullspace of C is given by linear combinations of these two vectors, i.e.,

$$N(C) = \operatorname{span}\left\{ \begin{bmatrix} -2 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -2 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Nullspace and solutions of $Ax = b$. For a given matrix $A \in \mathbb{R}^{m \times n}$, if $x \in N(A)$ and s be solution of $Ax = b$, then

$$A(s + \lambda x) = As + \lambda Ax = As = b, \tag{4.2}$$

which implies that $s + \lambda x$ is a solution of $Ax = b$.

Nullspace of $A \in \mathbb{R}^{m\times n}$ with $m < n$. For a given matrix $A \in \mathbb{R}^{m\times n}$, if $m < n$ (a short wide matrix), then the nullspace $N(A)$ is nontrivial.

Example 40. *Let us consider the matrix*

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 4 & 1 & 6 \end{bmatrix}$$

Then, the echelon form of this matrix is constructed as follows

$$\begin{bmatrix} 1 & 0 & 3 \\ 4 & 1 & 6 \end{bmatrix} \xrightarrow{r_1 \leftrightarrow r_2} \begin{bmatrix} 4 & 1 & 6 \\ 1 & 0 & 3 \end{bmatrix} \xrightarrow{r_1 \to 1/4 r_1} \begin{bmatrix} 1 & 1/4 & 3/2 \\ 1 & 0 & 3 \end{bmatrix} \xrightarrow{r_2 \to r_2 - r_1} \begin{bmatrix} 1 & 1/4 & 3/2 \\ 0 & -1/4 & 3/2 \end{bmatrix} \xrightarrow{r_2 \to -4r_2}$$
$$\begin{bmatrix} 1 & 1/4 & 3/2 \\ 0 & 1 & -6 \end{bmatrix} \xrightarrow{r_1 \to r_1 - 1/4 r_2} \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & -6 \end{bmatrix}.$$

Hence, the nullspace of A is given by

$$N(A) = \left\{ x \in \mathbb{R}^3 \mid x_1 = -3x_3,\ x_2 = 6x_3 \right\}.$$

Let us set $x_3 = 1$, i.e.,

$$N(A) = \operatorname{span}\left\{ \begin{bmatrix} -3 \\ 6 \\ 1 \end{bmatrix} \right\}.$$

Rank of a matrix $A \in \mathbb{R}^{m\times n}$. For a given matrix $A \in \mathbb{R}^{m\times n}$, the dimension of the column space of A (the number of vectors in a basis of column space $C(A)$) is called **rank** of A denoted by $\operatorname{rank}(A)$.

Example 41. *Let us consider the matrix*

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \\ 5 & 10 \\ 7 & 14 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

It is clear that $c_2^A = 2c_1^A$. Hence, the vector c_1^A is a basis for the column space of this matrix, which implies rank$(A) = 1$. Since $c_1^B = e_1$, $c_2^B = e_2$, and $c_3^B = e_3$ and these vectors are a basis for $\mathbb{R}^3$, the dimension of the column space of B is three, i.e., $\operatorname{rank}(B) = 3$.

Computing the rank of matrices. For a given matrix $A \in \mathbb{R}^{m\times n}$, we consider the following steps

(a) Compute the reduced (echelon) form of the matrix A;
(b) The number of pivots of A is the rank of A.

Example 42. *Let us consider the matrix*

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \\ 5 & 10 \\ 7 & 14 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Let us compute the echelon form of A as follows

$$\begin{bmatrix} 1 & 2 \\ 3 & 6 \\ 5 & 10 \\ 7 & 14 \end{bmatrix} \xrightarrow{r_1 \leftrightarrow r_4} \begin{bmatrix} 7 & 14 \\ 3 & 6 \\ 5 & 10 \\ 1 & 2 \end{bmatrix} \xrightarrow{r_1 \to 1/7 r_1} \begin{bmatrix} 1 & 2 \\ 3 & 6 \\ 5 & 10 \\ 1 & 2 \end{bmatrix} \xrightarrow{r_2 \to r_2 - 3r_1, r_3 \to r_3 - 5r_1, r_4 \to r_4 - r_1} \begin{bmatrix} 1 & 2 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Since the number of pivots of this matrix is one, we have $\operatorname{rank}(A) = 1$. *The number of pivots of B is three, we have* $\operatorname{rank}(B) = 3$.

Exercise 43. *Show the left nullspace $N(A)$ is a subspace of $\mathbb{R}^n$.*

4.3. Row space

In this subsection, we introduce the row space of an arbitrary matrix A and investigate some of its properties.

Row space. For a given matrix $A \in \mathbb{R}^{m\times n}$, the subspace spanned by the rows of A is named **row space**. In other words, row space of A is column space of A^T ($C(A^T)$).

Example 44. *Let us consider the matrix*

$$A = \begin{bmatrix} 1 & 4 \\ 3 & 12 \\ 3 & 5 \end{bmatrix},$$

where we aim to find its row space. The transpose of this matrix is given by

$$A^T = \begin{bmatrix} 1 & 3 & 3 \\ 4 & 12 & 5 \end{bmatrix}, \quad c_1 = \begin{bmatrix} 1 \\ 4 \end{bmatrix}, \quad c_2 = \begin{bmatrix} 3 \\ 12 \end{bmatrix}, \quad c_3 = \begin{bmatrix} 3 \\ 5 \end{bmatrix}.$$

From the definition of row space, we obtain

$$\alpha_1 c_1 + \beta c_2 + \gamma c_3 = \alpha_1 \begin{bmatrix} 1 \\ 4 \end{bmatrix} + \beta \begin{bmatrix} 3 \\ 12 \end{bmatrix} + \gamma \begin{bmatrix} 3 \\ 5 \end{bmatrix} = (\alpha_1 + 3\beta) \begin{bmatrix} 1 \\ 4 \end{bmatrix} + \gamma \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

which is a plane in $\mathbb{R}^3$.

Basis for column and row spaces. The pivot columns of A are a basis for its column space. The pivot rows of A are a basis for its row space.

Example 45. *Let us consider the matrix*

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix},$$

where its echelon form is given by

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \xrightarrow{r_1 \leftrightarrow r_2} \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix} \xrightarrow{r_1 \to 1/2 r_1} \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix} \xrightarrow{r_2 \to r_2 - r_1} \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}.$$

Therefore, the vectors $\begin{bmatrix} 1 & 2 \end{bmatrix}^T$ and $\begin{bmatrix} 1 & 2 \end{bmatrix}^T$ are bases for column and row spaces, respectively.

Exercise 46. *Show the row space $C(A^T)$ is a subspace of $\mathbb{R}^n$.*

4.4. Left nullspace

In this subsection, we introduce the left nullspace of an arbitrary matrix A and investigate some of its properties.

Left nullspace. For a given matrix $A \in \mathbb{R}^{m\times n}$, the set of all solutions of the homogeneous linear system $A^T x = 0$ is called **Left nullspace** of A, which is denoted by $N(A^T)$, i.e.,

$$N(A^T) = \left\{ x \in \mathbb{R}^m \mid A^T x = 0 \right\}. \tag{4.1}$$

It is clear that the zero vector $x = 0$ satisfies $A^T x = 0$, which implies that $0 \in N(A^T)$.

He asks this questions sometimes on the exam

Example 47. *Let us consider the matrix*

$$A = \begin{bmatrix} 1 & 4 \\ 2 & 7 \\ 3 & 5 \end{bmatrix},$$

where we aim to find its row space. The transpose of this matrix is given by

$$A^T = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 7 & 5 \end{bmatrix}.$$

From the definition of row space, we obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 4 & 7 & 5 & 0 \end{array}\right] \xrightarrow{r_1 \leftrightarrow r_2} \left[\begin{array}{ccc|c} 4 & 7 & 5 & 0 \\ 1 & 2 & 3 & 0 \end{array}\right] \xrightarrow{r_1 \to 1/4 r_1} \left[\begin{array}{ccc|c} 1 & 7/4 & 5/4 & 0 \\ 1 & 2 & 3 & 0 \end{array}\right] \xrightarrow{r_2 \to r_2 - r_1} \left[\begin{array}{ccc|c} 1 & 7/4 & 5/4 & 0 \\ 0 & 1/4 & 7/4 & 0 \end{array}\right]$$
$$\xrightarrow{r_2 \to 4r_2} \left[\begin{array}{ccc|c} 1 & 7/4 & 5/4 & 0 \\ 0 & 1 & 7 & 0 \end{array}\right] \xrightarrow{r_1 \to r_1 - 7/4 r_2} \left[\begin{array}{ccc|c} 1 & 0 & -11 & 0 \\ 0 & 1 & 7 & 0 \end{array}\right],$$

which leads to $x_1 = 11x_3$ and $x_2 = -7x_3$, i.e.,

$$N(A^T) = \left\{x \in \mathbb{R}^3 \mid,\ x_1 = 11x_3,\ x_2 = -7x_3\right\}.$$

Setting $x_3 = 1$ ensures that

$$N(A^T) = \operatorname{span}\left\{ \begin{bmatrix} 11 \\ -7 \\ 1 \end{bmatrix} \right\}.$$

Exercise 48. *Show the left nullspace $N(A^T)$ is a subspace of $\mathbb{R}^m$.*

4.5. Dimension of matrix four spaces

In this subsection, we investigate the dimension of column space, row space, nullspace, and left nullspace and study their relations. Let us begin with a summary of the last few sections.

Four fundamental subspaces. For a given matrix $A \in \mathbb{R}^{m\times n}$, the following four fundamental subspaces are recognized:
(1) The **row space** is $C(A^T)$, a subspace of $\mathbb{R}^n$;
(2) The **column space** is $C(A)$, a subspace of $\mathbb{R}^m$;
(3) The **nullspace** is $N(A)$, a subspace of $\mathbb{R}^n$;
(4) The **left nullspace** is $N(A^T)$, a subspace of $\mathbb{R}^m$.

Example 49. *Let us consider the echelon matrix*

$$A = \begin{bmatrix} 1 & 2 & 0 & 4 & 5 \\ 0 & 0 & 1 & 0 & 6 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Then, we have that

(1) The **row space** *is spanned by the first two rows (r_1 and r_2) because the third row adds nothing new to the row space. Hence, the dimension of the row space of A is 2;*
(2) The **column space** *is spanned by columns c_1 and c_3 because the other columns add nothing new to the column space. Hence, the dimension of the column space of A is 2;*
(3) From the system $Ax = 0$, we obtain that x_2, x_4 and x_5 are free variables, i.e.,

$$N(A) = \left\{x \in \mathbb{R}^5 \mid x_1 = -2x_2 - 4x_4 - 5x_5,\ x_3 = -6x_5\right\}$$

Setting $x_2 = 1,\ x_4 = x_5 = 0,\ x_2 = x_5 = 0,\ x_4 = 1,\ x_2 = x_4 = 0, x_5 = 1$, we respectively come to the special solutions

$$s_1 = \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad s_2 = \begin{bmatrix} -4 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad s_3 = \begin{bmatrix} -5 \\ 0 \\ -6 \\ 0 \\ 1 \end{bmatrix},$$

which form a basis for the **nullspace**. *Therefore, the dimension on $N(A) = 3$.*

(4) To identify the left nullspace, we proceed as follows

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 0 & 0 \\ 0 & 1 & 0 \\ 4 & 0 & 0 \\ 5 & 6 & 0 \end{bmatrix} \xrightarrow{r_1 \leftrightarrow r_5} \begin{bmatrix} 5 & 6 & 0 \\ 2 & 0 & 0 \\ 0 & 1 & 0 \\ 4 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \xrightarrow{r_1 \to 1/5 r_1} \begin{bmatrix} 1 & 6/5 & 0 \\ 2 & 0 & 0 \\ 0 & 1 & 0 \\ 4 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \xrightarrow{r_2 \to r_2 - 2r_1, r_4 \to r_4 - 4r_1, r_5 \to r_5 - r_1} \begin{bmatrix} 1 & 6/5 & 0 \\ 0 & -12/5 & 0 \\ 0 & 1 & 0 \\ 0 & -24/5 & 0 \\ 0 & -6/5 & 0 \end{bmatrix}$$
$$\xrightarrow{r_2 \leftrightarrow r_4} \begin{bmatrix} 1 & 6/5 & 0 \\ 0 & -24/5 & 0 \\ 0 & 1 & 0 \\ 0 & -12/5 & 0 \\ 0 & -6/5 & 0 \end{bmatrix} \xrightarrow{r_2 \to -5/24 r_2} \begin{bmatrix} 1 & 6/5 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & -12/5 & 0 \\ 0 & -6/5 & 0 \end{bmatrix} \xrightarrow{r_1 \to r_1 - 6/5 r_2, r_3 \to r_3 - r_2, r_4 \to r_4 + 12/5 r_2, r_5 \to r_5 + 6/5 r_2}$$
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Hence, the **left nullspace** *is given by*

$$N(A^T) = \left\{ y \in \mathbb{R}^3 \mid y_1 = y_2 = 0,\ y_3 \in \mathbb{R} \right\}.$$

Setting $y_3 = 1$, we get the special solution

$$s = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix},$$

which forms a basis for the left nullspace.

Therefore, we come to

(a) $\dim(C(A)) = \dim(C(A^T)) = 2 = r$*;*
(b) $\dim(N(A)) = n - r = 5 - 2 = 3$*;*
(c) $\dim(N(A^T)) = m - r = 3 - 2 = 1$*,*

which follows the **fundamental theorem of linear algebra** *that we present next.*

Fundamental theorem of linear algebra. For a given matrix $A \in \mathbb{R}^{m \times n}$, the following statements hold:

(a) The row space $C(A^T)$ and the column space $C(A)$ has the same dimension, i.e., $\dim(C(A)) = \dim(C(A^T)) = r$;
(b) The dimension of nullspace $N(A)$ is $n - r$, i.e., $\dim(N(A)) = n - r$;
(c) The dimension of left nullspace $N(A^T)$ is $m - r$, i.e., $\dim(N(A^T)) = m - r$.

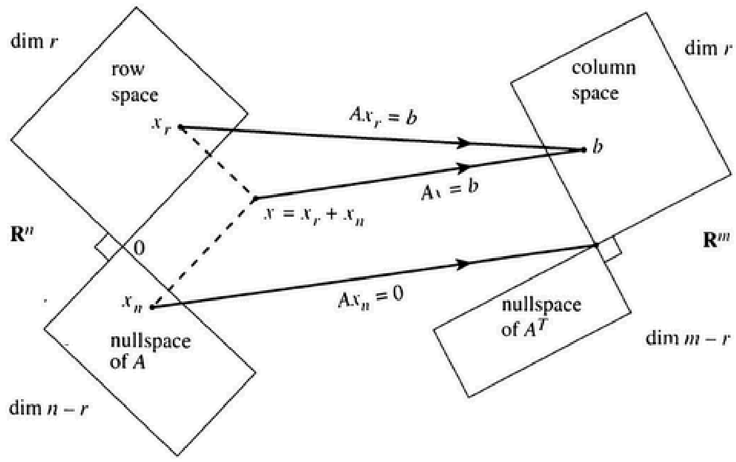


FIGURE 4.1. Matrix four subspaces and related dimensions (see, e.g., [**1**].

CHAPTER 5

Inverse Matrices

In this section, we describe invertible square matrices, which are a central topic in linear algebra. Then, we show how to calculate an inverse matrix by applying the Gauss-Jordan elimination procedure.

Our main objectives are to:

- introduce inverse matrices;
- find an inverse matrix by applying the Gauss-Jordan elimination procedure.

5.1. Invertible matrices

In this section, we introduce inverse matrices and present several instances where such inverse matrices exist.

Inverse matrix. Let A be an $n \times n$ matrix ($A \in \mathbb{R}^{n\times n}$). We say A is **invertible** if there is an $n \times n$ matrix A^{-1} such that

$$AA^{-1} = I_n \quad \text{and} \quad A^{-1}A = I_n. \tag{5.1}$$

If a matrix is invertible, then it is often called **non-singular**, and if a matrix is not invertible, then it is called **singular**.

Example 50. *Let us consider the matrices*

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 2 & 0 \\ 0 & 3 & 0 & 0 \\ 4 & 0 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 0 & 0 & 1/4 \\ 0 & 0 & 1/3 & 0 \\ 0 & 1/2 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}.$$

It is clear that

$$\begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 2 & 0 \\ 0 & 3 & 0 & 0 \\ 4 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 1/4 \\ 0 & 0 & 1/3 & 0 \\ 0 & 1/2 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 1/4 \\ 0 & 0 & 1/3 & 0 \\ 0 & 1/2 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 2 & 0 \\ 0 & 3 & 0 & 0 \\ 4 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

which implies that B is the inverse of A.

Example 51 (Not every matrix is invertible)**.** *Let us consider the matrix*

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix},$$

and we wish to find an inverse matrix for A. Since A is 2×2, then such an inverse matrix should be 2×2 as well. From the definition of inverse matrices, we obtain

$$AA^{-1} = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2.$$

Thus, we come to

$$\begin{bmatrix} b_{11} + 2b_{21} & b_{12} + 2b_{22} \\ 2b_{11} + 4b_{21} & 2b_{12} + 4b_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

that leads to the following system of equations

$$\begin{cases} b_{11} + 2b_{21} = 1 \\ b_{12} + 2b_{22} = 0 \\ 2b_{11} + 4b_{21} = 0 \\ 2b_{12} + 4b_{22} = 1 \end{cases}$$

or equivalently

$$\begin{bmatrix}1&2&0&0\\0&0&1&2\\2&4&0&0\\0&0&2&4\end{bmatrix}\begin{bmatrix}b_{11}\\b_{21}\\b_{12}\\b_{22}\end{bmatrix}=\begin{bmatrix}1\\0\\0\\1\end{bmatrix}.$$

Writing the augmented matrix and Gauss-Jordan elimination, we have

$$\left[\begin{array}{cccc|c}1&2&0&0&1\\0&0&1&1&0\\2&4&0&0&0\\0&0&2&4&1\end{array}\right]\xrightarrow{r_3\leftrightarrow r_1}\left[\begin{array}{cccc|c}2&4&0&0&0\\0&0&1&1&0\\1&2&0&0&1\\0&0&2&4&1\end{array}\right]\xrightarrow{1/2r_1\to r_1}\left[\begin{array}{cccc|c}1&2&0&0&0\\0&0&1&1&0\\1&2&0&0&1\\0&0&2&4&1\end{array}\right]\xrightarrow{r_3-r_1\to r_3}\left[\begin{array}{cccc|c}1&2&0&0&0\\0&0&1&2&0\\0&0&0&0&1\\0&0&2&4&1\end{array}\right]$$
$$\xrightarrow{r_2\leftrightarrow r_4}\left[\begin{array}{cccc|c}1&2&0&0&0\\0&0&2&4&1\\0&0&0&0&1\\0&0&1&2&0\end{array}\right]\xrightarrow{1/2r_2\to r_2}\left[\begin{array}{cccc|c}1&2&0&0&0\\0&0&1&2&1/2\\0&0&0&0&1\\0&0&1&2&0\end{array}\right]\xrightarrow{r_4-r_2\to r_4}\left[\begin{array}{cccc|c}1&2&0&0&0\\0&0&1&2&1/2\\0&0&0&0&1\\0&0&0&0&-1/2\end{array}\right].$$

This clearly implies that this system of equations is inconsistent. Therefore, **not all square matrices have inverses**.

In the next example, we investigate a condition that guarantees the existence of an inverse for 2×2 matrices.

Example 52 (Inverse of 2×2 matrices)**.** *Let us consider an arbitrary 2×2 matrix*

$$A=\begin{bmatrix}a_{11}&a_{12}\\a_{21}&a_{22}\end{bmatrix},$$

and we wish to find an inverse matrix for A. From the definition of inverse matrices, we obtain

$$AA^{-1}=\begin{bmatrix}a_{11}&a_{12}\\a_{21}&a_{22}\end{bmatrix}\begin{bmatrix}b_{11}&b_{12}\\b_{21}&b_{22}\end{bmatrix}=\begin{bmatrix}1&0\\0&1\end{bmatrix}=I_2.$$

Thus, we come to

$$\begin{bmatrix}a_{11}b_{11}+a_{12}b_{21}&a_{11}b_{12}+a_{12}b_{22}\\a_{21}b_{11}+a_{22}b_{21}&a_{21}b_{12}+a_{22}b_{22}\end{bmatrix}=\begin{bmatrix}1&0\\0&1\end{bmatrix}$$

that leads to the following system of equations

$$\begin{cases}a_{11}b_{11}+a_{12}b_{21}=1\\a_{11}b_{12}+a_{12}b_{22}=0\\a_{21}b_{11}+a_{22}b_{21}=0\\a_{21}b_{12}+a_{22}b_{22}=1\end{cases}$$

or equivalently

$$\begin{bmatrix}a_{11}&a_{12}&0&0\\0&0&a_{11}&a_{12}\\a_{21}&a_{22}&0&0\\0&0&a_{21}&a_{22}\end{bmatrix}\begin{bmatrix}b_{11}\\b_{21}\\b_{12}\\b_{22}\end{bmatrix}=\begin{bmatrix}1\\0\\0\\1\end{bmatrix}.$$

Writing the augmented matrix and Gauss-Jordan elimination, we have

$$
\left[\begin{array}{cccc|c} a_{11} & a_{12} & 0 & 0 & 1 \\ 0 & 0 & a_{11} & a_{12} & 0 \\ a_{21} & a_{22} & 0 & 0 & 0 \\ 0 & 0 & a_{21} & a_{22} & 1 \end{array}\right]
\xrightarrow{\frac{1}{a_{11}} r_1 \to r_1}
\left[\begin{array}{cccc|c} 1 & \frac{a_{12}}{a_{11}} & 0 & 0 & \frac{1}{a_{11}} \\ 0 & 0 & a_{11} & a_{12} & 0 \\ a_{21} & a_{22} & 0 & 0 & 0 \\ 0 & 0 & a_{21} & a_{22} & 1 \end{array}\right]
\xrightarrow{r_3 - a_{21} r_1 \to r_3}
$$

$$
\left[\begin{array}{cccc|c} 1 & \frac{a_{12}}{a_{11}} & 0 & 0 & \frac{1}{a_{11}} \\ 0 & 0 & a_{11} & a_{12} & 0 \\ 0 & a_{22} - \frac{a_{12}}{a_{11}} a_{21} & 0 & 0 & -\frac{a_{21}}{a_{11}} \\ 0 & 0 & a_{21} & a_{22} & 1 \end{array}\right]
\xrightarrow{r_2 \leftrightarrow r_3}
\left[\begin{array}{cccc|c} 1 & \frac{a_{12}}{a_{11}} & 0 & 0 & \frac{1}{a_{11}} \\ 0 & \frac{a_{11}a_{22} - a_{12}a_{21}}{a_{11}} & 0 & 0 & -\frac{a_{21}}{a_{11}} \\ 0 & 0 & a_{11} & a_{12} & 0 \\ 0 & 0 & a_{21} & a_{22} & 1 \end{array}\right]
\xrightarrow{\frac{a_{11}}{a_{11}a_{22} - a_{12}a_{21}} r_2 \to r_2}
$$

$$
\left[\begin{array}{cccc|c} 1 & \frac{a_{12}}{a_{11}} & 0 & 0 & \frac{1}{a_{11}} \\ 0 & 1 & 0 & 0 & -\frac{a_{21}}{a_{11}a_{22} - a_{12}a_{21}} \\ 0 & 0 & a_{11} & a_{12} & 0 \\ 0 & 0 & a_{21} & a_{22} & 1 \end{array}\right]
\xrightarrow{\frac{1}{a_{11}} r_3 \to r_3}
\left[\begin{array}{cccc|c} 1 & \frac{a_{12}}{a_{11}} & 0 & 0 & \frac{1}{a_{11}} \\ 0 & 1 & 0 & 0 & -\frac{a_{21}}{a_{11}a_{22} - a_{12}a_{21}} \\ 0 & 0 & 1 & \frac{a_{12}}{a_{11}} & 0 \\ 0 & 0 & a_{21} & a_{22} & 1 \end{array}\right]
\xrightarrow{r_4 - a_{21} r_3 \to r_4}
$$

$$
\left[\begin{array}{cccc|c} 1 & \frac{a_{12}}{a_{11}} & 0 & 0 & \frac{1}{a_{11}} \\ 0 & 1 & 0 & 0 & -\frac{a_{21}}{a_{11}a_{22} - a_{12}a_{21}} \\ 0 & 0 & 1 & \frac{a_{12}}{a_{11}} & 0 \\ 0 & 0 & 0 & \frac{a_{11}a_{22} - a_{12}a_{21}}{a_{11}} & 1 \end{array}\right]
\xrightarrow{\frac{a_{11}}{a_{11}a_{22} - a_{12}a_{21}} r_4 \to r_4}
\left[\begin{array}{cccc|c} 1 & \frac{a_{12}}{a_{11}} & 0 & 0 & \frac{1}{a_{11}} \\ 0 & 1 & 0 & 0 & -\frac{a_{21}}{a_{11}a_{22} - a_{12}a_{21}} \\ 0 & 0 & 1 & \frac{a_{12}}{a_{11}} & 0 \\ 0 & 0 & 0 & 1 & \frac{a_{11}}{a_{11}a_{22} - a_{12}a_{21}} \end{array}\right].
$$

Therefore, this system of equations has a unique solution if $a_{11}a_{22} - a_{12}a_{21} \neq 0$.

Invertibility of strictly diagonally dominant matrices. Let $A \in \mathbb{R}^{n \times n}$ be a strictly diagonally dominant matrix ($|a_{ii}| > \sum_{j \neq i} |a_{ij}|$ for all $i = 1, \dots, n$). Then, A is invertible.

Example 53. *Let us consider the matrices*

$$
A = \begin{bmatrix} 5 & 1 & 1 \\ 1 & 4 & 1 \\ 1 & 1 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 3 \end{bmatrix}, \quad C = \begin{bmatrix} 2 & 1 & 2 \\ 2 & 3 & 2 \\ 2 & 1 & 2 \end{bmatrix}.
$$

The matrix A is strictly diagonally dominant and therefore invertible; the matrix B is not strictly diagonally dominant but still invertible; and the matrix C is not strictly diagonally dominant and not invertible. Therefore, **being strictly diagonally dominant is a sufficient condition for invertibility**.

Inverse of diagonal matrices. Let us consider an $n \times n$ diagonal matrix

$$
A = \begin{bmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{bmatrix}, \quad a_{ii} \neq 0 \text{ for } i = 1, \dots, n.
$$

Then, it is clear that

$$
\begin{bmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{bmatrix}
\begin{bmatrix} \frac{1}{a_{11}} & 0 & \cdots & 0 \\ 0 & \frac{1}{a_{22}} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{a_{nn}} \end{bmatrix}
= \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix},
$$

leading to

$$
\begin{bmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{bmatrix}^{-1}
= \begin{bmatrix} \frac{1}{a_{11}} & 0 & \cdots & 0 \\ 0 & \frac{1}{a_{22}} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{a_{nn}} \end{bmatrix}.
$$

Example 54. *Let us consider the matrices*

$$A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix}.$$

Since $a_{ii} \neq 0$ for $i = 1, \ldots, 4$, the inverse of A is given by

$$A^{-1} = \begin{bmatrix} 1/2 & 0 & 0 & 0 \\ 0 & 1/3 & 0 & 0 \\ 0 & 0 & 1/4 & 0 \\ 0 & 0 & 0 & 1/5 \end{bmatrix},$$

Since B is a diagonal matrix with $a_{33} = 0$, this matrix is not invertible.

Inverse matrix is unique. Let us consider the marices $A, B, C \in \mathbb{R}^{n\times n}$ such that

$$AB = BA = I_n, \quad AC = CA = I_n.$$

Then, it holds that

$$B = BI_n = B(AC) = (BA)C = I_nC = C.$$

Example 55 (Invertibility and sum)**.** *Let us consider the matrices*

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

It is clear that A and B are invertible, while

$$A + B = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix},$$

which is not invertible. Therefore, **invertibility of matrices A and B does not imply the invertibility of $A + B$***. It is clear that C and D are not invertible; however, the matrix*

$$C + D = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

is invertible. Therefore, **the sum of two noninvertible matrices can be invertible***.*

5.2. Calculating inverse matrix by Gauss-Jordan elimination

We have already seen how to compute the inverse of diagonal matrices in the previous example; however, we need a systematic way to compute the inverse for a general square matrix (if it is invertible). Let us start with a 2×2 matrix.

Example 56 (Inverse of 2×2 matrices)**.** *We consider an arbitrary 2×2 matrix*

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix},$$

where $a_{11}a_{22} - a_{12}a_{21} \neq 0$ (as suggested in Example 52). Let us augment A by I_2 and apply the Gauss-Jordan elimination scheme as follows

$$\left[\begin{array}{cc|cc} a_{11} & a_{12} & 1 & 0 \\ a_{21} & a_{22} & 0 & 1 \end{array}\right] \xrightarrow{\frac{1}{a_{11}} r_1 \to r_1} \left[\begin{array}{cc|cc} 1 & \frac{a_{12}}{a_{11}} & \frac{1}{a_{11}} & 0 \\ a_{21} & a_{22} & 0 & 1 \end{array}\right] \xrightarrow{r_2 - a_{21} r_1 \to r_2} \left[\begin{array}{cc|cc} 1 & \frac{a_{12}}{a_{11}} & \frac{1}{a_{11}} & 0 \\ 0 & \frac{a_{11}a_{22}-a_{12}a_{21}}{a_{11}} & -\frac{a_{21}}{a_{11}} & 1 \end{array}\right]$$

$$\xrightarrow{\frac{a_{11}}{a_{11}a_{22}-a_{12}a_{21}} r_2 \to r_2} \left[\begin{array}{cc|cc} 1 & \frac{a_{12}}{a_{11}} & \frac{1}{a_{11}} & 0 \\ 0 & 1 & -\frac{a_{21}}{a_{11}a_{22}-a_{12}a_{21}} & \frac{a_{11}}{a_{11}a_{22}-a_{12}a_{21}} \end{array}\right]$$

$$\xrightarrow{r_1 - \frac{a_{12}}{a_{11}} r_2 \to r_1} \left[\begin{array}{cc|cc} 1 & 0 & \frac{a_{22}}{a_{11}a_{22}-a_{12}a_{21}} & -\frac{a_{12}}{a_{11}a_{22}-a_{12}a_{21}} \\ 0 & 1 & -\frac{a_{21}}{a_{11}a_{22}-a_{12}a_{21}} & \frac{a_{11}}{a_{11}a_{22}-a_{12}a_{21}} \end{array}\right].$$

Let us multiply the augmented part by the matrix A, i.e.,

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} \frac{a_{22}}{a_{11}a_{22}-a_{12}a_{21}} & -\frac{a_{12}}{a_{11}a_{22}-a_{12}a_{21}} \\ -\frac{a_{21}}{a_{11}a_{22}-a_{12}a_{21}} & \frac{a_{11}}{a_{11}a_{22}-a_{12}a_{21}} \end{bmatrix} = \begin{bmatrix} \frac{a_{22}}{a_{11}a_{22}-a_{12}a_{21}} & -\frac{a_{12}}{a_{11}a_{22}-a_{12}a_{21}} \\ -\frac{a_{21}}{a_{11}a_{22}-a_{12}a_{21}} & \frac{a_{11}}{a_{11}a_{22}-a_{12}a_{21}} \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

which consequently ensures

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}^{-1} = \frac{1}{a_{11}a_{22}-a_{12}a_{21}} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}.$$

We now describe a scheme for calculating the inverse matrix A^{-1} by the Gauss-Jordan elimination procedure.

Calculating A^{-1} by Gauss-Jordan Elimination. Let A be $n \times n$ matrix ($A \in \mathbb{R}^{n\times n}$). We consider the following steps to calculate the inverse matrix A^{-1}:

(1) Augment A by the identity matrix I_n, i.e.,

$$\begin{bmatrix} A & I_n \end{bmatrix} = \left[\begin{array}{cccc|cccc} a_{11} & a_{12} & \cdots & a_{1n} & 1 & 0 & \cdots & 0 \\ a_{21} & a_{22} & \cdots & a_{2n} & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} & 0 & 0 & \cdots & 1 \end{array}\right];$$

(2) Apply the Gauss-Jordan elimination procedure to the above augmented matrix;

(3) If the left-hand side matrix has n pivots (i.e., A is transformed to I_n), then A is invertible and the resulted right-hand side matrix is A^{-1}. If the left-hand side matrix has less than n pivots, then it is not invertible.

Example 57. *Let us consider the matrix*

$$A = \begin{bmatrix} 1 & 3 & 0 \\ 0 & 0 & 2 \\ 1 & 4 & 5 \end{bmatrix},$$

then we augment this matrix by I_3 and apply the Gauss-Jordan elimination procedure as follows

$$\left[\begin{array}{ccc|ccc} 1 & 3 & 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 & 1 & 0 \\ 1 & 4 & 5 & 0 & 0 & 1 \end{array}\right] \xrightarrow{r_3 - r_1 \to r_3} \left[\begin{array}{ccc|ccc} 1 & 3 & 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 & 1 & 0 \\ 0 & 1 & 5 & -1 & 0 & 1 \end{array}\right] \xrightarrow{r_3 \leftrightarrow r_2} \left[\begin{array}{ccc|ccc} 1 & 3 & 0 & 1 & 0 & 0 \\ 0 & 1 & 5 & -1 & 0 & 1 \\ 0 & 0 & 2 & 0 & 1 & 0 \end{array}\right] \xrightarrow{r_1 - 3r_2 \to r_1}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -15 & 4 & 0 & -3 \\ 0 & 1 & 5 & -1 & 0 & 1 \\ 0 & 0 & 2 & 0 & 1 & 0 \end{array}\right] \xrightarrow{1/2 r_3 \leftrightarrow r_3} \left[\begin{array}{ccc|ccc} 1 & 0 & -15 & 4 & 0 & -3 \\ 0 & 1 & 5 & -1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1/2 & 0 \end{array}\right] \xrightarrow{r_2 - 5r_3 \to r_2,\ r_1 + 15r_3 \to r_1}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 4 & 15/2 & -3 \\ 0 & 1 & 0 & -1 & -5/2 & 1 \\ 0 & 0 & 1 & 0 & 1/2 & 0 \end{array}\right].$$

Since the left-hand side of the augmented matrix is I_3, we have

$$\begin{bmatrix} 1 & 3 & 0 \\ 0 & 0 & 2 \\ 1 & 4 & 5 \end{bmatrix}^{-1} = \begin{bmatrix} 4 & 15/2 & -3 \\ -1 & -5/2 & 1 \\ 0 & 1/2 & 0 \end{bmatrix}.$$

Inverse of a matrix product. Let A and B be $n \times n$ matrices ($A, B \in \mathbb{R}^{n\times n}$). Then, we have

$$ABB^{-1}A^{-1} = A(BB^{-1})A^{-1} = A(I_n)A^{-1} = AA^{-1} = I_n,$$
$$B^{-1}A^{-1}AB = B^{-1}(A^{-1}A)B = B^{-1}(I_n)B = B^{-1}B = I_n,$$

leading to

$$(AB)^{-1} = B^{-1}A^{-1}. \tag{5.1}$$

Example 58. *Let us consider the matrices*

$$A = \begin{bmatrix} 1 & 3 & 0 \\ 0 & 0 & 2 \\ 1 & 4 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

Since B is diagonal, it follows from Example 57 that

$$\begin{bmatrix}1&3&0\\0&0&2\\1&4&5\end{bmatrix}^{-1}=\begin{bmatrix}4&15/2&-3\\-1&-5/2&1\\0&1/2&0\end{bmatrix}\quad\begin{bmatrix}5&0&0\\0&3&0\\0&0&2\end{bmatrix}^{-1}=\begin{bmatrix}1/5&0&0\\0&1/3&0\\0&0&1/2\end{bmatrix}.$$

Then, we come to

$$(AB)^{-1}=B^{-1}A^{-1}=\begin{bmatrix}1/5&0&0\\0&1/3&0\\0&0&1/2\end{bmatrix}\begin{bmatrix}4&15/2&-3\\-1&-5/2&1\\0&1/2&0\end{bmatrix}=\begin{bmatrix}4/5&3/2&-3/5\\-1/3&-5/6&1/3\\0&1/4&0\end{bmatrix}.$$

5.3. Solving linear system by inverse matrices

In this subsection, we discuss the solution of square systems with invertible coefficient matrices using a matrix inversion technique.

Solving linear system of equations using inverse matrices. Let us consider the system of equations

$$Ax=b,$$

where $A\in\mathbb{R}^{n\times n}$ is an invertible matrix. Multiplying both sides of this identity by A^{-1} from left, we come to

$$x=xI_n=A^{-1}Ax=A^{-1}b,$$

i.e.,

$$x=A^{-1}b \tag{5.1}$$

Example 59. *Let us consider the linear system of equations*

$$\begin{cases}2x_1-x_2=1,\\-x_1+2x_2-x_3=5,\\-x_2+2x+3=2,\end{cases}$$

which can be written in the matrix form $Ax=b$ with

$$A=\begin{bmatrix}2&-1&0\\-1&2&-1\\0&-1&2\end{bmatrix},\quad x=\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix},\quad b=\begin{bmatrix}1\\5\\2\end{bmatrix}.$$

Let us calculate the inverse matrix A^{-1} as

$$\left[\begin{array}{ccc|ccc}2&-1&0&1&0&0\\-1&2&-1&0&1&0\\0&-1&2&0&0&1\end{array}\right]\xrightarrow{1/2r_1\leftrightarrow r_1}\left[\begin{array}{ccc|ccc}1&-1/2&0&1/2&0&0\\-1&2&-1&0&1&0\\0&-1&2&0&0&1\end{array}\right]\xrightarrow{r_2+r_1\to r_2}$$

$$\left[\begin{array}{ccc|ccc}1&-1/2&0&1/2&0&0\\0&3/2&-1&1/2&1&0\\0&-1&2&0&0&1\end{array}\right]\xrightarrow{2/3r_2\leftrightarrow r_2}\left[\begin{array}{ccc|ccc}1&-1/2&0&1/2&0&0\\0&1&-2/3&1/3&2/3&0\\0&-1&2&0&0&1\end{array}\right]\xrightarrow{r_3+r_2\to r_3,\ r_1+1/2r_2\to r_1}$$

$$\left[\begin{array}{ccc|ccc}1&0&-1/3&2/3&1/3&0\\0&1&-2/3&1/3&2/3&0\\0&0&4/3&1/3&2/3&1\end{array}\right]\xrightarrow{3/4r_3\leftrightarrow r_3}\left[\begin{array}{ccc|ccc}1&0&-1/3&2/3&1/3&0\\0&1&-2/3&1/3&2/3&0\\0&0&1&1/4&1/2&3/4\end{array}\right]\xrightarrow{r_2+2/3r_2\to r_2,\ r_1+1/2r_3\to r_1}$$

$$\left[\begin{array}{ccc|ccc}1&0&0&3/4&1/2&1/4\\0&1&0&1/2&1&1/2\\0&0&1&1/4&1/2&3/4\end{array}\right],$$

which implies

$$A^{-1}=\begin{bmatrix}3/4&1/2&1/4\\1/2&1&1/2\\1/4&1/2&3/4\end{bmatrix},\quad x=A^{-1}b=\begin{bmatrix}3/4&1/2&1/4\\1/2&1&1/2\\1/4&1/2&3/4\end{bmatrix}\begin{bmatrix}1\\5\\2\end{bmatrix}=\begin{bmatrix}15/4\\13/2\\17/4\end{bmatrix}.$$

CHAPTER 6

Determinant

In this chapter, we introduce an important tool called **determinant** that we have seen for deriving the invertibility of 2×2 matrices. Indeed, a determinant is a number containing an amazing amount of information about the matrix, which immediately explains whether the matrix is invertible. In fact, if the determinant is zero, then the matrix has no inverse. We further study determinant rules.

Our main objectives are to:

- introduce minors, cofactors, and determinant, and some of its properties;
- present determinant rules which can be used for calculating the determinant of structured matrices.

6.1. A Criteria for Invertibility

In the previous section, we have seen that an arbitrary 2×2 matrix

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix},$$

has an inverse if and only if $a_{11}a_{22} - a_{12}a_{21} \neq 0$, which leads to the inverse

$$A^{-1} = \frac{1}{a_{11}a_{22}-a_{12}a_{21}} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}.$$

Hence, the term $a_{11}a_{22} - a_{12}a_{21}$ is called **determinant** (denoted by $\det(A)$) plays a key role in the existence and calculation of the inverse of 2×2 matrices. We next extend the above result to 3×3 matrices.

Example 60 (Inverse of 3×3 matrices)**.** *Let us consider an arbitrary* 3×3 *matrix*

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix},$$

and we wish to find an inverse matrix for A. From the definition of inverse matrices, we obtain

$$AA^{-1} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3.$$

Thus, we come to

$$\begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} & a_{11}b_{12} + a_{12}b_{22} + a_{13}b_{32} & a_{11}b_{13} + a_{12}b_{23} + a_{13}b_{33} \\ a_{21}b_{11} + a_{22}b_{21} + a_{23}b_{31} & a_{21}b_{12} + a_{22}b_{22} + a_{23}b_{32} & a_{21}b_{13} + a_{22}b_{23} + a_{23}b_{33} \\ a_{31}b_{11} + a_{32}b_{21} + a_{33}b_{31} & a_{31}b_{12} + a_{32}b_{22} + a_{33}b_{32} & a_{31}b_{13} + a_{32}b_{23} + a_{33}b_{33} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

that leads to the following system of equations

$$\begin{cases} a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} = 1 \\ a_{11}b_{12} + a_{12}b_{22} + a_{13}b_{32} = 0 \\ a_{11}b_{13} + a_{12}b_{23} + a_{13}b_{33} = 0 \\ a_{21}b_{11} + a_{22}b_{21} + a_{23}b_{31} = 0 \\ a_{21}b_{12} + a_{22}b_{22} + a_{23}b_{32} = 1 \\ a_{21}b_{13} + a_{22}b_{23} + a_{23}b_{33} = 0 \\ a_{31}b_{11} + a_{32}b_{21} + a_{33}b_{31} = 0 \\ a_{31}b_{12} + a_{32}b_{22} + a_{33}b_{32} = 0 \\ a_{31}b_{13} + a_{32}b_{23} + a_{33}b_{33} = 1 \end{cases}$$

or equivalently

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & a_{11} & a_{12} & a_{13} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & a_{21} & a_{22} & a_{23} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & a_{31} & a_{32} & a_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} b_{11} \\ b_{21} \\ b_{31} \\ b_{12} \\ b_{22} \\ b_{32} \\ b_{13} \\ b_{23} \\ b_{33} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

Solving this system of equations by the Gauss-Jordan elimination procedure (check it as an exercise!), it can be shown that this system has a unique solution if

$$a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{31}a_{23}) + a_{13}(a_{21}a_{32} - a_{31}a_{22}) \neq 0.$$

This term is the determinant of 3×3 *matrices, which can be written in the form*

$$\det(A) = a_{11} \det\left(\begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix}\right) - a_{12} \det\left(\begin{bmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{bmatrix}\right) + a_{13} \det\left(\begin{bmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}\right) \neq 0,$$

where the determinant of 2×2 *matrices are known to us.*

6.2. Calculating determinant by cofactors

We have already seen that the determinant has a key role in solving square systems of equations, and its computation with Gauss-Jordan elimination is too costly for large matrices. We now describe a systematic way to compute the determinant using **minors** and **cofactors**, which we present next.

Definition (Minors). Let A be a $n \times n$ square matrix, and let us to pick any element a_{ij} $(1 \leq i, j \leq n)$, i.e.,

$$A = \left[\begin{array}{ccc|c|cc} a_{11} & a_{12} & \cdots & a_{1j} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2j} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \hline a_{i1} & a_{i2} & \cdots & a_{ij} & \cdots & a_{in} \\ \hline \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nj} & \cdots & a_{nn} \end{array}\right]$$

The **minor** of a_{ij} (denoted by M_{ij}) is the determinant of the submatrix that is constructed by removing the ith row and jth column of A, i.e.,

$$M_{ij} = \det\left(\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1(j-1)} & a_{1(j+1)} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2(j-1)} & a_{1(j+1)} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \\ a_{(i-1)1} & a_{(i-1)2} & \cdots & a_{(i-1)(j+1)} & a_{1(j-1)} & \cdots & a_{(i-1)n} \\ a_{(i+1)1} & a_{(i+1)2} & \cdots & a_{(i+1)(j+1)} & a_{1(j-1)} & \cdots & a_{(i+1)n} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \\ a_{n1} & a_{n2} & \cdots & a_{n(j-1)} & a_{1(j+1)} & \cdots & a_{nn} \end{bmatrix}\right)$$

Example 61. *Let us consider the matrix*

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 8 & 7 & 6 & 5 \\ 3 & 1 & 2 & 7 \\ 5 & 6 & 9 & 3 \end{bmatrix}.$$

Then, we come to

$$M_{11} = \det\left(\begin{bmatrix} 7 & 6 & 5 \\ 1 & 2 & 7 \\ 6 & 9 & 3 \end{bmatrix}\right), \quad M_{23} = \det\left(\begin{bmatrix} 1 & 2 & 4 \\ 3 & 1 & 7 \\ 5 & 6 & 3 \end{bmatrix}\right), \quad M_{41} = \det\left(\begin{bmatrix} 2 & 3 & 4 \\ 7 & 6 & 5 \\ 1 & 2 & 7 \end{bmatrix}\right).$$

Definition (Cofactors). Let A be a $n \times n$ square matrix, and let us to pick any element a_{ij} $(1 \leq i, j \leq n)$, i.e.,

$$A = \left[\begin{array}{ccc|c|cc} a_{11} & a_{12} & \cdots & a_{1j} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2j} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \hline a_{i1} & a_{i2} & \cdots & a_{ij} & \cdots & a_{in} \\ \hline \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nj} & \cdots & a_{nn} \end{array}\right]$$

The **cofactor** of a_{ij} (denoted by C_{ij}) is the number $(-1)^{i+j} M_{ij}$.

Example 62. *Let us consider the matrix*

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 8 & 7 & 6 & 5 \\ 3 & 1 & 2 & 7 \\ 5 & 6 & 9 & 3 \end{bmatrix}.$$

Then,

$$C_{11} = (-1)^{1+1} \det\left(\begin{bmatrix} 7 & 6 & 5 \\ 1 & 2 & 7 \\ 6 & 9 & 3 \end{bmatrix}\right) = 7(6-63) - 6(3-42) + 5(9-12) = -180,$$

$$C_{23} = (-1)^{2+3} \det\left(\begin{bmatrix} 1 & 2 & 4 \\ 3 & 1 & 7 \\ 5 & 6 & 3 \end{bmatrix}\right) = -(1(3-42) - 2(9-35) + 4(18-5)) = -65,$$

$$C_{41} = (-1)^{4+1} \det\left(\begin{bmatrix} 2 & 3 & 4 \\ 7 & 6 & 5 \\ 1 & 2 & 7 \end{bmatrix}\right) = -(2(42-10) - 3(49-5) + 4(14-6)) = 36.$$

We are in a position to introduce the determinant of $n \times n$ matrices in the following definition.

Definition (Determinant). Let A be a $n \times n$ square matrix. Then, the determinant of A can be computed in one of the following ways:

(a) Selecting any row i $(1 \leq i \leq n)$, it holds that

$$\det(A) = a_{i1}C_{i1} + a_{i2}C_{i2} + \ldots + a_{in}C_{in}; \tag{6.1}$$

(b) Selecting any column j $(1 \leq j \leq n)$, it holds that

$$\det(A) = a_{1j}C_{1j} + a_{2j}C_{2j} + \ldots + a_{nj}C_{nj}. \tag{6.2}$$

Example 63. *We consider an arbitrary* 2×2 *matrix*

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}.$$

Then,

$$\det(A) = a_{11}C_{11} + a12C_{12} = a_{11}a_{22} - a_{12}a_{21},$$

which we have already derived in the former section (Example **??***). We also consider an arbitrary* 3×3 *matrix*

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix},$$

then we have

$$\begin{aligned} \det(A) &= a_{11}C_{11} + a12C_{12} + a13C_{13} \\ &= a_{11} \det\left(\begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix}\right) - a_{12} \det\left(\begin{bmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{bmatrix}\right) + a_{13} \det\left(\begin{bmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}\right), \end{aligned}$$

which we have already derived in Example 60.

Example 64. *We consider the matrix*

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 8 & 7 & 6 & 5 \\ 3 & 1 & 2 & 7 \\ 5 & 6 & 9 & 3 \end{bmatrix}.$$

Let us expand on the first row

$$\begin{aligned} \det(A) &= 1C_{11} + 2C_{12} + 3C_{13} + 4C_{14} \\ &= \det\left(\begin{bmatrix} 7 & 6 & 5 \\ 1 & 2 & 7 \\ 6 & 9 & 3 \end{bmatrix}\right) - 2\det\left(\begin{bmatrix} 8 & 6 & 5 \\ 3 & 2 & 7 \\ 5 & 9 & 3 \end{bmatrix}\right) + 3\det\left(\begin{bmatrix} 8 & 7 & 5 \\ 3 & 1 & 7 \\ 5 & 6 & 3 \end{bmatrix}\right) - 4\det\left(\begin{bmatrix} 8 & 7 & 6 \\ 3 & 1 & 2 \\ 5 & 6 & 9 \end{bmatrix}\right) \\ &= [7(6-63) - 6(3-42) + 5(9-12)] \\ &\quad - 2[8(6-63) - 6(9-35) + 5(27-10)] \\ &\quad + 3[8(3-42) - 7(9-35) + 5(18-5)] \\ &\quad - 4[8(9-12) - 7(27-10) + 6(18-5)] \\ &= 315. \end{aligned}$$

Let us also expand on the second column

$$\begin{aligned} \det(A) &= 2C_{12} + 7C_{22} + 1C_{32} + 6C_{42} \\ &= -2\det\left(\begin{bmatrix} 8 & 6 & 5 \\ 3 & 2 & 7 \\ 5 & 9 & 3 \end{bmatrix}\right) + 7\det\left(\begin{bmatrix} 1 & 3 & 4 \\ 3 & 2 & 7 \\ 5 & 9 & 3 \end{bmatrix}\right) - \det\left(\begin{bmatrix} 1 & 3 & 4 \\ 8 & 6 & 5 \\ 5 & 9 & 3 \end{bmatrix}\right) + 6\det\left(\begin{bmatrix} 1 & 3 & 4 \\ 8 & 6 & 5 \\ 3 & 2 & 7 \end{bmatrix}\right) \\ &= -2[8(6-63) - 6(9-35) + 5(27-10)] \\ &\quad + 7[1(6-63) - 3(9-35) + 4(27-10)] \\ &\quad - [1(18-45) - 3(24-25) + 4(72-30)] \\ &\quad + 6[1(42-10) - 3(56-15) + 4(16-18)] \\ &= 315. \end{aligned}$$

6.3. Determinant Rules

Rule 1 (Determinant of the identity matrix). The determinant of the identity matrix is 1, i.e.,

$$\det\left(\begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}\right) = 1;$$

Example 65. *Let us calculate the determinant I_5, I_4, I_3, I_2. If we expand on the first row, then*

$$\det\left(\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}\right) = \det\left(\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}\right) = \det\left(\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\right) = \det\left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right) = 1.$$

and similarly

$$\det\left(\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}\right) = 1, \quad \det\left(\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\right) = 1, \quad \det\left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right) = 1.$$

Rule 2 (Determinant of a matrix with exchanged rows). The determinant changes sign when two rows are exchanged, i.e.,

$$\det\left(\begin{bmatrix} a_{21} & a_{22} & \cdots & a_{2n} \\ a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}\right) = -\det\left(\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}\right);$$

Example 66. *Compute the following determinant*

$$\det\left(\begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}\right).$$

Let us first exchange the first and fourth rows, i.e.,

$$\det\left(\begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}\right) = -\det\left(\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}\right).$$

We next exchange the second and third rows, i.e.,

$$\det\left(\begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}\right) = \det\left(\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}\right) = \det(I_4) = 1.$$

Example 67. *Let us calculate the determinant*

$$\det\left(\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}\right) = -\det\left(\begin{bmatrix} 7 & 8 & 9 \\ 4 & 5 & 6 \\ 1 & 2 & 3 \end{bmatrix}\right).$$

and

$$\det\left(\begin{bmatrix} 1 & 2 & 3 \\ 4 & 0 & 0 \\ 7 & 8 & 9 \end{bmatrix}\right) = -\det\left(\begin{bmatrix} 4 & 0 & 0 \\ 1 & 2 & 3 \\ 7 & 8 & 9 \end{bmatrix}\right) = 4\det\left(\begin{bmatrix} 2 & 3 \\ 8 & 9 \end{bmatrix}\right) = -4 * 6 = -24.$$

Rule 3 (Determinant of a matrix with a scaled row). Multiplying a row with a scalar t, the determinant is multiplied by t:

$$\det\left(\begin{bmatrix} ta_{11} & ta_{12} & \cdots & ta_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}\right) = t\det\left(\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}\right);$$

Example 68. *Let us calculate the determinant*

$$\det\left(\begin{bmatrix} 5 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}\right) = 5\det\left(\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}\right) = 5.$$

and

$$\det\left(\begin{bmatrix} 3 & 6 & 9 & 12 \\ 5 & 5 & 6 & 8 \\ 9 & 0 & 1 & 2 \\ 3 & 4 & 0 & 1 \end{bmatrix}\right) = 3\det\left(\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 5 & 6 & 8 \\ 9 & 0 & 1 & 2 \\ 3 & 4 & 0 & 1 \end{bmatrix}\right).$$

Rule 4 (Determinant of the sum in one row).

$$\det\left(\begin{bmatrix} a_{11}+b_{11} & a_{12}+b_{12} & \cdots & a_{1n}+b_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}\right) = \det\left(\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}\right) + \det\left(\begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}\right);$$

Example 69. *Let us calculate the determinant*

$$\det\left(\begin{bmatrix} 3 & 6 & 9 & 12 \\ 5 & 5 & 6 & 8 \\ 9 & 0 & 1 & 2 \\ 3 & 4 & 0 & 1 \end{bmatrix}\right) = \det\left(\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 5 & 6 & 8 \\ 9 & 0 & 1 & 2 \\ 3 & 4 & 0 & 1 \end{bmatrix}\right) + \det\left(\begin{bmatrix} 2 & 4 & 6 & 8 \\ 5 & 5 & 6 & 8 \\ 9 & 0 & 1 & 2 \\ 3 & 4 & 0 & 1 \end{bmatrix}\right).$$

Rule 5 (Determinant of a matrix with two equal rows). If two rows of A are equal, then $\det(A) = 0$, i.e.,

$$\det\left(\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}\right) = 0;$$

Example 70. *Let us calculate the determinant*

$$\det\left(\begin{bmatrix} 3 & 6 & 9 & 12 \\ 5 & 5 & 6 & 8 \\ 3 & 4 & 0 & 1 \\ 3 & 4 & 0 & 1 \end{bmatrix}\right) = 0, \quad \det\left(\begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 7 & 8 & 9 \end{bmatrix}\right) = 0.$$

Rule 6 (Determinant of a matrix by subtracting a multiple of one row from another row). Subtracting a multiple of one row from another row leaves $\det(A)$ unchanged, i.e.,

$$\det\left(\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21}-ta_{11} & a_{22}-ta_{12} & \cdots & a_{2n}-ta_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}\right) = \det\left(\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}\right);$$

Example 71. *Let us calculate the determinant*

$$\det\left(\begin{bmatrix} 1 & 2 & 3 \\ 4 & 8 & 12 \\ 7 & 8 & 9 \end{bmatrix}\right).$$

If we multiply the first row by 3 and subtract it from the second row, then

$$\det\left(\begin{bmatrix} 1 & 2 & 3 \\ 4 & 8 & 12 \\ 7 & 8 & 9 \end{bmatrix}\right) = \det\left(\begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 7 & 8 & 9 \end{bmatrix}\right) = 0.$$

Rule 7 (Determinant of a matrix with a zero row). A matrix A with a row of zeros has $\det(A) = 0$, i.e.,

$$\det\left(\begin{bmatrix} a_{11} & ta_{12} & \cdots & a_{1n} \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}\right) = 0;$$

Example 72. *Let us calculate the determinant*

$$\det\left(\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 7 & 8 & 9 \end{bmatrix}\right) = 0.$$

Rule 8 (Determinant of a triangular matrix). If A is either upper or lower triangular then $\det(A) = a_{11}a_{22}\dots a_{nn}$, i.e.,

$$\det\left(\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{21} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{bmatrix}\right) = \det\left(\begin{bmatrix} a_{11} & 0 & \cdots & 0 \\ a_{21} & a_{21} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}\right) = a_{11}a_{22}\dots a_{nn};$$

Example 73. *Let us calculate the determinant of the upper triangular matrix*

$$\det\left(\begin{bmatrix} 3 & 6 & 9 & 12 \\ 0 & 5 & 6 & 8 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}\right) = 3*5*2*1 = 30, \quad \det\left(\begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 9 \end{bmatrix}\right) = 1*2*9 = 18,$$

and the determinant of the lower triangular matrix

$$\det\left(\begin{bmatrix} 3 & 0 & 0 & 0 \\ 2 & 5 & 0 & 0 \\ 1 & 7 & 2 & 0 \\ 4 & 3 & 0 & 1 \end{bmatrix}\right) = 3*5*2*1 = 30, \quad \det\left(\begin{bmatrix} 1 & 0 & 0 \\ 4 & 2 & 0 \\ 5 & 8 & 9 \end{bmatrix}\right) = 1*2*9 = 18.$$

Rule 9 (Determinant of a diagonal matrix).

$$\det\left(\begin{bmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{21} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{bmatrix}\right) = a_{11}a_{22}\dots a_{nn}.$$

Example 74. *Let us calculate the determinant*

$$\det\left(\begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}\right) = 3*5*2*1 = 30, \quad \det\left(\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 9 \end{bmatrix}\right) = 1*2*9 = 18.$$

Example 75. *Let us consider the* 3×3 *matrix*

$$A = \begin{bmatrix} 1 & 3 & 0 \\ 0 & 0 & 2 \\ 1 & 4 & 5 \end{bmatrix}.$$

Then,

$$\det(A) = 1(0-8) - 3(0-2) + 0(0-0) = -2$$

and

$$\det\left(\begin{bmatrix} 0 & 0 & 2 \\ 1 & 3 & 0 \\ 1 & 4 & 5 \end{bmatrix}\right) = -\det\left(\begin{bmatrix} 1 & 3 & 0 \\ 0 & 0 & 2 \\ 1 & 4 & 5 \end{bmatrix}\right) = 2,$$

$$\det\left(\begin{bmatrix} 3 & 9 & 0 \\ 0 & 0 & 2 \\ 1 & 4 & 5 \end{bmatrix}\right) = 3\det\left(\begin{bmatrix} 1 & 3 & 0 \\ 0 & 0 & 2 \\ 1 & 4 & 5 \end{bmatrix}\right) = -6,$$

$$\det\left(\begin{bmatrix} 2 & 7 & 5 \\ 1 & 3 & 0 \\ 1 & 4 & 5 \end{bmatrix}\right) = \det\left(\begin{bmatrix} 1 & 3 & 0 \\ 0 & 0 & 2 \\ 1 & 4 & 5 \end{bmatrix}\right) + \det\left(\begin{bmatrix} 1 & 4 & 5 \\ 0 & 0 & 2 \\ 1 & 4 & 5 \end{bmatrix}\right) = -2 + 0 = -2,$$

$$\det\left(\begin{bmatrix} 0 & 0 & 0 \\ 1 & 4 & 2 \\ 3 & 9 & 7 \end{bmatrix}\right) = 0, \quad \det\left(\begin{bmatrix} 2 & 2 & 2 \\ 1 & 4 & 3 \\ 2 & 2 & 2 \end{bmatrix}\right) = 0,$$

$$\det\left(\begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 2 \\ 0 & 0 & 7 \end{bmatrix}\right) = 1 * 4 * 7 = 28, \quad \det\left(\begin{bmatrix} 2 & 0 & 0 \\ 3 & 8 & 0 \\ 5 & 1 & 6 \end{bmatrix}\right) = 2 * 8 * 6 = 96.$$

Rule 10 (Determinant of an invertible matrix). The matrix A is invertible if and only if $\det(A) \neq 0$.

Example 76. *Show that the following matrices are not invertible*

$$A = \begin{bmatrix} 3 & 0 & 0 & 6 \\ 2 & 5 & 0 & 1 \\ 1 & 7 & 0 & 5 \\ 4 & 3 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 1 & 1/2 \\ 4 & 2 & 1 \\ 5 & 8 & 9 \end{bmatrix}.$$

Let us consider the determinant of A and B, i.e.,

$$\det(A) = \det\left(\begin{bmatrix} 3 & 0 & 0 & 6 \\ 2 & 5 & 0 & 1 \\ 1 & 7 & 0 & 5 \\ 4 & 3 & 0 & 1 \end{bmatrix}\right) = 0, \quad \det(B) = \det\left(\begin{bmatrix} 2 & 1 & 1/2 \\ 4 & 2 & 1 \\ 5 & 8 & 9 \end{bmatrix}\right) = 0,$$

implying that A and B are not invertible.

Rule 11 (Determinant of matrix products). Let A, B be $n \times n$ matrix ($A, B \in \mathbb{R}^{n\times n}$). Then, $\det(AB) = \det(A)\det(B)$.

Example 77. *Let us consider the 3×3 matrix*

$$A = \begin{bmatrix} 7 & 2 & 1 \\ 0 & 1 & 3 \\ 1 & 4 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 8 & 3 \\ 4 & 5 & 6 \\ 2 & 0 & 1 \end{bmatrix}$$

Let us to calculate determinants of A and B are given by

$$\det(A) = 7(5-12) - 2(0-3) + 1(0-1) = 64, \quad \det(B) = 1(5-0) - 8(4-12) + 3(0-10) = -89.$$

Then, we have

$$\det(AB) = \det(A)\det(B) = -64 * 89 = -5696.$$

Example 78. *Let A be a $n \times n$ invertible matrix ($AA^{-1} = A^{-1}A = I_n$). Then, it follows from the product rule for determinant that*

$$\det(A)\det(A^{-1}) = \det(AA^{-1}) = \det(I_n) = 1,$$

which leads to

$$\det(A^{-1}) = \tfrac{1}{\det(A)}.$$

Let c be a scalar. Then,

$$\det(tA) = \det\left(\begin{bmatrix} ta_{11} & ta_{12} & \cdots & ta_{1n} \\ ta_{21} & ta_{22} & \cdots & ta_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ ta_{n1} & ta_{n2} & \cdots & ta_{nn} \end{bmatrix}\right) = t * t * \ldots * t \det(A) = t^n \det(A).$$

> **Rule 12 (Determinant of matrix transpose).** Let A, B be $n \times n$ matrix ($A, B \in \mathbb{R}^{n\times n}$). Then, $\det(A^T) = \det(A)$.

Example 79. *Let us consider the 3×3 matrix*

$$A = \begin{bmatrix} 7 & 2 & 1 \\ 0 & 1 & 3 \\ 1 & 4 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 8 & 3 \\ 4 & 5 & 6 \\ 2 & 0 & 1 \end{bmatrix}$$

Let us calculate the determinants of A and B, which are given by

$$\det(A) = 7(5-12) - 2(0-3) + 1(0-1) = 64, \quad \det(B) = 1(5-0) - 8(4-12) + 3(0-10) = -89.$$

Then, we have

$$\det(A^T) = 64, \quad \det(B^T) = -89.$$

6.4. Cramer's rule and inverse matrices

In this section, we present a procedure to solve nonsingular square systems of equations and find the inverse of nonsingular square matrices using determinants.

Let us consider $A \in \mathbb{R}^{3\times3}$, $b \in \mathbb{R}^3$, and the system if equation $Ax = b$. Then, it is clear that

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 & 0 & 0 \\ x_2 & 1 & 0 \\ x_3 & 0 & 1 \end{bmatrix} = \begin{bmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{bmatrix} = B_1$$

Getting the determinant from both sides of this identity implies

$$\det(A)x_1 = \det(B_1) \quad \Rightarrow \quad x_1 = \tfrac{\det(B_1)}{\det(A)}.$$

In the same way, we get

$$x_2 = \tfrac{\det(B_2)}{\det(A)}, \quad x_3 = \tfrac{\det(B_3)}{\det(A)},$$

for

$$B_2 = \begin{bmatrix} a_{11} & b_1 & a_{13} \\ a_{11} & b_2 & a_{23} \\ a_{11} & b_3 & a_{33} \end{bmatrix}, \quad B_3 = \begin{bmatrix} a_{11} & a_{12} & b_1 \\ a_{11} & a_{12} & b_2 \\ a_{11} & a_{12} & b_3 \end{bmatrix}.$$

We next generalize the above result for arbitrary square systems $Ax = b$ with $\det(A) \neq 0$.

> **Cramer's Rule.** Let $Ax = b$ for $b \in \mathbb{R}^n$ and $A \in \mathbb{R}^{n\times n}$ with $\det(A) \neq 0$. Then, this system is solved by determinants, i.e.,
>
> $$x_i = \tfrac{\det(B_i)}{\det(A)} \quad \text{for } i = 1, 2, \ldots, n,$$
>
> where the matrix B_i has the ith column of A replaced by b, i.e.,
>
> $$B_i = \left[c_1^A, \ldots, c_{i-1}^A, b, c_{i+1}^A, \ldots, c_n^A\right].$$

Let us consider a 2×2 matrix A with $\det(A) \neq 0$. Then, we aim to compute its inverse matrix B, i.e.,

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

In order to compute the inverse matrix, we solve the following systems of equations

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} \\ b_{21} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{12} \\ b_{22} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

By Cramer's rule, we come to

$$b_{11} = \frac{\det\left(\begin{bmatrix} 1 & a_{12} \\ 0 & a_{22} \end{bmatrix}\right)}{\det(A)} = \frac{a_{22}}{\det(A)}, \quad b_{21} = \frac{\det\left(\begin{bmatrix} a_{11} & 1 \\ a_{21} & 0 \end{bmatrix}\right)}{\det(A)} = -\frac{a_{21}}{\det(A)}$$

and

$$b_{12} = \frac{\det\left(\begin{bmatrix} 0 & a_{12} \\ 1 & a_{22} \end{bmatrix}\right)}{\det(A)} = -\frac{a_{12}}{\det(A)}, \quad b_{22} = \frac{\det\left(\begin{bmatrix} a_{11} & 0 \\ a_{21} & 1 \end{bmatrix}\right)}{\det(A)} = \frac{a_{11}}{\det(A)}.$$

Therefore,

$$A^{-1} = B = \frac{1}{\det(A)} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}.$$

Let us consider 3×3 matrix A with $\det(A) \neq 0$. Then, for finding the first column of the inverse matrix, we need to solve following system

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} b_{11} \\ b_{21} \\ b_{31} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

Thus,

$$\det(B_1) = \det\left(\begin{bmatrix} 1 & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & a_{32} & a_{33} \end{bmatrix}\right) = \det\left(\begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix}\right) = C_{11},$$

$$\det(B_2) = \det\left(\begin{bmatrix} a_{11} & 1 & a_{13} \\ a_{21} & 0 & a_{23} \\ a_{31} & 0 & a_{33} \end{bmatrix}\right) = \det\left(\begin{bmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{bmatrix}\right) = C_{12},$$

$$\det(B_3) = \det\left(\begin{bmatrix} a_{11} & a_{12} & 1 & \\ a_{21} & a_{22} & 0 & \\ a_{31} & a_{32} & 0 & a_{33} \end{bmatrix}\right) = \det\left(\begin{bmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}\right) = C_{13}.$$

Therefore, the determinant of each B_i in Cramer's Rule is a cofactor of A.

Calculating inverse with Cramer's Rule. Let $A \in \mathbb{R}^{n \times n}$ with $\det(A) \neq 0$. Then,

$$\left(A^{-1}\right)_{ij} = \frac{C_{ji}}{\det(A)}, \quad A^{-1} = \frac{C^T}{\det(A)}.$$

Example 80. *For $A \in \mathbb{R}^{3 \times 3}$, we have*

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} C_{11} & C_{21} & C_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{23} & a_{33} \end{bmatrix}.$$

CHAPTER 7

Eigenvalue and Eigenvector

In this chapter, we will introduce eigenvalues and eigenvectors of square matrices, which play a central role in linear algebra and are useful for many applications.

Our main objectives are to:

- introduce eigenvalues and eigenvectors of square matrices;
- study the relations of eigenvalues with trace and determinant.

7.1. Eigenvalues and eigenvectors

Let us begin with some explanation about eigenvectors. Although the matrix-vector multiplication Ax may change the vector x in a variety of directions, it is possible to have certain exceptional vectors x such that the direction of Ax is in the same direction as x. These vectors are called **eigenvectors**. Hence, an eigenvector of A is a vector x such that Ax is a number λ multiply by x.

Example 81. *Let us consider the matrix*

$$A = \begin{bmatrix} 0 & -4 \\ 1 & 5 \end{bmatrix}$$

and vectors

$$x = \begin{bmatrix} -4 \\ 1 \end{bmatrix}, \quad y = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \quad z = \begin{bmatrix} -2 \\ 2 \end{bmatrix}, \quad w = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Let us calculate the matrix-vector product Ax, i.e.,

$$Ax = \begin{bmatrix} 0 & -4 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} -4 \\ 1 \end{bmatrix} = \begin{bmatrix} -4 \\ 1 \end{bmatrix},$$

which implies that Ax is just x for the above selection of x. Let us see if we replace x by y, i.e.,

$$Ay = \begin{bmatrix} 0 & -4 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} -4 \\ 8 \end{bmatrix} \neq \lambda \begin{bmatrix} 3 \\ 1 \end{bmatrix},$$

where there is no scalar λ such that $[-4, 8]^T = \lambda[3, 1]$. For z, we have

$$Az = \begin{bmatrix} 0 & -4 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} -2 \\ 2 \end{bmatrix} = \begin{bmatrix} -8 \\ 8 \end{bmatrix} = 4 \begin{bmatrix} -2 \\ 2 \end{bmatrix},$$

which implies $Az = 4z$. Moreover, for w, we have

$$Aw = \begin{bmatrix} 0 & -4 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \neq \lambda \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

where clearly there is no scalar λ such that $Aw = \lambda w$.

Definition (Eigenvalues and eigenvectors). Let A be a $n \times n$ square matrix ($A \in \mathbb{R}^{n\times n}$). A scalar λ is called **eigenvalue** of A if we have the **basic equation**

$$Ax = \lambda x, \tag{7.1}$$

for the nonzero vector x ($x \neq 0$) which is called **eigenvector** of A corresponding to the eigenvalue λ, i.e.,

$$Ax = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} \lambda x_1 \\ \lambda x_2 \\ \vdots \\ \lambda x_n \end{bmatrix} = \lambda x. \tag{7.2}$$

Example 82. *Let us consider the matrix*

$$A = \begin{bmatrix} 2 & 0 \\ 3 & 7 \end{bmatrix},$$

and we want to show that 2 *and* 7 *are its eigenvalues and find corresponding eigenvectors. Since eigenvectors should be nonzero, we are searching for a nontrivial solution of the equations*

$$Ax = 2x, \quad Ax = 7x,$$

leading to

$$(A - 2I_2)x = 0, \quad (A - 7I_2)x = 0.$$

Let us begin with the first system as follows

$$(A - 2I_2)x = \left(\begin{bmatrix} 2 & 0 \\ 3 & 7 \end{bmatrix} - 2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix},$$

which clearly implies that this system has infinitely many solutions $x_1 = \frac{5}{3} x_2$.

We also consider the second system, i.e.,

$$(A - 7I_2)x = \left(\begin{bmatrix} 2 & 0 \\ 3 & 7 \end{bmatrix} - 7 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -5 & 0 \\ 3 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix},$$

which indicates that this system has infinitely many solutions $x_1 = 0$ *for arbitrary* x_2. *Therefore, both* 2 *and* 7 *are eigenvalues of* A *with corresponding eigenvectors* $[5/3x_2, x_2]^T$ *and* $[0, x_2]^T$ *for* ($x_2 \neq 0$), *respectively.*

Simple and multiple eigenvalues. Let A be an $n \times n$ symmetric matrix with the eigenvalues $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_n$. Then,

(a) **Simple eigenvalue.** An eigenvalue λ is **simple** if it is only repeated once among $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_n$;

(b) **Multiple eigenvalue.** An eigenvalue λ is called **multiple** if it is repeated more than once among $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_n$, and the number of its multiplicity is denoted by M.

Example 83. *Let us consider the matrices*

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix}.$$

It is clear that 1, 2, 3, 4 *are eigenvalues of* A *that are simple eigenvalues; however,* 2 *and* 5 *are multiple eigenvalues of* B *with multiplicity* 2.

Rayleigh quotient. Let A be an $n \times n$ symmetric matrix with the eigenvalues

$$\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_n.$$

Then, for any $x \neq 0$, the quotient

$$R_A(x) = \frac{\langle x, Ax \rangle}{\|x\|^2} \tag{7.3}$$

is called **Rayleigh quotient** satisfying

$$\lambda_1 \geq R_A(x) \geq \lambda_n, \tag{7.4}$$

for any $x \neq 0$. Moreover, if v_1 and v_n be eigenvector corresponding to λ_1 and λ_n, respectively, then

$$\begin{aligned} R_A(v_1) &= \frac{\langle v_1, Av_1 \rangle}{\|v_1\|^2} = \frac{\lambda_1 \langle v_1, v_1 \rangle}{\|v_1\|^2} = \lambda_1, \\ R_A(v_n) &= \frac{\langle v_n, Av_n \rangle}{\|v_n\|^2} = \frac{\lambda_n \langle v_n, v_n \rangle}{\|v_n\|^2} = \lambda_n. \end{aligned} \tag{7.5}$$

Example 84. *Let us consider the matrix*

$$Aw = \begin{bmatrix} 2 & 0 \\ 3 & 7 \end{bmatrix}.$$

Then, in Example 82, we showed that 7 *and* 2 *are its eigenvalues with corresponding eigenvectors* $v_1 = [5/3x_2, x_2]^T$ *and* $v_2[0, x_2]^T$ *for* $(x_2 \neq 0)$*, respectively. Therefore, from the Rayleigh quotient, we obtain*

$$7 \geq \frac{\langle x, Ax\rangle}{\|x\|^2} \leq 2, \quad \forall x \neq 0,$$

and

$$R_A(v_1) = \frac{\langle v_1, Av_1\rangle}{\|v_1\|^2} = 7, \quad R_A(v_2) = \frac{\langle v_n, Av_n\rangle}{\|v_n\|^2} = 2.$$

7.2. Computing eigenvalues and eigenvectors

In this section, we discuss systematic ways for computing eigenvalues and eigenvectors of square matrices. In general, there are two ways to compute an eigenvalue λ of a matrix A:

- **Geometric approach.** A given scalar λ is an eigenvalue of A if the system of equations $Ax = \lambda x$ has a nontrivial solution x.
- **Algebraic approach.** A scalar λ is an eigenvalue of A if it is a root of the characteristic equation $\det(A - \lambda I_n) = 0$.

We have already studied the geometric approach in the previous section for some examples; however, we study this with the algebraic approach, which is a more systematic way of computing eigenvalues of matrices.

Computing eigenvalues. We now verify a systematic way to compute eigenvalues of the matrix A. It is easy to see that the equation (7.2) is equivalent to

$$Ax - \lambda x = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} - \begin{bmatrix} \lambda & 0 & \cdots & 0 \\ 0 & \lambda & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix},$$

leading to the linear system of equations

$$(A - \lambda I_n)x = \begin{bmatrix} a_{11} - \lambda & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} - \lambda & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} - \lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix},$$

where λ and $x_1, x_2, \ldots, x_n$ are $n + 1$ unknowns. Since x should be nonzero, this system should have nontrivial solutions, which ensures that

$$\det(A - \lambda I_n) = 0. \tag{7.1}$$

This equation is called **characteristic equation** of A.

Example 85 (eigenvalues of an upper triangular matrix)**.** *Let us consider the upper triangular matrix*

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{bmatrix}$$

Then,

$$\det(A - \lambda I_n) = \det\left(\begin{bmatrix} a_{11} - \lambda & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} - \lambda & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} - \lambda \end{bmatrix}\right) = (a_{11} - \lambda)(a_{22} - \lambda)\ldots(a_{nn} - \lambda).$$

Hence, finding eigenvalues of the matrix A leads to

$$\det(A - \lambda I_n) = (a_{11} - \lambda)(a_{22} - \lambda)\ldots(a_{nn} - \lambda) = 0,$$

leading to the eigenvalues $\lambda_1 = a_{11},\ \lambda_2 = a_{22}, \dots,\ \lambda_n = a_{nn}$.

Example 86 (eigenvalues of a lower triangular matrix)**.** *Let us consider the lower triangular matrix*

$$A = \begin{bmatrix} a_{11} & 0 & \cdots & 0 \\ a_{21} & a_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

Then,

$$\det(A - \lambda I_n) = \det\left(\begin{bmatrix} a_{11}-\lambda & 0 & \cdots & 0 \\ a_{21} & a_{22}-\lambda & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn}-\lambda \end{bmatrix}\right) = (a_{11}-\lambda)(a_{22}-\lambda)\dots(a_{nn}-\lambda).$$

Hence, finding eigenvalues of the matrix A *leads to*

$$\det(A - \lambda I_n) = (a_{11}-\lambda)(a_{22}-\lambda)\dots(a_{nn}-\lambda) = 0,$$

leading to the eigenvalues $\lambda_1 = a_{11},\ \lambda_2 = a_{22}, \dots,\ \lambda_n = a_{nn}$.

Example 87 (eigenvalues of a diagonal matrix)**.** *Let us consider the lower triangular matrix*

$$A = \begin{bmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{bmatrix}$$

Then,

$$\det(A - \lambda I_n) = \det\left(\begin{bmatrix} a_{11}-\lambda & 0 & \cdots & 0 \\ 0 & a_{22}-\lambda & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn}-\lambda \end{bmatrix}\right) = (a_{11}-\lambda)(a_2-\lambda)\dots(a_{nn}-\lambda).$$

Hence, finding eigenvalues of the matrix A *leads to*

$$\det(A - \lambda I_n) = (a_{11}-\lambda)(a_2-\lambda)\dots(a_{nn}-\lambda) = 0,$$

leading to the eigenvalues $\lambda_1 = a_{11},\ \lambda_2 = a_{22}, \dots,\ \lambda_n = a_{nn}$.

Example 88 (eigenvalues of AB and $A+B$)**.** *Let us consider the lower triangular matrix*

$$A = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix},$$

where eigenvalues of A *and* B *are* $\lambda_1 = \lambda_2 = 0$. *Further,*

$$AB = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix},$$

where eigenvalue of AB *are* $\lambda_1 = 0$ *and* $\lambda_2 = 1$. *Hence,* **eigenvalues of AB are not equal to the product of eigenvalues of A and** b. *In addition,*

$$A + B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix},$$

where eigenvalue of $A+B$ *are given by the characteristic equation* $\lambda^2 = 1$ *(i.e.,* $\lambda_1 = -1$ *and* $\lambda_2 = 1$*). Hence,* **eigenvalues of $A+B$ are not equal to the sum of eigenvalues of A and** B.

Computing eigenvectors. In order to compute, eigenvectors corresponding to the eigenvalue λ of matrix A, the next three steps are recognized:

(1) **Compute the determinant of $A - \lambda I_n$.** With λ subtracted along the diagonal of A, this determinant starts with λ^n or $-\lambda^n$, i.e., the characteristic equation is a polynomial in λ of degree n.
(2) **Find the roots of this characteristic polynomial, by solving f $\det(A - \lambda I_n) = 0$.** Then roots are the n eigenvalues of A. They make $A - \lambda I_n$ singular.
(3) **Find an eigenvector by solving the basic equation.** For each eigenvalue λ, we solve the system of equations $(A - \lambda I_n)x = 0$ to find an eigenvector x.

Example 89. *Let us consider the* 3×3 *upper triangular matrix*

$$A = \begin{bmatrix} 1 & 4 & 5 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Since this is upper triangular, it has the eigenvalues $\lambda_1 = 1$, $\lambda_2 =$, *and* $\lambda_3 = 3$. *For* $\lambda_1 = 1$, *the system of equations*

$$(A - \lambda_1 I_3)x = \begin{bmatrix} 0 & 4 & 5 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix},$$

leading to

$$\begin{aligned} &4x_2 + 5x_3 = 0, \\ &x_2 = 0, \\ &2x_3 = 0, \end{aligned} \qquad .$$

This consequently ensures that

$$x = \begin{bmatrix} x_1 \\ 0 \\ 0 \end{bmatrix}, \quad x_1 \neq 0$$

is an eigenvector of A *corresponding to* $\lambda_1 = 1$. *For* $\lambda_2 = 2$, *the system of equations*

$$(A - \lambda_2 I_3)x = \begin{bmatrix} -1 & 4 & 5 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix},$$

implying

$$\begin{aligned} &-x_1 + 4x_2 + 5x_3 = 0, \\ &x_3 = 0, \end{aligned} \quad ,$$

i.e., $x_1 = 4x_2, x_3 = 0$. *This consequently ensures that*

$$x = \begin{bmatrix} 4x_2 \\ x_2 \\ 0 \end{bmatrix}, \quad x_2 \neq 0$$

is an eigenvector of A *corresponding to* $\lambda_2 = 2$. *For* $\lambda_3 = 3$, *the system of equations*

$$(A - \lambda_3 I_3)x = \begin{bmatrix} -2 & 4 & 5 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix},$$

implying

$$\begin{aligned} &-2x_1 + 4x_2 + 5x_3 = 0, \\ &x_2 = 0, \end{aligned} \quad ,$$

i.e., $x_1 = 5/2x_3, x_2 = 0$. *This consequently ensures that*

$$x = \begin{bmatrix} \frac{5}{2}x_x \\ 0 \\ x_3 \end{bmatrix}, \quad x_3 \neq 0$$

is an eigenvector of A *corresponding to* $\lambda_3 = 3$.

7.3. Relation of eigenvalues with determinant and trace

Let us start with the following example.

Example 90. *Let us consider the matrix*

$$A = \begin{bmatrix} 0 & -4 \\ 1 & 5 \end{bmatrix}.$$

Then, solving the characteristic equation

$$\det(A - \lambda I_2) = \det\left(\begin{bmatrix} -\lambda & -4 \\ 1 & 5-\lambda \end{bmatrix}\right) = \lambda^2 - 5\lambda + 4 = 0$$

leads to $\lambda_1 = 1$ *and* $\lambda_2 = 4$. *Note that* $\det(A) = 4 = \lambda_1\lambda_2$ *and* $\mathrm{tr}(A) = 5 = \lambda_1 + \lambda_2$.

Determinant and eigenvalues. Let A be a $n \times n$ matrix with the eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$. Then, the determinant of A is equal to the product of its eigenvalues, i.e.,

$$\det(A) = \lambda_1\lambda_2 \ldots \lambda_n.$$

Example 91. *Let us consider the matrix*

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 5 & 0 \\ 3 & 4 & 4 \end{bmatrix}.$$

Then, solving the characteristic equation

$$\det(A - \lambda I_3) = \det\left(\begin{bmatrix} 2-\lambda & 0 & 0 \\ 1 & 5-\lambda & 0 \\ 3 & 4 & 4-\lambda \end{bmatrix}\right) = (2-\lambda)(5-\lambda)(4-\lambda) = 0$$

leads to $\lambda_1 = 2$, $\lambda_2 = 4$, *and* $\lambda = 5$. *We notice that* $\det(A) = \lambda_1\lambda_2\lambda_3 = 40$.

Eigenvalues, determinant, invertibility, and system of equations. Let us consider the square matrix $A \in \mathbb{R}^{n\times n}$, $b \in \mathbb{R}^n$, and

$$Ax = b.$$

If $\lambda_1, \lambda_2, \ldots, \lambda_n$ are eigenvalues of A, then the following statements hold:

(a) If $\lambda_i \neq 0$ for all $i = 1, 2, \ldots, n$, then $\det(A) = \lambda_1\lambda_2 \ldots \lambda_n \neq 0$;
(b) If $\lambda_i \neq 0$ for all $i = 1, 2, \ldots, n$, then A is invertible;
(c) If $\lambda_i \neq 0$ for all $i = 1, 2, \ldots, n$, then the system of equations $Ax = b$ has a unique solution $x = A^{-1}b$;
(d) If at least one of einvalues of A is zero, then $\det(A) = 0$ and therefore A is singular.

Example 92. *Let us consider the matrices*

$$A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 7 & 0 & 0 \\ 0 & 0 & 9 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & -5 & 0 & 0 \\ 2 & 2 & 8 & 0 \\ 1 & 7 & 0 & -6 \end{bmatrix}, \quad C = \begin{bmatrix} -4 & 0 & 3 & 8 \\ 0 & -1 & 2 & 7 \\ 0 & 0 & 0 & -9 \\ 0 & 0 & 0 & 3 \end{bmatrix}.$$

Then, we have

- *Since A is a diagonal matrix, the eigenvalues of A are $\lambda_1 = 2$, $\lambda_2 = 7$, $\lambda_3 = 9$, and $\lambda_4 = 1$. Hence, $\det(A) = 2 * 7 * 9 * 1 = 126$ and therefore A is invertible and the system of equation $Ax = b$ has the unique solution*

$$x_1 = b_1/2, \quad x_2 = b_2/7, \quad x_3 = b_3/9, \quad x_4 = b_4.$$

- *The matrix B is lower triangular with the eigenvalues $\lambda_1 = 3$, $\lambda_2 = -5$, $\lambda_3 = 8$, and $\lambda_4 = -6$, i.e., $\det(B) \neq 0$ and B is invertible and the system $Ax = b$ has a unique solution.*

- *Finally, the matrix C is upper triangular with the eigenvalues $\lambda_1 = -4$, $\lambda_2 = -1$, $\lambda_3 = 0$, and $\lambda_4 = 3$. Therefore, $\det(C) = 0$, C is not invertible, and the system $Ax = b$ does not have a unique solution.*

Trace and eigenvalues. Let A be a $n \times n$ matrix with the eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$. Then, the trace of A is equal to the sum of its eigenvalues, i.e.,

$$\mathrm{tr}(A) = \lambda_1 + \lambda_2 + \ldots + \lambda_n.$$

Example 93. *Let us consider the matrix*

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 5 & 0 \\ 3 & 4 & 4 \end{bmatrix}.$$

Then, solving the characteristic equation

$$\det(A - \lambda I_3) = \det\left(\begin{bmatrix} 2-\lambda & 0 & 0 \\ 1 & 5-\lambda & 0 \\ 3 & 4 & 4-\lambda \end{bmatrix}\right) = (2-\lambda)(5-\lambda)(4-\lambda) = 0$$

leads to $\lambda_1 = 2$, $\lambda_2 = 4$, and $\lambda = 5$. We notice that $\mathrm{tr}(A) = \lambda_1 + \lambda_2 + \lambda_3 = 11$.

Example 94. *Determine a and b in the matrix*

$$A = \begin{bmatrix} 2 & 5 & 7 \\ 0 & a & 8 \\ 0 & 0 & b \end{bmatrix}$$

if $det(A) = 24$ and $\mathrm{tr}(A) = 9$. We have the following equations

$$\det(A) = 2ab = 24, \quad \mathrm{tr}(A) = 2 + a + b = 9.$$

Hence,

$$a = \tfrac{12}{b}, \quad \tfrac{12}{b} + b = 7 \quad \Rightarrow \quad b^2 - 7b + 12 = (b-3)(b-4) = 0$$

leading to $b = 3$ or $b = 4$. If we set $b = 3$, then $a = 4$, and if we set $b = 4$, then $a = 3$.

CHAPTER 8

Diagonalizable and Similar Matrices

In this chapter, we introduce diagonalizable matrices that are matrices turning into diagonal matrices when we use the eigenvectors properly. In addition, we present similar matrices and verify some of their properties.

Our main objectives are to:

- introduce diagonalizable matrices and verify conditions guaranteeing the diagonalizability of a matrix;
- introduce similar matrices and their properties.

8.1. Matrix diagonalization

In this section, we introduce a class of matrices that can turn to a diagonal matrix using a proper eigenvector matrix. Let us begin with the following example.

Example 95 (Matrix diagonalization). *Let us consider the matrix*

$$A = \begin{bmatrix} 2 & 0 \\ 5 & 7 \end{bmatrix},$$

where $\lambda_1 = 2$ and $\lambda_2 = 7$ are its distinct eigenvalues. Then, for $\lambda_1 = 2$, we have

$$(A - \lambda_1 I_2)x = \begin{bmatrix} 0 & 0 \\ 5 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

implying $x_2 = -x_1$, i.e.,

$$v_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

is an eigenvector of A corresponding to $\lambda_1 = 2$. Moreover, for $\lambda_2 = 7$, we have

$$(A - \lambda_2 I_2)x = \begin{bmatrix} -5 & 0 \\ 5 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

implying $x_1 = 0$, i.e.,

$$v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

is an eigenvector of A corresponding to $\lambda_1 = 7$. It is evident that v_1 and v_2 are two independent eigenvectors of A. Let us define the eigenvalue and eigenvector matrices

$$X = [v_1\ v_2] = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}, \quad \Lambda = \begin{bmatrix} 2 & 0 \\ 0 & 7 \end{bmatrix}.$$

It is clear that

$$X^{-1} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \quad \Rightarrow \quad X^{-1}AX = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 7 & 7 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 7 \end{bmatrix} = \Lambda.$$

This clearly implies that the matrix A turns into the diagonal matrix Λ by applying the eigenvector matrix X and its inverse X^{-1}.

Definition (Matrix diagonalization). Let A be an $n \times n$ square matrix ($A \in \mathbb{R}^{n\times n}$), let $\lambda_1, \lambda_2, \ldots, \lambda_n$ be its n eigenvalues, and let $x_1, x_2, \ldots, x_n$ be n linearly independent eigenvectors of A. Let us define

$$X = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nn} \end{bmatrix}, \quad \Lambda = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}. \tag{8.1}$$

Then, we have

$$X^{-1}AX = \Lambda, \tag{8.2}$$

which equivalently leads to the following decomposition

$$A = X\Lambda X^{-1}. \tag{8.3}$$

We next present a sufficient condition for diagonalizability of an arbitrary matrix. Let A be an $n \times n$ matrix ($A \in \mathbb{R}^{n\times n}$) with n **distinct** eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$ and n corresponding eigenvectors $x_1, x_2, \ldots, x_n$. Then, we consider the linear combination

$$\alpha_1 x_1 + \alpha_2 x_2 + \ldots + \alpha_n x_n = 0. \tag{8.4}$$

Multiplying this identity by A leads to

$$\alpha_1 Ax_1 + \alpha_1 Ax_2 + \ldots + \alpha_1 Ax_n = \alpha_1\lambda_1 x_1 + \alpha_2\lambda_2 x_2 + \ldots + \alpha_n\lambda_n x_n = 0.$$

Multiplying (8.4) by λ_n implies

$$\alpha_1\lambda_n x_1 + \alpha_2\lambda_n x_2 + \ldots + \alpha_n\lambda_n x_n = 0.$$

Subtracting the last two identities indicates

$$\alpha_1(\lambda_1 - \lambda_n)x_1 + \alpha_2(\lambda_2 - \lambda_n)x_2 + \ldots + \alpha_{n-1}(\lambda_{n-1} - \lambda_n)x_{n-1} = 0.$$

Multiplying by A and by λ_{n-1} and subtracting the results lead to

$$\alpha_1(\lambda_1 - \lambda_n)(\lambda_1 - \lambda_{n-1})x_1 + \alpha_2(\lambda_2 - \lambda_n)(\lambda_2 - \lambda_{n-1})x_2 + \ldots + \alpha_{n-2}(\lambda_{n-2} - \lambda_n)(\lambda_{n-2} - \lambda_{n-1})x_{n-2} = 0.$$

Continuing this procedure will imply

$$\alpha_1(\lambda_1 - \lambda_n)(\lambda_1 - \lambda_{n-1})\cdots(\lambda_1 - \lambda_2)x_1 = 0,$$

leading to $\alpha_1 = 0$. Repeating this procedure, we can show that

$$\alpha_1 = \alpha_2 = \ldots = \alpha_n = 0,$$

which says that the eigenvectors $x_1, x_2, \ldots, x_n$ are independent, i.e, the eigenvector matrix

$$X = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nn} \end{bmatrix}$$

is invertible. This consequently suggests the following sufficient result for diagonalizability of square matrices.

Sufficient condition for diagonalizability of matrices. Let A be an $n \times n$ matrix ($A \in \mathbb{R}^{n\times n}$). If A has $\lambda_1, \lambda_2, \ldots, \lambda_n$ as n **distinct** eigenvalues, and $x_1, x_2, \ldots, x_n$ be its eigenvectors corresponding to $\lambda_1, \lambda_2, \ldots, \lambda_n$, respectively, then, A must be diagonalizable.

Example 96. *Let us consider the matrices*

$$A_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}, \quad B_1 = \begin{bmatrix} 2 & 3 & 0 & 6 \\ 0 & 3 & 7 & 0 \\ 0 & 0 & 6 & 5 \\ 0 & 0 & 0 & 8 \end{bmatrix}, \quad C_1 = \begin{bmatrix} 9 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 3 & 1 & 5 & 0 \\ 0 & 7 & 0 & 0 \end{bmatrix},$$

and

$$A_2 = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}, \quad B_2 = \begin{bmatrix} 2 & 3 & 0 & 6 \\ 0 & 0 & 7 & 0 \\ 0 & 0 & 8 & 5 \\ 0 & 0 & 0 & 8 \end{bmatrix}, \quad C_2 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 3 & 1 & 0 & 0 \\ 0 & 7 & 0 & 0 \end{bmatrix}.$$

It is clear that

- A_1 *is a diagonal matrix with 4 distinct eigenvalues* $1, 2, 3, 4$, *and therefore it is diagonalizable;*
- B_1 *is an upper triangular matrix with 4 distinct eigenvalues* $2, 3, 6, 8$, *and therefore it is diagonalizable;*
- C_1 *a lower triangular matrix with 4 distinct eigenvalues* $9, 2, 5, 0$, *and therefore it is diagonalizable;*

and

- A_2 *is a diagonal matrix with 3 distinct eigenvalues* $2, 3, 4$, *and therefore we are not sure that is diagonalizable;*
- B_2 *is an upper triangular matrix with 3 distinct eigenvalues* $2, 0, 8$, *and therefore we are not sure that is diagonalizable;*
- C_2 *is a lower triangular matrix with an eigenvalue* 0. *Hence, eigenvectors corresponding* $\lambda = 0$ *satisfy* $Ax = 0$, *which is the rank of the null space of* A. *Since the rank of* A *is 2, we have* $\dim(N(A)) = 2$, *i.e.,* A *has at most 2 independent eigenvectors. Therefore,* A *is not diagonalizable.*

In the next section, we introduce a necessary condition that can be used to identify the diagonalizability of A_2, *and* B_2.

Calculating kth power of a diagonalizable matrix. Let A be an $n \times n$ diagonalizable matrix ($A \in \mathbb{R}^{n \times n}$), let $\lambda_1, \lambda_2, \ldots, \lambda_n$ be its n eigenvalues, and let $x_1, x_2, \ldots, x_n$ be n linearly independent eigenvectors of A. Then, kth power of A is given by

$$A^k = (X\Lambda X^{-1})(X\Lambda X^{-1}) \ldots (X\Lambda X^{-1}) = X\Lambda^k X^{-1}$$

or equivalently

$$A^k = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nn} \end{bmatrix} \begin{bmatrix} \lambda_1^k & 0 & \cdots & 0 \\ 0 & \lambda_2^k & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n^k \end{bmatrix} \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nn} \end{bmatrix}^{-1}.$$

Example 97 (Power k of a matrix)**.** *Let us consider the matrix*

$$A = \begin{bmatrix} 2 & 0 \\ 5 & 7 \end{bmatrix},$$

where $\lambda_1 = 2$ *and* $\lambda_2 = 7$ *are its eigenvalues. From Example 95, we obtain*

$$X = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}, \quad X^{-1} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \quad \Lambda = \begin{bmatrix} 2 & 0 \\ 0 & 7 \end{bmatrix}.$$

In addition,

$$A^k = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 2^k & 0 \\ 0 & 7^k \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}.$$

For $k = 3$, *we obtain*

$$A^3 = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 2^3 & 0 \\ 0 & 7^3 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 8 & 0 \\ 0 & 343 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 8 & 0 \\ 335 & 343 \end{bmatrix}.$$

For $k = 100$, *we have*

$$A^{100} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 2^{100} & 0 \\ 0 & 7^{100} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2^{100} & 0 \\ -2^{100} & 7^{100} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2^{100} & 0 \\ 7^{100} - 2^{100} & 7^{100} \end{bmatrix}.$$

We can clearly see that calculating the power of a diagonalizable matrix can be done very fast, avoiding multiplying such a matrix many times.

Example 98. *Let A be an $n \times n$ diagonalizable matrix ($A \in \mathbb{R}^{n\times n}$) with $|\lambda_i| < 1$ ($i = 1, 2, \ldots, n$). Then,*

$$\lim_{k\to\infty} A^k = \lim_{k\to\infty} X\Lambda^k X^{-1}$$

because

$$\lim_{k\to\infty} \Lambda^k = \lim_{k\to\infty} \begin{bmatrix} \lambda_1^k & 0 & \cdots & 0 \\ 0 & \lambda_2^k & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n^k \end{bmatrix} = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix}.$$

8.2. Diagonalizability of matrices

In this section, we present a necessary condition that can be used to check if an arbitrary matrix is diagonalizable.

Definition (Nondiagonalizable matrices). Let A be an $n \times n$ matrix ($A \in \mathbb{R}^{n\times n}$). Then, A is nondiagonalizable if it is not a diagonalizable matrix.

Example 99 (Nondiagonalizable matrix)**.** *Let us consider the matrix*

$$A = \begin{bmatrix} 0 & 0 \\ 5 & 0 \end{bmatrix},$$

which had two eigenvalues $\lambda_1 = 0$ and $\lambda_2 = 0$, i.e., 0 is a multiple eigenvalue by multiplicity $M = 2$. However, for any invertible matrix

$$X = \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix}, \quad \Lambda = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix},$$

we have

$$X^{-1}\Lambda X = \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix}^{-1} \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \neq \begin{bmatrix} 0 & 0 \\ 5 & 0 \end{bmatrix} = A,$$

which shows that A is nondiagonalizable.

Definition (Geometric and algebraic multiplicity). Let A be an $n \times n$ matrix ($A \in \mathbb{R}^{n\times n}$) and let λ be its eigenvlaue. Then,

- **Geometric multiplicity.** The number of independent eigenvectors corresponding to λ is called *geometric multiplicity* and denoted by M_g, which is the dimension of the null space of $A - \lambda I_n$.
- **Algebraic multiplicity.** The number of repetition of λ among its eigenvalues is called *algebraic multiplicity* and denoted by M_a, given by solving the characteristic equation $\det(A - \lambda I_n) = 0$.

Example 100 (Nondiagonalizable matrix)**.** *Let us consider the matrix*

$$A = \begin{bmatrix} 1 & 4 & 1 \\ 0 & 2 & 3 \\ 0 & 0 & 2 \end{bmatrix}.$$

It is clear that $\lambda_1 = 1$, $\lambda_2 = 2$, $\lambda_3 = 2$ are its eigenvalues (i.e., 2 has the algebraic multiplicity $M_a = 2$). For $\lambda_1 = 1$, we have $M_a = M_g = 1$ because the null space of

$$A - \lambda_1 I_3 = \begin{bmatrix} 0 & 4 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

has one dimension. Moreover, for $\lambda_2 = 2$, A has the algebraic multiplicity $M_a = 2$ and we have that the matrix

$$A - \lambda_2 I_3 = \begin{bmatrix} -1 & 4 & 1 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

has rank 2 and its null space has one dimension. Hence, the number of independent eigenvectors corresponding to $\lambda_2 = 2$ is one, i.e., $M_g = 1$.

Checking diagolizability of a matrix. Let A be an $n \times n$ matrix ($A \in \mathbb{R}^{n\times n}$) and let $\lambda_1, \lambda_2, \ldots, \lambda_m$ be its distinct eigenvlaues (i.e., $\mathcal{N}_{\text{val}}(A) = \sum_{i=1}^m M_a^i = n$). Then, if for all λ_i we have that $M_g^i = M_a^i$, then the number of distinct eigenvectors of A is denoted by $\mathcal{N}_{\text{vec}}$ and given by

$$\mathcal{N}_{\text{vec}}(A) = \sum_{i=1}^m M_g^i = \sum_{i=1}^m M_a^i = n,$$

which implies that we have n distinct eignvectors, i.e., A is diagonalizable. If there is $i \in \{1, 2, \ldots, m\}$ such that $M_g^i < M_a^i$, then the matrix A is nondiagolizable.

Example 101 (Nondiagonalizable matrix)**.** *Let us consider the matrices*

$$A = \begin{bmatrix} 3 & 5 & 6 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 4 & 1 \\ 0 & 2 & 3 \\ 0 & 0 & 2 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 0 \\ 5 & 0 \end{bmatrix}.$$

For the matrix A, it is clear that $\lambda_1 = 3$, $\lambda_2 = 4$, $\lambda_3 = 4$ are its eigenvalues (i.e., 4 has the algebraic multiplicity $M_a^2 = 2$). For $\lambda_1 = 3$, we have $M_a^1 = M_g^1 = 1$ because the null space of

$$A - \lambda_1 I_3 = \begin{bmatrix} 0 & 5 & 6 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

has one dimension. Moreover, for $\lambda_2 = 4$, B has the algebraic multiplicity $M_a^2 = 2$ and we have that the matrix

$$A - \lambda_2 I_3 = \begin{bmatrix} -1 & 5 & 6 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

has rank one and its null space is two-dimensional. Hence, the number of independent eigenvectors corresponding to $\lambda_2 = 2$ is two, i.e., $M_g^2 = M_a^2 = 2$. Thus,

$$\mathcal{N}_{\text{vec}}(A) = \mathcal{N}_{\text{val}}(A) = 2 + 1 = 3$$

which implies that A is diagonalizable.

For the matrix B, it follows from Example 100 that for $\lambda = 2$ we have $M_g^2 = 1 < 2 = M_a^2$. Therefore, the matrix A is nondiagonalizable. For the matrix C, we have $\lambda = 0$, $M_a^1 = 2$, and null space of

$$C - \lambda I_2 = \begin{bmatrix} 0 & 0 \\ 5 & 0 \end{bmatrix}$$

has rank one, i.e., $M_g^1 = 1$. Therefore, $M_g^1 = 1 < 2 = M_a^1$ and the matrix A is nondiagonalizable.

Diagonalizability and invertibility. Note that while diagonalizability is about the existence of n distinct eigenvectors, invertibility is about having nonzero eigenvalues. Moreover, the following statements are true:

(a) **Diagonalizability does not imply invertibility.** Let us consider the matrix

$$A = \begin{bmatrix} 1 & 0 \\ 5 & 0 \end{bmatrix},$$

where its eigenvalues are $\lambda_1 = 1$ and $\lambda_1 = 0$ that are diagonalizable (since there are two distinct eigenvalues); however, A is not invertible.

(b) **Invertibility does not imply diagonalizability.** Let us consider the matrix

$$A = \begin{bmatrix} 1 & 0 \\ 5 & 1 \end{bmatrix},$$

where its eigenvalue is $\lambda_1 = 1$ ($M_a^1 = 2$), i.e., A is invertible. On the other hand, we have

$$A - \lambda_1 I_2 = \begin{bmatrix} 0 & 0 \\ 5 & 0 \end{bmatrix},$$

which has rank one, i.e., its null space has rank one too. Hence, there is only one distinct eigenvector ($1 = M_g^1 < 2 = M_a^1$), i.e., A is not diagonalizable.

8.3. Similar matrices

In this section, we introduce similar matrices, describe some of their properties, and verify their connection to diagonalizable matrices.

Definition (Similar matrices). Let A, B, and C be three $n \times n$ matrices ($A, B, C \in \mathbb{R}^{n \times n}$), and let B be invertible. Then, the matrices A and C are **similar matrices** if

$$A = BCB^{-1}. \tag{8.1}$$

Example 102. *Let us consider the* 3×3 *matrices*

$$A = \begin{bmatrix} 5 & 2 & -1 \\ 0 & 2 & 0 \\ 3 & 0 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 4 & 5 \\ 0 & 2 & 0 \\ 1 & 0 & 3 \end{bmatrix}$$

Let us first compute the inverse of B*. To do so, we need to compute its determinant, i.e.,*

$$\det(B) = 3 \det\left(B = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}\right) = 3 * 2 = 6,$$

which implies that B *is invertible and* B^{-1} *is given by*

$$B^{-1} = \frac{1}{\det(B)} \begin{bmatrix} C_{11} & C_{21} & C_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{23} & C_{33} \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 6 & 0 & -8 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -\frac{4}{3} \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{3} \end{bmatrix}.$$

We now have

$$BDB^{-1} = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 4 & 5 \\ 0 & 2 & 0 \\ 1 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & -\frac{4}{3} \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{3} \end{bmatrix} = \begin{bmatrix} 5 & 4 & 17 \\ 0 & 4 & 0 \\ 3 & 0 & 9 \end{bmatrix} \begin{bmatrix} 1 & 0 & -\frac{4}{3} \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{3} \end{bmatrix} = \begin{bmatrix} 5 & 2 & -1 \\ 0 & 2 & 0 \\ 3 & 0 & -1 \end{bmatrix} = A.$$

Therefore, A *and* D *are similar matrices.*

Similar matrices have the same eigenvalues. Let A, B, and C be three $n \times n$ matrices ($A, B, C \in \mathbb{R}^{n\times n}$), and let B be invertible. Let us also assume that A and C are similar matrices, i.e., $A = BCB^{-1}$. If $Cx = \lambda x$, then

$$A(Bx) = BCB^{-1}(Bx) = BCx = \lambda(Bx), \tag{8.2}$$

which means that λ is an eigenvalue of A and Bx is the corresponding eigenvector. Therefore, **similar matrices share the same eigenvalues**.

Example 103. *Let us consider the* 3×3 *matrices*

$$A = \begin{bmatrix} 5 & 2 & -1 \\ 0 & 2 & 0 \\ 3 & 0 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 4 & 5 \\ 0 & 2 & 0 \\ 1 & 0 & 3 \end{bmatrix}.$$

In Example 102, we have shown that A and C are similar. Therefore, A and C share the same eigenvalues.

Diagonalizable and similar matrices. Let A be an $n \times n$ matrix ($A \in \mathbb{R}^{n\times n}$), and let $\lambda_1, \lambda_2, \ldots, \lambda_n$ be its n distinct eigenvalues, and let $x_1, x_2, \ldots, x_n$ be n corresponding eigenvectors of A. Then, considering the matrices X and Λ given in (8.1), we have

$$A = X\Lambda X^{-1},$$

implying that the digonalizable matrix A and Λ are similar matrices.

Example 104 (Matrix diagonalization and similar matrices)**.** *Let us consider the matrix*

$$A = \begin{bmatrix} 2 & 0 \\ 5 & 7 \end{bmatrix},$$

where $\lambda_1 = 2$ *and* $\lambda_2 = 7$ *are its eigenvalues. From Example 95, we obtain*

$$X = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}, \quad X^{-1} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \quad \Lambda = \begin{bmatrix} 2 & 0 \\ 0 & 7 \end{bmatrix}.$$

In Example 97, We have already shown that A *is diagonalizable* ($X^{-1}AX = \Lambda$)*, which clearly implies that* A *and* Λ *are similar.*

CHAPTER 9

Orthogonality

In this chapter, we study orthogonality of vectors and subspaces, present projection onto a line and onto a subspace, describe the least square approximation, and discuss the construction of orthonormal bases by the Gram-Schmidt process.

Our main objectives are to:

- define orthogonal vectors and subspaces;
- discuss projection onto a line and onto a subspace;
- present the least square approximation for a linear system;
- study construction of orthonormal bases by the Gram-Schmidt process.

9.1. Orthogonal subspaces

In this section, we define orthogonality of vectors and subspaces. In particular, we discuss orthogonality in matrix four spaces.

Definition (Orthogonal vectors). Two vectors $a \in \mathbb{R}^n$ and $b\mathbb{R}^n$ are orthogonal if

$$\langle a, b\rangle = 0$$

and consequently

$$\|a+b\|^2 = \|a\|^2 + \|b\|^2 + \langle a, b\rangle = \|a\|^2 + \|b\|^2.$$

If a vector a is orthogonal to itself, then

$$\langle a, a\rangle = \|a\|^2 = 0,$$

which implies that a is the zero vector.

Example 105. *Let us consider vectors*

$$a = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad b = \begin{bmatrix} -13 \\ 2 \\ 3 \end{bmatrix}, \quad c = \begin{bmatrix} -1 \\ 2 \\ -3 \end{bmatrix}.$$

Then, we have

$$\langle a, b\rangle = -13 + 4 + 9 = 0, \quad \langle a, c\rangle = -1 + 4 - 9 = -6, \quad \langle b, c\rangle = 13 + 4 - 9 = 8,$$

which implies that while a and b are orthogonal, the vecors a and c and the vectors b and c are not orthogonal.

Definition (Orthogonal subspaces). Let U and V be two subspaces of $\mathbb{R}^n$. Then, we say that U and V are orthogonal if

$$\langle u, v\rangle = 0$$

for all $u \in U$ and all $v \in V$.

Example 106. *Let us set*

$$U = \operatorname{span}\{u_1, u_2\} = \operatorname{span}\left\{ \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -3 \\ 0 \end{bmatrix} \right\}, \quad V = \operatorname{span}\{v_1\} = \operatorname{span}\left\{ \begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix} \right\}.$$

Then, we have

$$\langle v_1, u_1 \rangle = 0, \quad \langle v_1, u_2 \rangle = 0,$$

which clearly implies that U *and* V *are orthogonal subspaces.*

Orthogonality of four subspaces. Let $A \in \mathbb{R}^{m \times n}$, and let $C(A)$, $C(A^T)$, $N(A)$ and $N(A^T)$ show its column space, row space, null space and left null space, respectively. Then, the following statements are true:

(a) **The column space $C(A)$ and the left null space $N(A^T)$ are orthogonal.** If $Ay \in C(A)$ and $x \in N(A^T)$, then

$$\langle x, Ay \rangle = \langle A^T x, y \rangle = \langle 0, y \rangle = 0,$$

which can also be seen by

$$A^T y = \begin{bmatrix} c_1^T \\ c_2^T \\ \vdots \\ c_n^T \end{bmatrix} y = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

(b) **The row space $C(A^T)$ and the null space $N(A)$ are orthogonal.** If $A^T y \in C(A^T)$ and $x \in N(A)$, then

$$\langle x, A^T y \rangle = \langle Ax, y \rangle = \langle 0, y \rangle = 0,$$

which can also be seen by

$$Ax = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_m \end{bmatrix} x = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

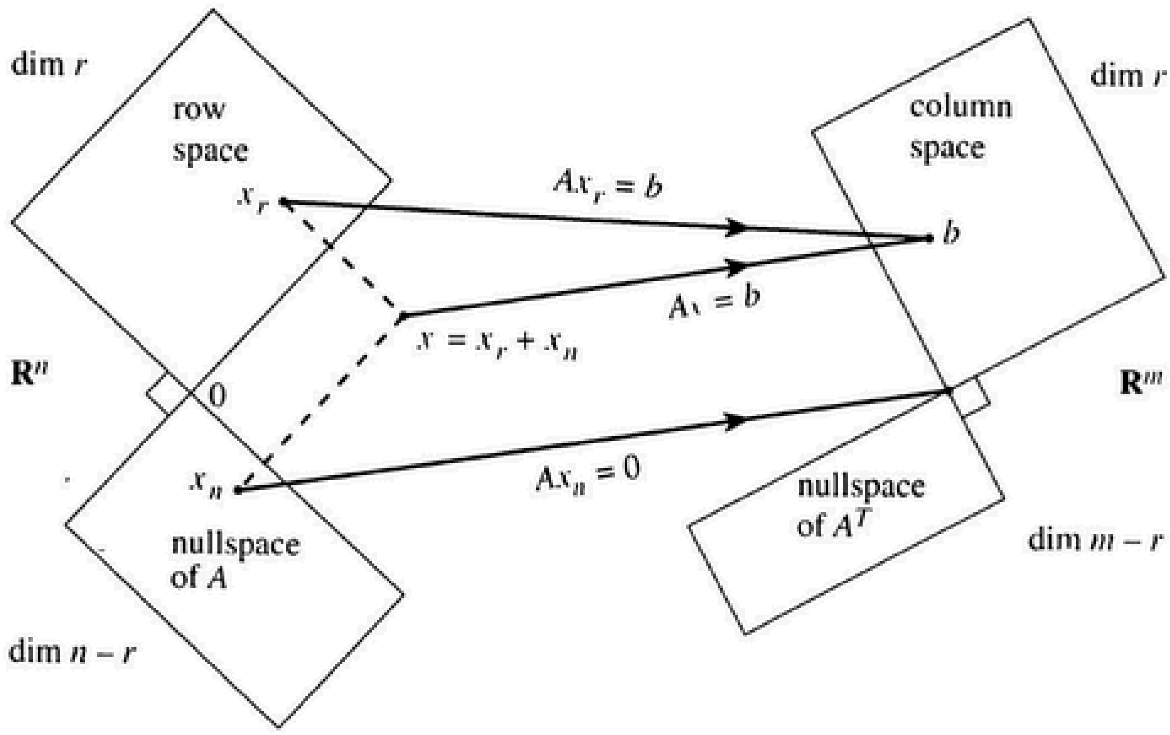


FIGURE 9.1. Matrix four subspaces and related dimensions (see, e.g., [**1**]. The row space $C(A^T)$ and the null space $N(A)$ are orthogonal, and the column space $C(A)$ and the left null space $N(A^T)$ are orthogonal.

Definition (Orthogonal complement). Let V be a subspace of $\mathbb{R}^n$. Then, the orthogonal complement of V is a set of all vectors that are perpendicular to V, which is denoted by $V^\perp$, i.e.,

$$V^\perp = \{u \in \mathbb{R}^n \mid \langle u, v\rangle = 0 \quad \forall v \in V\}.$$

Let $A \in \mathbb{R}^{m\times n}$, and let $C(A)$, $C(A^T)$, $N(A)$, and $N(A^T)$ show its column space, row space, null space, and left null space, respectively. Then,

(a) **The column space $C(A)$ is the orthogonal complement to the left null space $N(A^T)$.**
(b) **The row space $C(A^T)$ is the orthogonal complement to the null space $N(A)$.**

9.2. Orthogonal projections

In this section, we discuss the projection of a point onto a line and also onto a subspace.

Definition (Orthogonal projection onto a line). For $a, b \in \mathbb{R}^n$, the **orthogonal projection** of b onto the line through a (a line passes through the origin and a) is a point p on this line such that

$$\langle a, b - p\rangle = 0.$$

(a) p is point on the line through a, i.e., there exists $\alpha \in \mathbb{R}$ such that $p = \alpha a$;
(b) The error is given by $e = b - p = b - \alpha a$;
(c) The vectors a and e are orthogonal, i.e.,

$$\langle a, e\rangle = \langle a, b - \alpha a\rangle = \langle a, b\rangle - \alpha\langle a, a\rangle = 0 \quad \Rightarrow \quad \alpha = \tfrac{\langle a,b\rangle}{\langle a,a\rangle},$$

which consequently implies that the **orthogonal projection** of b onto the line through a is given by

$$p = \alpha a = \tfrac{\langle a,b\rangle}{\langle a,a\rangle} a.$$

d Therefore, the **projection matrix** P is given by

$$p = \alpha a = \tfrac{\langle a,b\rangle}{\langle a,a\rangle} a = \tfrac{aa^T}{\langle a,a\rangle} b \quad \Rightarrow \quad p = Pb = \tfrac{aa^T}{\langle a,a\rangle} b, \quad P = \tfrac{aa^T}{\langle a,a\rangle}.$$

Special cases are as follows:

- If $b = a$, then $p = \frac{\langle a,b\rangle}{\langle a,a\rangle} a = \frac{\langle a,a\rangle}{\langle a,a\rangle} a = a$;
- If b is perpendicular to a (i.e., $\langle a, b\rangle = 0$), then $p = \frac{\langle a,b\rangle}{\langle a,a\rangle} a = 0$.

Example 107. *Let us consider the vectors*

$$a = \begin{bmatrix}1\\2\\3\end{bmatrix}, \quad b = \begin{bmatrix}-3\\7\\1\end{bmatrix}.$$

We aim at computing the projection of a onto the line through b, which is given by

$$p = \tfrac{\langle a,b\rangle}{\langle a,a\rangle} a = \tfrac{14}{14}\begin{bmatrix}1\\2\\3\end{bmatrix} = \begin{bmatrix}1\\2\\3\end{bmatrix}, \quad p = Pb = \tfrac{aa^T}{\langle a,a\rangle} b = \begin{bmatrix}1/14 & 2/14 & 3/14\\2/14 & 4/14 & 6/14\\3/14 & 6/14 & 9/14\end{bmatrix}\begin{bmatrix}-3\\7\\1\end{bmatrix} = \begin{bmatrix}1\\2\\3\end{bmatrix}.$$

Invertibility of A^TA for full column rank A. Let $A \in \mathbb{R}^{m\times n}$. Then, we first show that $N(A^TA) = N(A)$. Indeed, if $x \in N(A)$ ($Ax = 0$), then $A^TAx = 0$, i.e., $N(A) \subseteq N(A^TA)$. If $x \in N(A^TA)$ ($A^TAx = 0$), then

$$\left\langle x, A^TAx\right\rangle = \langle Ax, Ax\rangle = \|Ax\|^2 = 0 \quad \Rightarrow \quad Ax = 0,$$

which implies that $x \in N(A)$ ($N(A^TA) \subseteq N(A)$), i.e., $N(A) = N(A^TA)$. Now, if A has full column rank ($N(A) = \{0\}$), then $N(A^TA) = \{0\}$ and therefore $\mathrm{rank}(A^TA) = n$, A^TA is invertible. Conversely, if A^TA is invertible ($N(A^TA) = \{0\}$), then $N(A) = \{0\}$ implying $\dim(C(A)) = n$. Therefore, A^TA **is invertible if and only if A has full column rank**.

Definition (Orthogonal projection onto a subspace). For $a_1, a_2, \ldots, a_n, b \in \mathbb{R}^m$ be some vectors in $\mathbb{R}^m$ such that $a_1, a_2, \ldots, a_n$ are linearly independent. Then, we can span a subspace S from these vectors, i.e., $S = \operatorname{span}\{a_1, a_2, \ldots, a_n\}$. The point p on S is called the **orthogonal projection** of b onto the the subspace S if it is a closest point on S to b.

(a) p is point on the subspace S, i.e., there exists $\alpha_1, \alpha_2, \ldots, \alpha_n \in \mathbb{R}$ such that

$$p = \alpha_1 a_1 + \alpha_2 a_2 + \ldots + \alpha_n a_n = Ax,$$

where

$$A = \begin{bmatrix} a_1 & a_2 & \ldots & a_n \end{bmatrix}, \quad x = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{bmatrix}.$$

(b) The error is given by $e = b - p = b - Ax$;

(c) The vectors e is perpendicular to the subspace S (nearest point is given when b is orthogonal to S), i.e.,

$$\begin{cases} \langle a_1, e\rangle = \langle a_1, b - Ax\rangle = 0 \\ \langle a_2, e\rangle = \langle a_2, b - Ax\rangle = 0 \\ \qquad \vdots \\ \langle a_n, e\rangle = \langle a_n, b - Ax\rangle = 0 \end{cases} \quad \Rightarrow \quad \begin{bmatrix} a_1^T \\ a_2^T \\ \vdots \\ a_n^T \end{bmatrix} \begin{bmatrix} b - Ax \\ b - Ax \\ \vdots \\ b - Ax \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad \Rightarrow \quad A^T(b - Ax) = 0.$$

which consequently leads to the linear system

$$A^T Ax = A^T b.$$

Since $a_1, a_2, \ldots, a_n$ are independent the matrix $A^T A$ is invertible and it is also symmetric, i.e.,

$$x = \left(A^T A\right)^{-1} A^T b,$$

i.e.,

$$p = Ax = A\left(A^T A\right)^{-1} A^T b.$$

d Therefore, the **projection matrix** P is given by

$$p = Ax = A\left(A^T A\right)^{-1} A^T b \quad \Rightarrow \quad P = A\left(A^T A\right)^{-1} A^T.$$

Important facts:

- $S = C(A)$;
- The error vector $b - Ax$ is perpendicular to the column space $C(A)$;
- Since $A^T(b - Ax) = 0$, the vector $b - Ax$ are in the left null space $N(A^T)$.

Example 108. *Let us consider the vectors*

$$a_1 = \begin{bmatrix} 0 \\ 2 \\ 0 \\ 0 \end{bmatrix}, \quad a_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 3 \end{bmatrix}, \quad a_3 = \begin{bmatrix} -1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad b = \begin{bmatrix} 4 \\ 1 \\ -3 \\ 5 \end{bmatrix}.$$

We aim at finding projection of b onto the subspace $S = \operatorname{span}\{\mathrm{a}_1, \mathrm{a}_2, \mathrm{a}_3\}$. Note that the vectors a_1, a_2, and a_3 are linearly independent, and therefore the matrix

$$A = \begin{bmatrix} 0 & 0 & -1 \\ 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 0 \end{bmatrix}$$

has full column rank, which implies that the matrix $A^T A$ is invertible and

$$A^T A = \begin{bmatrix} 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 3 \\ -1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & -1 \\ 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 0 \end{bmatrix} = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad (A^T A)^{-1} = \begin{bmatrix} 1/4 & 0 & 0 \\ 0 & 1/10 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

This implies that

$$
\begin{aligned}
p = A(A^TA)^{-1}A^Tb &= \begin{bmatrix} 0 & 0 & -1 \\ 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 0 \end{bmatrix} \begin{bmatrix} 1/4 & 0 & 0 \\ 0 & 1/10 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 3 \\ -1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \\ -3 \\ 5 \end{bmatrix} \\
&= \begin{bmatrix} 0 & 0 & -1 \\ 1/2 & 0 & 0 \\ 0 & 1/10 & 0 \\ 0 & 3/10 & 0 \end{bmatrix} \begin{bmatrix} 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 3 \\ -1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \\ -3 \\ 5 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1/10 & 3/10 \\ 0 & 0 & 3/10 & 9/10 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \\ -3 \\ 5 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \\ 6/5 \\ 18/5 \end{bmatrix},
\end{aligned}
$$

which is the projection of b onto the subspace S.

9.3. Least squares approximation

For $A \in \mathbb{R}^{m\times n}$, it often happens that the linear system $Ax = b$ is overdetermined ($m > n$), which usually has no solution. Since $\dim(C(A^T) = \dim(C(A) = \operatorname{rank}(A) \leq \min\{m, n\}$, if $m > n$ (too many equations), then $\dim(C(A) \leq n$. Then, the columns span a small part of m-dimensional space. Unless all measurements are perfect, b is outside the column space of A, which implies that we often have no solution because the elimination procedure reaches an impossible equation and stops. We can always get the error $e = b - Ax$ down to zero. If the error e is zero, then x is the exact solution, i.e., $Ax = b$. In cases that the length of e is as small as possible, x is a least squares solution. This consequently leads to the minimization problem

$$\min_x \ \tfrac{1}{2}\|Ax - b\|^2,$$

which is an unconstrained optimization problem. Let us define the function $f : \mathbb{R}^n \to \mathbb{R}$ given by

$$f(x) = \tfrac{1}{2}\|Ax - b\|^2 = \tfrac{1}{2}\langle Ax, Ax\rangle + \tfrac{1}{2}\|b\|^2 - \langle b, Ax\rangle = \tfrac{1}{2}\langle x, A^TAx\rangle + \tfrac{1}{2}\|b\|^2 - \langle b, Ax\rangle,$$

which is a quadratic function. The first-order optimality function for this problem implies that solutions of this minimization problem can be obtained by solving the gradient system

$$\nabla f(x) = 0.$$

Therefore, the first-order optimality condition for the above problem leads to

$$\nabla f(x) = A^TAx - A^Tb = 0, \quad \Rightarrow \quad A^TAx = A^Tb,$$

which we have seen it in projection of b onto the column space of A (i.e., $C(A)$).

For $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$ with $m < n$, we come to the underdetermined systems of equations $Ax = b$, which have infinitely many solutions. Therefore, we some times prefer to find those solutions that have a minimum norm $\|x\|$, which leads to the least squares problem with the Tikhonov regularization

$$\min_x \ \tfrac{1}{2}\|Ax - b\|^2 + \tfrac{\lambda}{2}\|x\|^2,$$

where $\lambda > 0$ is the regularization parameter. Moreover, if we aim to find the sparsest solution for this underdetermind system of equations, we may solve the ℓ_1-regularized least square problem

$$\min_x \ \tfrac{1}{2}\|Ax - b\|^2 + \lambda\|x\|_1,$$

where $\|x\|_1 = \sum_{i=1}^n |x_i|$ is the ℓ_1-norm. This problem is also called LASSO and has many applications in science and engineering.

9.4. Orthonormal bases and Gram-Schmidt process

In this section, we describe orthogonal bases for subspaces, show that the projection has a simple form in such cases, and describe the Gram-Schmidt process for producing an orthogonal basis.

Definition (Orthonormal vectors). Let $q_1, q_2, \ldots, q_n \in \mathbb{R}^m$. Then, these vectors are said to be **orthonormal** if

$$\langle q_i, q_j \rangle = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j. \end{cases}$$

Setting

$$Q = \begin{bmatrix} q_1 & q_2 & \ldots & q_n \end{bmatrix},$$

this is equivalent to

$$Q^T Q = \begin{bmatrix} q_1^T \\ q_2^T \\ \vdots \\ q_n^T \end{bmatrix} \begin{bmatrix} q_1 & q_2 & \ldots & q_n \end{bmatrix} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} = I_n.$$

We note that if $m = n$, then Q will be a square matrix and orthonormality is equivalent to invertibility, i.e.,

$$Q^T Q = I \quad \Leftrightarrow \quad Q^{-1} = Q^T.$$

Example 109. *for the fixed angle theta, let us consider the vectors*

$$q_1 = \begin{bmatrix} \cos(\theta) \\ \sin(\theta) \end{bmatrix}, \quad q_2 = \begin{bmatrix} -\sin(\theta) \\ \cos(\theta) \end{bmatrix}.$$

Then, we form the matrix Q by

$$Q = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix},$$

where

$$Q^T Q = \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2,$$

which clearly implies that q_1 and q_2 are orthonormal and therefore

$$Q^{-1} = Q^T = \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix},$$

which form an orthonormal basis for $\mathbb{R}^2$.

Let $q_1, q_2, \ldots, q_n \in \mathbb{R}^m$ be orthonormal vectors and $Q = \begin{bmatrix} q_1 & q_2 & \ldots & q_n \end{bmatrix}$. Then, we have

$$\|Qx\|^2 = \langle Qx, Qx \rangle = \left\langle x, Q^T Q x \right\rangle = \langle x, x \rangle = \|x\|^2,$$

which implies that **orthonormal matrices leave length unchanged**.

Projection using orthogonal bases. Let $q_1, q_2, \ldots, q_n \in \mathbb{R}^m$ be orthonormal vectors and $Q = \begin{bmatrix} q_1 & q_2 & \ldots & q_n \end{bmatrix}$. Then, the projection of b onto $S = \operatorname{span}\{q_1, q_2, \ldots, q_n\}$ is given by

$$p = Q(Q^T Q)^{-1} Q^T b = Q(I_n)^{-1} Q^T b = Q Q^T b, \quad P = Q Q^T.$$

Example 110. *Let us consider the vectors*

$$a_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad a_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad a_3 = \begin{bmatrix} -1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}.$$

We aim at finding projection of b *onto the subspace* $S = \mathrm{span}\{\mathrm{a}_1, \mathrm{a}_2, \mathrm{a}_3\}$. *Note that the vectors* a_1, a_2, *and* a_3 *are orthonormal bases for* S, *and therefore the matrix* $Q = \begin{bmatrix} a_1 & a_2 & a_3 \end{bmatrix}$ *satisfies* $Q^T Q = I_4$, *i.e.,*

$$p = QQ^T b = \begin{bmatrix} 0 & 0 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \\ -3 \\ 5 \end{bmatrix}$$
$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \\ -3 \\ 5 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \\ -3 \\ 0 \end{bmatrix},$$

which is the projection of b *onto the subspace* S.

Gram-Schmidt process. Let $a_1, a_2, \ldots, a_n \in \mathbb{R}^m$ be independent vectors. We aim at generating orthonormal bases from them. To this end, we first generate orthogonal vectors and then divide them by their length to make them orthonormal. We present this process as follows:

(a) Set $b_1 = a_1$;

(b) Set $b_2 = a_2 - \frac{\langle b_1, a_2 \rangle}{\langle b_1, b_1 \rangle} b_1$, where it is clear that

$$\langle b_1, b_2 \rangle = \langle b_1, a_1 \rangle - \langle b_1, a_1 \rangle = 0,$$

In the same way, we set

$$b_i = a_i - \frac{\langle b_1, a_i \rangle}{\langle b_1, b_1 \rangle} b_1 - \ldots - \frac{\langle b_{i-1}, a_i \rangle}{\langle b_{i-1}, b_{i-1} \rangle} b_{i-1} = a_i - \sum_{j=1}^{i-1} \frac{\langle b_j, a_i \rangle}{\langle b_j, b_j \rangle} b_j;$$

(c) Set $q_i = \frac{b_i}{\|b_i\|}$.

Example 111. *Let us consider vectors*

$$a_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \quad a_2 = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}, \quad a_3 = \begin{bmatrix} 3 \\ -3 \\ 3 \end{bmatrix}.$$

Then, by the Gram-Schmidt process, we get

$$b_1 = a_1, \quad b_2 = a_2 - \frac{\langle b_1, a_2 \rangle}{\langle b_1, b_1 \rangle} b_1 = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}, \quad b_3 = a_2 - \frac{\langle b_1, a_3 \rangle}{\langle b_1, b_1 \rangle} b_1 - \frac{\langle b_2, a_3 \rangle}{\langle b_2, b_2 \rangle} b_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

This implies that

$$q_1 = \frac{1}{\|b_1\|} b_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \quad q_2 = \frac{1}{\|b_2\|} b_2 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}, \quad q_3 = \frac{1}{\|b_3\|} b_3 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix},$$

that are orthonormal vectors.

CHAPTER 10

Symmetric and positive definite matrices

In this chapter, we describe symmetric matrices and present some of their properties. Then, we define positive definite and positive semidefinite matrices and their fundamental features.

Our main objectives are to:

- introduce symmetric matrices and some of their features;
- present positive definite and positive semidefinite matrices and some of their properties.

10.1. Symmetric matrices

In this section, we investigate fundamental properties of symmetric matrices A.

Definition (Symmetric matrices). Let A be an arbitrary $n \times n$ matrix. Then, A is said to be symmetric if $A = A^T$, where the set of all such matrices is denoted by $\mathbb{S}^{n\times n}$, i.e., $(A \in \mathbb{S}^{n\times n})$.

Example 112. *Let us consider the matrices*

$$A = \begin{bmatrix} 0 & 1 & 2 & 3 \\ 1 & 1 & 0 & 5 \\ 2 & 0 & -5 & -1 \\ 3 & 5 & -1 & 7 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -5 & 0 \\ 0 & 0 & 0 & 7 \end{bmatrix}, \quad C = \begin{bmatrix} -1 & 1 & 2 & 3 \\ 1 & 1 & 0 & 5 \\ 2 & 0 & -2 & -1 \\ -3 & 5 & -1 & 0 \end{bmatrix}.$$

It is clear that $A = A^T$, $B = B^T$, and $C \neq C^T$, which imply that A and B are symmetric and C is not symmetric.

Property 1 (Real eigenvalues). Let A be an arbitrary $n \times n$ symmetric matrix $(A \in \mathbb{S}^{n\times n})$. Then, all eigenvalues of A are real.

Example 113. *Let us verify the positive definiteness of the following matrices:*

$$A = \begin{bmatrix} 0 & 3 & 1 \\ 0 & -5 & 9 \\ 0 & 0 & 7 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 4 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 3 & 1 \\ 3 & 2 & 4 \\ 1 & 4 & 5 \end{bmatrix}.$$

Since A is an upper triangular matrix, the eigenvalues of A are $\lambda_1 = 0$, $\lambda_1 = -5$, and $\lambda_1 = 7$. The matrix B is a diagonal matrix with eigenvalues $\lambda_1 = -1$, $\lambda_1 = -3$, and $\lambda_1 = 4$. Moreover, the matrix C is symmetric and its eigenvalues are given by solving the characteristic equation

$$\det(C - \lambda I_3) = \det\left(\begin{bmatrix} 1-\lambda & 0 & 1 \\ 0 & 2-\lambda & 0 \\ 1 & 0 & 5-\lambda \end{bmatrix}\right)$$
$$= (1-\lambda)(2-\lambda)(5-\lambda) - (2-\lambda) = (2-\lambda)[(1-\lambda)(5-\lambda) - 1] = 0,$$

which leads to

$$2 - \lambda = 0, \quad \lambda^2 - 6\lambda + 4 = 0,$$

i.e.,

$$\lambda_1 = 2, \quad \lambda_2 = \tfrac{6+\sqrt{20}}{2}, \quad \lambda_2 = \tfrac{6-\sqrt{20}}{2}.$$

Since A, B, and C are symmetric matrices, all of their eigenvalues are real, as verified in the above discussion.

Property 2 (Orthogonal eigenvectors). Let A be an arbitrary $n \times n$ symmetric matrix ($A \in \mathbb{S}^{n\times n}$). Then, the eigenvectors of A corresponding to different eigenvalues are perpendicular.

Example 114. *We consider the matrix*

$$A = \begin{bmatrix} 1 & \frac{\sqrt{21}}{2} \\ \frac{\sqrt{21}}{2} & 3 \end{bmatrix}.$$

Then,

$$\det(A - \lambda I_2) = \det\left(\begin{bmatrix} 1-\lambda & \frac{\sqrt{21}}{2} \\ \frac{\sqrt{21}}{2} & 3-\lambda \end{bmatrix}\right) = (1-\lambda)(3-\lambda) - \tfrac{21}{4} = \lambda^2 - 4\lambda - \tfrac{9}{4} = 0,$$

implying

$$\lambda_1 = \tfrac{4-5}{2} = -\tfrac{1}{2}, \quad \lambda_2 = \tfrac{4+5}{2} = \tfrac{9}{2}$$

that both of them are real. Hence, for λ_1, we have

$$(A - \lambda_1 I_2)x = \begin{bmatrix} \frac{3}{2} & \frac{\sqrt{21}}{2} \\ \frac{\sqrt{21}}{2} & \frac{7}{2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

By Gauss-Jordan elimination, we get

$$x_1 = -\frac{\sqrt{21}}{3} x_2.$$

For λ_1, we get

$$(A - \lambda_2 I_2)x = \begin{bmatrix} -\frac{7}{2} & \frac{\sqrt{21}}{2} \\ \frac{\sqrt{21}}{2} & -\frac{3}{2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

By Gauss-Jordan elimination, we get

$$x_1 = \frac{\sqrt{21}}{7} x_2.$$

Setting $x_2 = 1$, it is clear that

$$v_1 = \begin{bmatrix} -\frac{\sqrt{21}}{3} \\ 1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} \frac{\sqrt{21}}{7} \\ 1 \end{bmatrix}$$

are eigenvalues of A corresponding to $\lambda_1 = -\frac{1}{2}$ and $\lambda_2 = \frac{9}{2}$. Then, the inner product

$$\langle v_1, v_2 \rangle = -\frac{\sqrt{21}}{3} * \frac{\sqrt{21}}{7} + 1 * 1 = 1 - 1 = 0,$$

which implies that v_1 and v_2 are prependicular.

Property 3 (Spectral decomposition). Let A be an arbitrary $n \times n$ symmetric matrix ($A \in \mathbb{S}^{n\times n}$). Then, A has the factorization

$$A = Q\Lambda Q^T, \tag{10.1}$$

where

$$Q = \begin{bmatrix} q_1 & q_2 & \cdots & q_n \end{bmatrix}, \quad \Lambda = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix} \tag{10.2}$$

such that

$$q_i \in \mathbb{R}^n, \quad \langle q_i, q_j \rangle = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

Example 115. *We consider the matrix*

$$A = \begin{bmatrix} 1 & \frac{\sqrt{21}}{2} \\ \frac{\sqrt{21}}{2} & 3 \end{bmatrix}.$$

From Example 114, we obtain

$$v_1 = \begin{bmatrix} -\frac{\sqrt{21}}{3} \\ 1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} \frac{\sqrt{21}}{7} \\ 1 \end{bmatrix} \Rightarrow q_1 = \begin{bmatrix} -\frac{\sqrt{7}}{\sqrt{10}} \\ \frac{\sqrt{3}}{\sqrt{10}} \end{bmatrix}, \quad q_2 = \begin{bmatrix} \frac{\sqrt{3}}{\sqrt{10}} \\ \frac{\sqrt{7}}{\sqrt{10}} \end{bmatrix}$$

Hence, we set

$$\Lambda = \begin{bmatrix} -\frac{1}{2} & 0 \\ 0 & \frac{9}{2} \end{bmatrix}, \quad Q = \begin{bmatrix} -\frac{\sqrt{7}}{\sqrt{10}} & \frac{\sqrt{3}}{\sqrt{10}} \\ \frac{\sqrt{3}}{\sqrt{10}} & \frac{\sqrt{7}}{\sqrt{10}} \end{bmatrix},$$

which leads to

$$\begin{aligned} Q\Lambda Q^T &= \begin{bmatrix} -\frac{\sqrt{7}}{\sqrt{10}} & \frac{\sqrt{3}}{\sqrt{10}} \\ \frac{\sqrt{3}}{\sqrt{10}} & \frac{\sqrt{7}}{\sqrt{10}} \end{bmatrix} \begin{bmatrix} -\frac{1}{2} & 0 \\ 0 & \frac{9}{2} \end{bmatrix} \begin{bmatrix} -\frac{\sqrt{7}}{\sqrt{10}} & \frac{\sqrt{3}}{\sqrt{10}} \\ \frac{\sqrt{3}}{\sqrt{10}} & \frac{\sqrt{7}}{\sqrt{10}} \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{7}}{2\sqrt{10}} & 9\frac{3}{2\sqrt{10}} \\ -\frac{\sqrt{3}}{2\sqrt{10}} & \frac{9\sqrt{7}}{2\sqrt{10}} \end{bmatrix} \begin{bmatrix} -\frac{\sqrt{7}}{\sqrt{10}} & \frac{\sqrt{3}}{\sqrt{10}} \\ \frac{\sqrt{3}}{\sqrt{10}} & \frac{\sqrt{7}}{\sqrt{10}} \end{bmatrix} \\ &= \begin{bmatrix} 1 & \frac{\sqrt{21}}{2} \\ \frac{\sqrt{21}}{2} & 3 \end{bmatrix} = A, \end{aligned}$$

which implies that $Q\Lambda Q^T$ is a spectral factorization for A.

Property 4 (Diagonalizability). Let A be an arbitrary $n \times n$ symmetric matrix ($A \in \mathbb{S}^{n\times n}$). If Λ and Q be eigenvalue and eigenvector matrices of A given in (10.2), then A is diagonalizable, i.e.,

$$A = Q\Lambda Q^T, \quad Q^T Q = I_n, \quad Q^{-1} = Q^T.$$

Example 116. *We consider the matrix*

$$A = \begin{bmatrix} 1 & \frac{\sqrt{21}}{2} \\ \frac{\sqrt{21}}{2} & 3 \end{bmatrix}.$$

From Example 115, we obtain

$$\Lambda = \begin{bmatrix} -\frac{1}{2} & 0 \\ 0 & \frac{9}{2} \end{bmatrix}, \quad Q = \begin{bmatrix} -\frac{\sqrt{7}}{\sqrt{10}} & \frac{\sqrt{3}}{\sqrt{10}} \\ \frac{\sqrt{3}}{\sqrt{10}} & \frac{\sqrt{7}}{\sqrt{10}} \end{bmatrix},$$

Hence,

$$Q^T Q = \begin{bmatrix} -\frac{\sqrt{7}}{\sqrt{10}} & \frac{\sqrt{3}}{\sqrt{10}} \\ \frac{\sqrt{3}}{\sqrt{10}} & \frac{\sqrt{7}}{\sqrt{10}} \end{bmatrix} \begin{bmatrix} -\frac{\sqrt{7}}{\sqrt{10}} & \frac{\sqrt{3}}{\sqrt{10}} \\ \frac{\sqrt{3}}{\sqrt{10}} & \frac{\sqrt{7}}{\sqrt{10}} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2,$$

which implies $Q^T = Q^{-1}$, i.e., $A = Q\Lambda Q^T = Q\Lambda Q^{-1}$ that shows the diagonalizability of A.

10.2. Positive definite matrices

This section focuses on symmetric matrices that have positive eigenvalues, which appears in many real-world applications. This class of matrices is called positive definite, and the first problem is to recognize them. One way is to compute all of its eigenvalues and check if all of them are positive; however, this is very costly, and we want to avoid it. Here, we investigate fast approaches that tell us if a matrix A is positive definite.

Definition (Positive definite matrix). An $n \times n$ symmetric matrix ($A \in \mathbb{S}^{n\times n}$) is called positive definite if all of its eigenvalues are positive.

Example 117. *Let us verify the positive definiteness of the following matrices:*

$$A_1 = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 7 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 7 \end{bmatrix}, \quad A_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}, \quad A_4 = \begin{bmatrix} 8 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 9 \end{bmatrix},$$

and

$$B_1 = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -5 & 0 \\ 0 & 0 & 7 \end{bmatrix}, \quad B_2 = \begin{bmatrix} -7 & 0 & 0 \\ 0 & -7 & 0 \\ 0 & 0 & -7 \end{bmatrix}, \quad B_3 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -4 \end{bmatrix}, \quad B_4 = \begin{bmatrix} 8 & 0 & 0 \\ 0 & -4 & 0 \\ 0 & 0 & 9 \end{bmatrix}.$$

For matrices $A_1 - A_4$, it is evident that

- A_1 *a diagonal matrix with eigenvalues* $\lambda_1 = 2$, $\lambda_2 = 5$, *and* $\lambda_3 = 7$, *i.e.,* A_1 *is positive definite.*
- A_2 *a scalar matrix with eigenvalues* $\lambda_1 = 7$, $\lambda_2 = 7$, *and* $\lambda_3 = 7$, *i.e.,* A_2 *is positive definite.*
- A_3 *a diagonal matrix with eigenvalues* $\lambda_1 = 1$, $\lambda_2 = 3$, *and* $\lambda_3 = 4$, *i.e.,* A_3 *is positive definite.*
- A_4 *a diagonal matrix with eigenvalues* $\lambda_1 = 8$, $\lambda_2 = 4$, *and* $\lambda_3 = 9$, *i.e.,* A_4 *is positive definite.*

On the other hand, for matrices $B_1 - B_4$, it is clear that

- B_1 *a diagonal matrix with eigenvalues* $\lambda_1 = 2$, $\lambda_2 = -5$, *and* $\lambda_3 = 7$, *i.e.,* B_1 *is not positive definite.*
- B_2 *a scalar matrix with eigenvalues* $\lambda_1 = -7$, $\lambda_2 = -7$, *and* $\lambda_3 = -7$, *i.e.,* B_2 *is not positive definite.*
- B_3 *a diagonal matrix with eigenvalues* $\lambda_1 = -1$, $\lambda_2 = 3$, *and* $\lambda_3 = -4$, *i.e.,* B_3 *is not positive definite.*
- B_4 *a diagonal matrix with eigenvalues* $\lambda_1 = 8$, $\lambda_2 = -4$, *and* $\lambda_3 = 9$, *i.e.,* B_4 *is not positive definite.*

Example 118. *Let us consider the symmetric 2×2 matrix*

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{bmatrix},$$

and let us calculate its eigenvalues. Then, writing its characteristic equation leads to

$$\begin{aligned} \det(A - \lambda I_2) = \det\left(\begin{bmatrix} a_{11} - \lambda & a_{12} \\ a_{12} & a_{22} - \lambda \end{bmatrix}\right) &= (a_{11} - \lambda)(a_{22} - \lambda) - a_{12}^2 \\ &= \lambda^2 - (a_{11} + a_{22})\lambda + a_{11}a_{22} - a_{12}^2. \end{aligned}$$

Thus, this one-dimensional quadratic equation has solution if the discriminant Δ of this quadratic equation is nonnegative, i.e.,

$$\Delta = (a_{11} + a_{22})^2 - 4(a_{11}a_{22} - a_{12}^2) \geq 0.$$

If $\Delta = 0$, then it has one solution, i.e.,

$$\lambda = \tfrac{a_{11}+a_{22}}{2},$$

which is positive if $a_{11} + a_{22} > 0$. Note that $\Delta = 0$ implies

$$a_{11}a_{22} - a_{12}^2 = \tfrac{1}{4}(a_{11} + a_{22})^2 > 0.$$

If $\Delta > 0$, then it has two solutions that are given by

$$\lambda_1 = \tfrac{a_{11}+a_{22}+\sqrt{\Delta}}{2}, \quad \lambda_2 = \tfrac{a_{11}+a_{22}-\sqrt{\Delta}}{2}.$$

If $\lambda_1, \lambda_2 > 0$, we add λ_1 to λ_2 leading to $a_{11} + a_{22} > 0$. This guarantees λ_1 to be positive. In order to guarantee the positivity of λ_2, it is required that $a_{11} + a_{22} > \sqrt{\Delta}$, i.e.,

$$a_{11}a_{22} - a_{12}^2 > 0,$$

implying $a_{11}a_{22} > a_{12}^2$. Overall, in both cases, one needs the inequalities $a_{11} > 0$ and $a_{11}a_{22} > a_{12}^2$ to guarantee the positivity of λ_1 and λ_2.

Example 119. *Let us consider the symmetric 2×2 matrices*

$$A = \begin{bmatrix} 4 & -3 \\ -3 & 7 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 8 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 5 \\ 5 & 2 \end{bmatrix}.$$

For the matrix A, it is clear that $4+7 > 0$ and $4*7-(-3)^2 > 0$, which ensures that A is positive definite.

For the matrix B, it is clear that $1+8 > 0$ and $1*8-0^2 > 0$, which ensures that B is positive definite. In this case, we can directly see that the matrix B has the eigenvalues $\lambda_1 = 1 > 0$ and $\lambda_1 = 8 > 0$, i.e., B is positive definite.

For the matrix C, it is clear that $1 + 2 > 0$ but $1 * 2 - 5^2 < 0$, which implies that C is not positive definite.

Definition (Positive definite matrix (energy-based definition)). An $n \times n$ matrix ($A \in \mathbb{R}^{n\times n}$) is called positive definite if $\langle x, Ax\rangle > 0$ all nonzero vector $x \in \mathbb{R}^n$.

Example 120. *Let us consider the symmetric* 2×2 *matrix*

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{bmatrix},$$

and let us verify a condition guarantees the positive definiteness by the energy-based definition, i.e.,

$$\begin{aligned}\langle x, Ax\rangle &= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} a_{11}x_1 + a_{12}x_2 \\ a_{12}x_1 + a_{22}x_2 \end{bmatrix} \\ &= (a_{11}x_1 + a_{12}x_2)x_1 + (a_{12}x_1 + a_{22}x_2)x_2 \\ &= a_{11}x_1^2 + a_{22}x_2^2 + 2a_{12}x_1x_2 > 0,\end{aligned}$$

which is our desired inequality.

Positive definiteness of A^TA. Let A be an $m \times n$ matrix ($A \in \mathbb{R}^{m\times n}$) that its columns are independent (i.e., $Ax \neq 0$ if $x \neq 0$). Then, the matrix A^TA is symmetric and

$$\left\langle x, (A^TA)x \right\rangle = \langle Ax, Ax\rangle = \|Ax\|^2 > 0 \quad \forall\ 0 \neq x \in \mathbb{R}^n,$$

i.e., A^TA is positive definite.

Example 121. *Let us consider the* 4×3 *matrix*

$$A = \begin{bmatrix} 2 & 0 & 0 \\ -2 & 2 & 0 \\ 0 & -2 & 2 \\ 0 & 0 & -2 \end{bmatrix}$$

We first show the columns of A are independent, i.e.,

$$\alpha_1 c_1 + \alpha_2 c_2 + \alpha_3 c_3 = \alpha_1 \begin{bmatrix} 2 \\ -2 \\ 0 \\ 0 \end{bmatrix} + \alpha_2 \begin{bmatrix} 0 \\ 2 \\ -2 \\ 0 \end{bmatrix} + \alpha_2 \begin{bmatrix} 0 \\ 0 \\ 2 \\ -2 \end{bmatrix} = \begin{cases} 2\alpha_1 = 0, \\ -2\alpha_1 + 2\alpha_2 = 0, \\ -2\alpha_2 + 2\alpha_3 = 0, \\ -2\alpha_3 = 0, \end{cases}$$

which clearly implies $\alpha_1 = \alpha_2 = \alpha_3 = 0$, i.e., columns of A are independent. Let us consider the matrix B as

$$B = A^TA = \begin{bmatrix} 2 & -2 & 0 & 0 \\ 0 & 2 & -2 & 0 \\ 0 & 0 & 2 & -2 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ -2 & 2 & 0 \\ 0 & -2 & 2 \\ 0 & 0 & -2 \end{bmatrix} = \begin{bmatrix} 8 & -4 & 0 \\ -4 & 8 & -4 \\ 0 & -4 & 8 \end{bmatrix}.$$

Since columns of A are independent, B is a positive definite matrix.

Sum of positive definite matrices. Let A and B be two $n \times n$ matrix ($A, B \in \mathbb{R}^{n\times n}$) that are positive definite. Then, it follows from the energy-based definition that

$$\langle x, (A + B)x\rangle = \langle x, Ax\rangle + \langle x, Bx\rangle > 0 \quad \forall\ 0 \neq x \in \mathbb{R}^n,$$

i.e., $A + B$ is positive definite.

Example 122. *Let us show that the matrix*

$$C = \begin{bmatrix} 13 & -4 & 0 \\ -4 & 11 & -4 \\ 0 & -4 & 9 \end{bmatrix}.$$

is positive definite. To do so, we consider the symmetric 3×3 *matrices*

$$A = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 8 & -4 & 0 \\ -4 & 8 & -4 \\ 0 & -4 & 8 \end{bmatrix}.$$

The matrix A is diagonal with positive elements; therefore, they are eigenvalues of A, which ensures that A is positive definite. On the other hand, in Example 121, we have shown that B is positive definite as well. Therefore, their summation

$$A + B = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 8 & -4 & 0 \\ -4 & 8 & -4 \\ 0 & -4 & 8 \end{bmatrix} = \begin{bmatrix} 13 & -4 & 0 \\ -4 & 11 & -4 \\ 0 & -4 & 9 \end{bmatrix} = C.$$

Since C is sum of two positive definite matrices, it is positive definite as well.

Positive definiteness of strictly diagonally dominant matrices with positive diagonal. Let A be an $n \times n$ symmetric matrix. If the matrix A is **strictly diagonally dominant** with positive diagonal elements, i.e.,

$$a_{ii} \geq \sum_{j \neq i} |a_{ij}|, \quad a_{ii} > 0, \quad \forall i = 1, 2, \ldots, n, \tag{10.1}$$

then A is positive definite.

Example 123. *Let us consider the matrix*

$$C = \begin{bmatrix} 13 & -4 & 0 \\ -4 & 11 & -4 \\ 0 & -4 & 9 \end{bmatrix}.$$

Since A is strictly diagonally dominant, it is positive definite.

10.3. Positive semidefinite matrices

Definition (Positive semidefinite matrix). An $n \times n$ symmetric matrix ($A \in \mathbb{S}^{n \times n}$) is called *positive semidefinite* if 0 is its smallest eigenvalue.

Example 124. *Let us verify the positive definiteness of the following matrices:*

$$A_1 = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 4 \end{bmatrix}, \quad A_3 = \begin{bmatrix} 8 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

and

$$B_1 = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -5 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad B_2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -4 \end{bmatrix}, \quad B_3 = \begin{bmatrix} 8 & 0 & 0 \\ 0 & -4 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

For matrices $A_1 - A_3$, it is evident that

- *A_1 a diagonal matrix with eigenvalues $\lambda_1 = 2$, $\lambda_2 = 5$, and $\lambda_3 = 0$, i.e., A_1 is positive semidefinite.*
- *A_3 a digonal matrix with eigenvalues $\lambda_1 = 0$, $\lambda_2 = 0$, and $\lambda_3 = 4$, i.e., A_3 is positive semidefinite.*
- *A_4 a diagonal matrix with eigenvalues $\lambda_1 = 8$, $\lambda_2 = 4$, and $\lambda_3 = 0$, i.e., A_4 is positive semidefinite.*

On the other hand, for matrices $B_1 - B_3$, it is clear that

- *B_1 a diagonal matrix with eigenvalues $\lambda_1 = 2$, $\lambda_2 = -5$, and $\lambda_3 = 0$, i.e., B_1 is not positive semidefinite.*
- *B_3 a diagonal matrix with eigenvalues $\lambda_1 = 0$, $\lambda_2 = 0$, and $\lambda_3 = -4$, i.e., B_3 is not positive semidefinite.*
- *B_4 a diagonal matrix with eigenvalues $\lambda_1 = 8$, $\lambda_2 = -4$, and $\lambda_3 = 0$, i.e., B_4 is not positive semidefinite.*

Definition (Positive semidefinite matrix (energy-based definition)). An $n \times n$ matrix ($A \in \mathbb{S}^{n\times n}$) is called positive *semidefinite* if $\langle x, Ax\rangle \geq 0$ all nonzero vector $x \in \mathbb{R}^n$.

Example 125. *Let us consider the symmetric* 2×2 *matrix*

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{bmatrix}.$$

To guarantee the positive semidefinitness of A*, we need the inequality*

$$\langle x, Ax\rangle = a_{11}x_1^2 + a_{22}x_2^2 + 2a_{12}x_1x_2 \geq 0$$

to be satisfied for all $0 \neq x \in \mathbb{R}^2$*.*

Positive semidefiniteness of A^TA. Let A be an $m \times n$ matrix ($A \in \mathbb{R}^{m\times n}$) that its columns are dependent. Then, the matrix A^TA is symmetric and

$$\left\langle x, (A^TA)x\right\rangle = \langle Ax, Ax\rangle = \|Ax\|^2 \geq 0 \quad \forall\ 0 \neq x \in \mathbb{R}^n,$$

i.e., A^TA is positive semidefinite.

Example 126. *Let us consider the* 4×3 *matrix*

$$A = \begin{bmatrix} 2 & 1 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

We first show the columns of A *are independent, i.e.,*

$$\alpha_1 c_1 + \alpha_2 c_2 + \alpha_3 c_3 = \alpha_1 \begin{bmatrix} 2 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \alpha_2 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \alpha_2 \begin{bmatrix} 3 \\ 0 \\ 0 \\ 0 \end{bmatrix} = 2\alpha_1 + \alpha_2 + 3\alpha_3 = 0,$$

which clearly implies $\alpha_1 = -1/2(\alpha_2 + 3\alpha_3)$*, i.e., columns of* A *are dependent. Let us now consider the matrix* B *given by*

$$B = A^TA = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 2 & 1 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 4 & 2 & 6 \\ 2 & 1 & 3 \\ 6 & 3 & 9 \end{bmatrix}.$$

Since columns of A *are dependent,* B *is a positive semidefinite matrix.*

Positive semidefiniteness of diagonally dominant matrices with nonnegative diagonal. Let A be an $n \times n$ symmetric matrix. If the matrix A is **diagonally dominant** with nonnegative diagonal elements, i.e.,

$$a_{ii} \geq \sum_{j\neq i} |a_{ij}|, \quad a_{ii} \geq 0, \quad \forall i = 1, 2, \ldots, n, \tag{10.1}$$

then A is positive semidefinite.

Example 127. *Let us consider the symmetric matrix*

$$A = \begin{bmatrix} 10 & -1 & 6 \\ -1 & 11 & -6 \\ 6 & -6 & 12 \end{bmatrix}.$$

Since A *is diagonally dominant, it is positive semidefinite. Moreover, since it is not strictly diagonally dominant, it is not positive definite.*

CHAPTER 11

Singular Value Decomposition

In the previous chapter, we discussed *spectral decomposition* of *square* matrices, which factorizes the matrix A as $A = Q\Lambda Q^{-1}$, for the eigenvalue matrix Λ and eigenbasis matrix Q. The spectral decomposition has several issues:

(i) the eigenvectors of A are usually not orthogonal;
(ii) there are not always enough eigenvectors;
(iii) the basic equation $Ax = \lambda x$ requires A to be a square matrix.

This chapter generalizes such decomposition to any $m \times n$ *rectangular* matrix with **singular value decomposition**, which is the highlight of linear algebra. This decomposition technique leads to computing the right bases for the four subspaces of a matrix A

The main objectives of this chapter are as follows:

- What are singular values and left and right singular vectors?
- How to decompose a rectangular matrix singular value decomposition?
- How to compute the right bases for the four subspaces of a matrix A?

11.1. Singular values and singular vectors

Let A be an arbitrary $m \times n$ matrix ($A \in \mathbb{R}^{m\times n}$). We already know that A^TA and AA^T are $n \times n$ and $m \times m$ symmetric matrices that are both orthononally diagonalizable. Let $\{v_1, v_2, \ldots, v_n\}$ be orthonormal eigenvectors of (A^TA) corresponding to eigenvectors $\lambda_1, \lambda_2, \ldots, \lambda_n$. Hence,

$$\lambda_i = \lambda_i \|v_i\|^2 = \langle v_i, \lambda_i v_i \rangle = \langle v_i, A^TAv_i \rangle = \langle Av_i, Av_i \rangle = \|Av_i\|^2 \geq 0,$$

which means that the eigenvalues of A^TA are nonnegative and can be reordered such that

$$\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_n \geq 0,$$

which we use to the following definition. We also note that the positive eigenvalues of A^TA and AA^T are the same.

Definition (Singular values). Let A be an arbitrary $m \times n$ matrix ($A \in \mathbb{R}^{m\times n}$), and let $\lambda_1, \lambda_2, \ldots, \lambda_n$ be nonnegative eigenvalues of A^TA. Then, the scalars

$$\sigma_i = \sqrt{\lambda_i} \quad i = 1, 2, \ldots, n, \tag{11.1}$$

are called **singular values** of A.

Example 128. *We consider the matrix*

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 2 & 0 & 0 \\ 0 & 0 & 7 \end{bmatrix}, \quad B = \begin{bmatrix} 7 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 6 & 0 \end{bmatrix}$$

Then, for the matrix A, we have

$$A^TA = \begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & 0 & 7 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 2 & 0 & 0 \\ 0 & 0 & 7 \end{bmatrix} = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 25 & 0 \\ 0 & 0 & 49 \end{bmatrix},$$

with eigenvalues $\lambda_1 = 5$, $\lambda_2 = 25$, and $\lambda_3 = 49$ of A^TA. This consequently leads to singular values

$$\sigma_1 = \sqrt{\lambda_1} = \sqrt{5}, \quad \sigma_2 = \sqrt{\lambda_2} = 5, \quad \sigma_3 = \sqrt{\lambda_3} = 7.$$

For the matrix B, we get

$$B^TB = \begin{bmatrix} 7 & 1 & 0 & 0 \\ 0 & 0 & 1 & 6 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 7 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 6 & 0 \end{bmatrix} = \begin{bmatrix} 50 & 0 & 0 \\ 0 & 37 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

with eigenvalues $\lambda_1 = 50$, $\lambda_2 = 37$, and $\lambda_3 = 0$ of B^TB. This consequently leads to singular values

$$\sigma_1 = \sqrt{\lambda_1} = \sqrt{50}, \quad \sigma_2 = \sqrt{\lambda_2} = \sqrt{37}, \quad \sigma_3 = \sqrt{\lambda_3} = 0,$$

which are nonnegative.

Definition (left and right singular vectors). Let A be an arbitrary $m \times n$ matrix ($A \in \mathbb{R}^{m\times n}$), and let $\sigma_1, \sigma_2, \ldots, \sigma_r > 0$ be singular values of A for $r \leq \min\{m, n\}$. Then,

- the eigenvectors u_i ($i = 1, 2, \ldots, m$) of AA^T are called **left singular vectors** of A;
- the eigenvalues v_j ($j = 1, 2, \ldots, n$) of A^TA are called **right singular vectors** of A.

Since AA^T is symmetric, u_i ($i = 1, 2, \ldots, m$) are orthogonal, and similarly v_j ($j = 1, 2, \ldots, n$) are orthogonal too. Hence, we can normalize them ($u_i = \frac{u_i}{\|u_i\|}$ ($i = 1, 2, \ldots, m$) and $v_i = \frac{v_i}{\|v_i\|}$ ($j = 1, 2, \ldots, n$)) to produce the orthonormal eigenvectors. Therefore, in the above definition, we can deduce that

(a) from $A(A^Tu_i) = AA^Tu_i = \sigma_i^2 u_i$ and $\sigma_i > 0$ ($i = 1, 2, \ldots, r$), we obtain that $u_i \in C(A)$. Together with the independence of $\{u_1, u_2, \ldots, u_r\}$, this implies that *the set of vectors $\{u_1, u_2, \ldots, u_r\}$ is an orthonormal basis for the column space*;

(b) by $AA^Tu_i = \sigma_i^2 u_i = 0$ for $\sigma_i = 0$ ($i = r+1, r+2, \ldots, m$), we get that

$$\langle u_i, AA^Tu_i \rangle = \|A^Tu_i\| = 0 \quad \Rightarrow \quad A^Tu_i = 0,$$

i.e., $u_i \in N(A^T)$. Together with the independence of $\{u_{r+1}, u_{r+2}, \ldots, u_m\}$, this implies that *the set of vectors $\{u_{r+1}, u_{r+2}, \ldots, u_m\}$ is an orthonormal basis for the left nullspace $\mathcal{N}(A^T)$*;

(c) from $A^T(Av_i) = A^TAv_i = \sigma_i^2 v_i$ and $\sigma_i > 0$ ($i = 1, 2, \ldots, r$), we obtain that $v_i \in C(A^T)$. Together with the independence of $\{v_1, v_2, \ldots, v_r\}$, this implies that *the set of vectors $\{v_1, v_2, \ldots, v_r\}$ is an orthonormal basis for the row space*;

(d) by $A^TAv_i = \sigma_i^2 v_i = 0$ for $\sigma_i = 0$ ($i = r+1, r+2, \ldots, n$), we get that

$$\langle v_i, A^TAv_i \rangle = \|Av_i\| = 0 \quad \Rightarrow \quad Av_i = 0,$$

i.e., $v_i \in N(A)$. Together with the independence of $\{v_{r+1}, v_{r+2}, \ldots, v_n\}$, this implies that *the set of vectors $\{v_{r+1}, v_{r+2}, \ldots, v_n\}$ is an orthonormal basis for the nullspace $\mathcal{N}(A)$.*

Example 129. *We consider the matrix*

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 2 & 0 & 0 \\ 0 & 0 & 7 \end{bmatrix}.$$

Then,

$$AA^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 2 & 0 & 0 \\ 0 & 0 & 7 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & 0 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 25 & 0 & 0 \\ 2 & 0 & 4 & 0 \\ 0 & 0 & 0 & 49 \end{bmatrix},$$

i.e.,

$$\det(AA^T - \lambda I_4) = \det\left(\begin{bmatrix} 1-\lambda & 0 & 2 & 0 \\ 0 & 25-\lambda & 0 & 0 \\ 2 & 0 & 4-\lambda & 0 \\ 0 & 0 & 0 & 49-\lambda \end{bmatrix}\right) = (49-\lambda)(25-\lambda)[(4-\lambda)(1-\lambda) - 4] = 0,$$

leading to the eigenvalues $\lambda_1 = 5$, $\lambda_2 = 25$, $\lambda_3 = 49$, *and* $\lambda_4 = 0$. *For* $\lambda_1 = 5$, *we have*

$$(A^T A - \lambda_1 I_4)x = \begin{bmatrix} -4 & 0 & 2 & 0 \\ 0 & 20 & 0 & 0 \\ 2 & 0 & -1 & 0 \\ 0 & 0 & 0 & 44 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix},$$

implies $x_1 = 1/2x_3$ *and* $x_2 = x_4 = 0$, *i.e.,* $\bar{u}_1 = \begin{bmatrix} 1/2 & 0 & 1 & 0 \end{bmatrix}^T$. *For* $\lambda_2 = 25$, *we have*

$$(A^T A - \lambda_2 I_4)x = \begin{bmatrix} -24 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 2 & 0 & -21 & 0 \\ 0 & 0 & 0 & 24 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix},$$

implies $x_1 = 1/12x_3$, $x_1 = 21/2x_3$ *and* $x_4 = 0$, *i.e.,* $\bar{u}_2 = \begin{bmatrix} 0 & 1 & 0 & 0 \end{bmatrix}^T$. *For* $\lambda_3 = 49$, *we have*

$$(A^T A - \lambda_3 I_4)x = \begin{bmatrix} -48 & 0 & 2 & 0 \\ 0 & -24 & 0 & 0 \\ 2 & 0 & -45 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix},$$

implies $x_1 = 1/24x_3$, $x_1 = 45/2x_3$ *and* $x_2 = 0$, *i.e.,* $\bar{u}_3 = \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix}^T$. *For* $\lambda_4 = 0$, *we have*

$$(A^T A - \lambda_3 I_4)x = \begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 25 & 0 & 0 \\ 2 & 0 & 4 & 0 \\ 0 & 0 & 0 & 49 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix},$$

implies $x_1 = 2x_3$ *and* $x_2 = x_4 = 0$, *i.e.,* $\bar{u}_4 = \begin{bmatrix} 2 & 0 & 1 & 0 \end{bmatrix}^T$. *In addition,*

$$A^T A = \begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & 0 & 7 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 2 & 0 & 0 \\ 0 & 0 & 7 \end{bmatrix} = \begin{bmatrix} 5 & 0 & 8 \\ 0 & 25 & 0 \\ 0 & 0 & 49 \end{bmatrix},$$

which leads to eigenvalues $\lambda_1 = 5$, $\lambda_2 = 25$, *and* $\lambda_3 = 49$. *For* $\lambda_1 = 5$, *we have*

$$(A^T A - \lambda_1 I_3)x = \begin{bmatrix} 0 & 0 & 8 \\ 0 & 20 & 0 \\ 0 & 0 & 44 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix},$$

leading to the eigenvector $v_1 = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}^T$. *For* $\lambda_2 = 25$, *we have*

$$(A^T A - \lambda_2 I_3)x = \begin{bmatrix} -20 & 0 & 8 \\ 0 & 0 & 0 \\ 0 & 0 & 24 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix},$$

leading to the eigenvector $v_2 = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}^T$. *For* $\lambda_3 = 49$, *we have*

$$(A^T A - \lambda_2 I_3)x = \begin{bmatrix} -44 & 0 & 8 \\ 0 & -24 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix},$$

leading to the eigenvector $v_3 = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}^T$. *Therefore, the singular values are*

$$\sigma_1 = \sqrt{5}, \quad \sigma_2 = 5, \quad \sigma_3 = 7,$$

and left and right eigenvalues of A *are*

$$u_1 = \frac{\bar{u}_1}{\|\bar{u}_1\|} = \begin{bmatrix} \frac{1}{\sqrt{5}} \\ 0 \\ \frac{2}{\sqrt{5}} \\ 0 \end{bmatrix}, \quad u_2 = \frac{\bar{u}_2}{\|\bar{u}_2\|} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad u_3 = \frac{\bar{u}_3}{\|\bar{u}_3\|} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \quad u_4 = \frac{\bar{u}_4}{\|\bar{u}_4\|} = \begin{bmatrix} \frac{2}{\sqrt{5}} \\ 0 \\ \frac{1}{\sqrt{5}} \\ 0 \end{bmatrix},$$

and

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix},$$

where u_i $(i = 1, \ldots, 4)$ *are normalized version of* $\bar{u}_i$ $(i = 1, \ldots, 4)$.

Let us define the matrices

$$\Sigma = \begin{bmatrix} \sqrt{5} & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 7 \\ 0 & 0 & 0 \end{bmatrix}, \quad U = \begin{bmatrix} \frac{1}{\sqrt{5}} & 0 & 0 & \frac{2}{\sqrt{5}} \\ 0 & 1 & 0 & 0 \\ \frac{2}{\sqrt{5}} & 0 & 0 & \frac{1}{\sqrt{5}} \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad V = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

where U and V are orthonormal matrices ($U^TU = I_4$ and $V^TV = I_3$). Moreover, we have

$$U\Sigma V^T = \begin{bmatrix} \frac{1}{\sqrt{5}} & 0 & 0 & \frac{2}{\sqrt{5}} \\ 0 & 1 & 0 & 0 \\ \frac{2}{\sqrt{5}} & 0 & 0 & \frac{1}{\sqrt{5}} \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \sqrt{5} & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 7 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 2 & 0 & 0 \\ 0 & 0 & 7 \end{bmatrix} = A,$$

which decomposes the matrix A to three factors.

11.2. Singular values decomposition

Definition (Singular value decomposition). Let A be an arbitrary $m\times n$ matrix ($A \in \mathbb{R}^{m\times n}$), and let $\sigma_1, \sigma_2, \ldots, \sigma_r > 0$ be its singular values, u_i ($i = 1, \ldots, m$) be its orthonormal left singular vectors, and let v_i ($i = 1, \ldots, n$) be its orthonormal right singular vectors. Then, A accepts the decomposition

$$A = U\Sigma V^T = \sum_{i=1}^{r} \sigma_i u_i v_i^T, \tag{11.1}$$

where

$$\Sigma = \begin{bmatrix} \sigma_1 & 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & \cdots & \sigma_r & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \cdots & \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & \cdots & 0 & 0 & \cdots & 0 \end{bmatrix}_{m\times n}$$

$$U = \begin{bmatrix} u_1 & u_2 & \ldots & u_m \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} & \cdots & u_{1m} \\ u_{21} & u_{22} & \cdots & u_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ u_{m1} & u_{m2} & \cdots & u_{mm} \end{bmatrix},$$

$$V = \begin{bmatrix} v_1 & v_2 & \ldots & v_n \end{bmatrix} = \begin{bmatrix} v_{11} & v_{12} & \cdots & v_{1n} \\ v_{21} & v_{22} & \cdots & v_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ v_{n1} & v_{n2} & \cdots & v_{nn} \end{bmatrix},$$

where Σ has $m - r$ zero rows and $n - 1$ zero columns.

Example 130. *Let us consider the matrix*

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

It is clear that all eigenvalues of this matrix are zero ($\lambda_1 = \ldots = \lambda_4 = 0$) and

$$(A - \lambda I_4)x = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix},$$

leading to $x_2 = x_3 = x_4 = 0$, i.e., $x = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix}^T$ is the only distinct eigenvector. However,

$$AA^T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 9 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad A^T A = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 9 \end{bmatrix},$$

which implies that $\sigma_1 = 3$, $\sigma_1 = 2$, and $\sigma_1 = 1$ are its singular values. For $\sigma_1 = 3$ ($\lambda_1 = 9$ of AA^T), we have

$$(AA^T - \lambda_1 I_4)x = \begin{bmatrix} -8 & 0 & 0 & 0 \\ 0 & -5 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix},$$

implying $x_1 = x_2 = x_4 = 0$ that leads to $u_1 = \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix}^T$. Continuing this process for other singular values, we come to

$$U = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad V = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

leading to the decomposition $A = U\Sigma V^T$.

Definition (Diagonalizability and singular value decomposition). For a matrix A, the diagonalizibility ($A = X\Lambda X^{-1}$) is defined if A is a square matrix; however, the singular value decomposition $A = U\Sigma V^T$ is defined for any matrix. We now verify that when are these two decompositions the same? Since singular values are positive numbers, the eigenvalues of A should be nonnegative, i.e., A is a symmetric positive semidefinite matrix. In this case, we have $A = X\Lambda X^{-1} = Q\Lambda Q^T$, which is the same as the singular value decomposition $A = U\Sigma V^T$.

Example 131. *Let us consider the matrix*

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix},$$

Then,

$$\det(A - \lambda I_3) = \det\left(\begin{bmatrix} 2-\lambda & 0 & 1 \\ 0 & 2-\lambda & 0 \\ 1 & 0 & 2-\lambda \end{bmatrix}\right) = (2-\lambda)^3 - (2-\lambda) = (2-\lambda)[\lambda(2-\lambda) - 1] = 0$$

which implies $\lambda_1 = 3$ and $\lambda_2 = 2$, $\lambda_3 = 1$. For $\lambda_1 = 3$, we have

$$(A - \lambda_1 I_3)y = \begin{bmatrix} -1 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix},$$

leading to $x_1 = \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}^T$. For $\lambda_2 = 2$, we get

$$(A - \lambda_2 I_3)y = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix},$$

leading to $x_2 = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}^T$. For $\lambda_3 = 1$, we get

$$(A - \lambda_2 I_3)y = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix},$$

leading to $x_2 = \begin{bmatrix}-1 & 0 & 1\end{bmatrix}^T$*. Normalizing these eigenvectors yields*

$$X = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}, \quad \Lambda = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad X^{-1} = X^T = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}}. \end{bmatrix}$$

Moreover,

$$A^TA = AA^T = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 0 & 4 \\ 0 & 4 & 0 \\ 4 & 0 & 5 \end{bmatrix}.$$

Then,

$$\det(AA^T - \lambda I_3) = \det\left(\begin{bmatrix} 5-\lambda & 0 & 4 \\ 0 & 4-\lambda & 0 \\ 4 & 0 & 5-\lambda \end{bmatrix}\right) = (4-\lambda)(5-\lambda)^2 - 16(4-\lambda) = (4-\lambda)[(5-\lambda)^2 - 16] = 0$$

leading to $\sigma_1 = 3$*,* $\sigma_2 = 2$*, and* $\sigma_3 = 1$*. Computing the corresponding singular vectors, we come to*

$$U = V = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}, \quad \Sigma = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

which implies that for this matrix the decomposition $A = X\Lambda X^{-1}$*, spectral decomposition* $A = Q\Lambda Q^T$*, and singular value decomposition* $A = U\Sigma V^T$ *are the same.*

CHAPTER 12

Applications of Linear Algebra

In this chapter, we discuss some applications of linear algebra in graph theory and optimization. Our main objectives are to:

- describe some applications of linear algebra in graph theory;
- describe some applications of linear algebra in optimization.

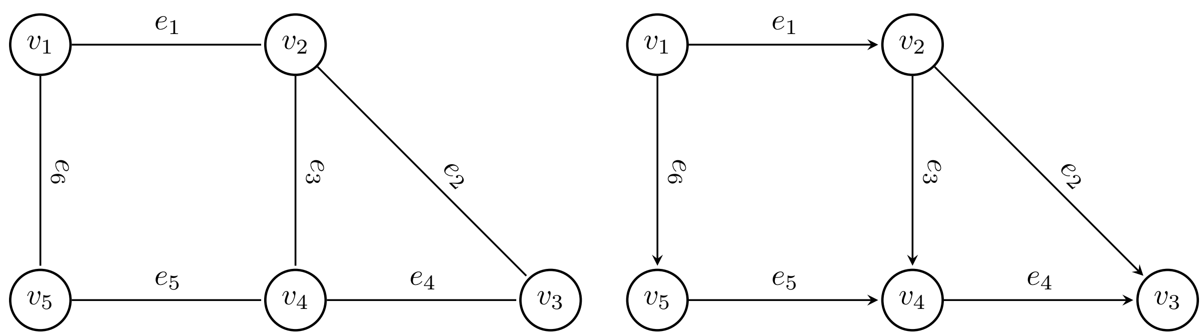
12.1. Graphs and networks

A **graph** (network) is a structure that is used to represent **relationships** among some **objects**, where these objects can be social network actors, web pages, chemical entities, and so on. The objects in a graph are referred to as **vertices** (nodes), and the relationships among them are referred to as **edges**. Hence, a graph G is denoted by the pair $(V(G), E(G))$ in which V is the set of all vertices and E is the set of all edges. Therefore, for a graph G with n vertices and m edges, we have

$$V(G) = \{v_1, v_2, \dots, v_n\}, \quad E(G) = \{e_1, e_2, \dots, e_m\} \subseteq \{\{v_i, v_j\} \mid v_i, v_j \in V(G)\},$$

where v_i, v_j is the edge between v_i and v_j. If the edges of our graphs have directions, then the graph is called **directed graph**; otherwise, it is called **undirected graph**. In the edge $\{v_i, v_j\}$, v_i is called **tail** and v_j is called **head**.

Example 132. *Let G be a graph with five vertices $V(G) = \{v_1, v_2, \dots, v_5\}$ and six edges $E(G) = \{e_1, e_2, \dots, e_6\}$ (see Subfigures (A) and (B) of Figure 12.1 for udirected and directed graphs, respectively).*



(A) Undirected graph.

(B) Directed graph.

FIGURE 12.1. Undirected and directed graphs.

Definition (Incidence matrix). Let us consider a directed graph G with the vertex set $V(G) = \{v_1, v_2, \ldots, v_n\}$ and the edge set $E(G) = \{e_1, e_2, \ldots, e_m\} \subseteq \{\{v_i, v_j\} \mid v_i, v_j \in V(G)\}$. Then, an $m \times n$ matrix I_G is called **indices matrix** if

$$I_G(i,j) = \begin{cases} -1 & \text{if } v_j \text{ is a tail of } e_i, \\ 1 & \text{if } v_j \text{ is a head of } e_i, \\ 0 & \text{otherwise,} \end{cases}$$

which is a matrix that

(1) its elements are -1, 1, and 0;
(2) each row has exactly one -1 and one 1 elements and the rest are zero;
(3) the number of nonzero elements of each column is the same as the number of edges accepted by the corresponding vertex;
(4) has no loops.

Example 133. *The incidence matrix of the graph G given in Example 132 is given by*

$$I_G = \begin{bmatrix} -1 & 1 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & -1 \\ -1 & 0 & 0 & 0 & 1 \end{bmatrix} \tag{12.1}$$

which is a 6×5 matrix.

Four spaces of an incidence matrix. Let us consider a directed graph G with the vertex set $V(G) = \{v_1, v_2, \ldots, v_n\}$ and the edge set $E(G) = \{e_1, e_2, \ldots, e_m\} \subseteq \{\{v_i, v_j\} \mid v_i, v_j \in V(G)\}$. If I_G is the incidence matrix of G, then

(1) **Nullspace of I_G (i.e., $N(I_G)$).** Since in each row of I_G there are only one -1 and one 1, we have

$$I_G \mathbf{1} = 0,$$

where $\mathbf{1}$ is the vector of all ones (i.e., $\mathbf{1} = \begin{bmatrix} 1 & 1 & \ldots & 1 \end{bmatrix}$). This clearly implies that $\mathbf{1}$ is the only vector nullspace, that is,

$$N(I_G) = \{\mathbf{1}\},$$

i.e., $\dim(N(I_G)) = 1$;

(2) **Row space of I_G (i.e., $C(I_G^T)$).** Since the nullspace and the row space are orthogonal, the vector $v \in \mathbb{R}^n$ is in the row space $A^T y = v$ if

$$\langle v, \mathbf{1} \rangle = 0,$$

i.e., $\dim(C(I_G^T)) = n - 1$;

(3) **Column space of I_G (i.e., $C(I_G)$).** Since $\operatorname{rank}(I_G) = \dim(C(I_G^T)) = n - 1$, we have that $\dim(C(I_G)) = n - 1$. One way to find if $b \in \mathbb{R}^m$ is in the column space is to find a solution of $Ax = b$, but this misses all the insight. On the other hand, if we add differences around a closed loop in the graph, they cancel to leave zero. **Kirchhoff's Voltage Law**: The components of $Ax = b$ add to zero around every loop. Note that by testing each loop, the Voltage Law decides whether b is in the column space. The system $Ax = b$ can be solved exactly when the components of b satisfy all the same dependencies as the rows of A. Then elimination leads to $O = 0$, and $Ax = b$ is consistent.

(4) **Left nullspace of I_G (i.e., $N(I_G^T)$).** Since $\dim(C(I_G)) = n-1$, we have that $\dim(N(I_G^T)) = m - n + 1$. Moreover, $y \in N(I_G^T)$ if $A^T y = 0$, which is **Kirchhoff's Current Law** (the conservation law): Flow in equals flow out at each node $A^T y = \text{flow in} - \text{flow out} = 0$.

Example 134. *Let us consider the graph G given in Example 132 with the incidence matrix* (12.3)*. It is clear that*

$$I_G x = \begin{bmatrix} -1 & 1 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & -1 \\ -1 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} x_2 - x_1 \\ x_2 - x_3 \\ x_4 - x_2 \\ x_3 - x_4 \\ x_4 - x_5 \\ x_5 - x_1 \end{bmatrix} \tag{12.2}$$

which clearly implies that

- $I_G x = 0$ *if* $x_1 = \ldots = x_5$*, i.e.,* $N(I_G) = \{\mathbf{1}\}$*;*
- *any member v of the row space $C(I_G^T)$ satisfies $\langle v, \mathbf{1}\rangle = 0$ (e.g., $v = \begin{bmatrix} 1 & 2 & 3 & 4 & -10 \end{bmatrix}$);*
- *the left nullspace of I_G has dimension $6 - 5 + 1 = 2$ which is spanned by the vectors*

$$y_1 = \begin{bmatrix} 0 \\ -1 \\ 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad y_2 = \begin{bmatrix} -1 \\ 0 \\ -1 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \tag{12.3}$$

  that are given by loops $v_2 - v_3 - v_4 - v_2$ and $v_1 - v_2 - v_4 - v_5 - v_1$ in which elements of y_i $(i = 1, 2)$ is 1 if the direction is right and -1 otherwise;
- *any member u of the column space $C(I_G)$ satisfies $\langle u, y_1\rangle = 0$ and $\langle u, y_2\rangle = 0$(e.g., $u = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$).*

12.2. Application to Optimization

One of the main applications of linear algebra is for determining minima, maxima, and saddle points of a given function, which is the main topic of this section.

Definition 135 (Fréchet differentiability)**.** *Let $f : U \to \mathbb{R}$ be a function defined on an open set $U \subseteq \mathbb{R}^n$. We say that f is Fréchet differentiable at x if there exists the vector $g \in \mathbb{R}^n$ such that*

$$\lim_{d \to 0} \frac{f(x+d) - f(x) - \langle g, d\rangle}{\|d\|} = 0. \tag{12.1}$$

If such g exists, it is called gradient of f at x which is denoted by

$$\nabla f(x) = \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{pmatrix}. \tag{12.2}$$

The function f is called **continuously differentiable** *over U if all the partial derivatives $\frac{\partial f}{\partial x_i}$ exist and are continuous on U.*
In addition, a function $f : U \to \mathbb{R}$ is said to be **twice continuously differentiable** *if all the second-order partial derivatives exist and are continuous over U. Let us assume that the function $f : \mathbb{R}^n \to \mathbb{R}$ is twice continuously (Fréchet) differentiable ($f \in \mathcal{C}^2$). Then, the Hessian of f at x is the $n \times n$ matrix given by*

$$\nabla^2 f(x) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{pmatrix}. \tag{12.3}$$

Notice that the twice continuous differentiability of f leads to **second-order expansion**

$$f(y) = f(x) + \langle \nabla f(x), y - x\rangle + \left\langle y - x, \nabla^2 f(x)(y - x)\right\rangle + o(\|y - x\|^2). \tag{12.4}$$

Example 136. *Let us consider the three-dimensional function*

$$f(x_1, x_2, x_3) = 3x_1x_2^2 - 5x_2^7x_3 + x_1x_2x_3.$$

Then, its gradient and Hessian are given by

$$\nabla f(x_1, x_2, x_3) = \begin{pmatrix} 3x_2^2 + x_2x_3 \\ 6x_1x_2 - 35x_2^6x_3 + x_1x_3 \\ -5x_2^7 + x_1x_2 \end{pmatrix},$$

$$\nabla^2 f(x_1, x_2, x_3) = \begin{pmatrix} 0 & 6x_2 + x_3 & x_2 \\ 6x_2 + x_3 & 6x_1 - 210x_2^5x_3 & -35x_2^6 + x_1 \\ x_2 & -35x_2^6 + x_1 & 0 \end{pmatrix},$$

where $\nabla^2 f(\cdot)$ is a symmetric matrix.

THEOREM 137 (first-order necessary optimality conditions). *Let $f : \mathbb{R}^n \to \mathbb{R}$ be a continuously differentiable function on an open neighborhood $B(x^*, r)$ ($f \in \mathcal{C}^1(B(x^*, r))$), for a local minimizer x^* and $r > 0$. Then, we have that*

$$\nabla f(x^*) = 0. \tag{12.5}$$

Definition 138 (Critical points)**.** *Let us consider a continuously differentiable function $f : \mathbb{R}^n \to \mathbb{R}$. A point x^* is called a* **critical point (stationary point)** *of f if $\nabla f(x^*) = 0$.*

Definition 139 (Saddle points)**.** *Let us consider a continuously differentiable function $f : \mathbb{R}^n \to \mathbb{R}$. A point x^* is called a* **saddle (minimax) point** *of f if $\nabla f(x^*) = 0$ but neither minimizer nor maximizer. In other words, x^* is a minimizer along some directions and is a maximizer for some other directions.*

Different types of critical points. Let $f : \mathbb{R}^n \to \mathbb{R}$ be a $\mathcal{C}^2$ function (twice differentiable with continuous derivatives), and let x^* be its critical point. Then, the point x^* could be
(a) a local (global) minimizer;
(b) a local (global) maximizer;
(c) a saddle (minimax) point,
of this optimization problem.

Example 140. *Let us consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ given by*

$$f(x, y) = x^2 - y^2.$$

The gradient and Hessian of f at x are

$$\nabla f(x, y) = \begin{pmatrix} 2x \\ -2y \end{pmatrix}, \quad \nabla^2 f(x, y) = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}$$

It is clear that $\nabla f(0, 0) = (0, 0)^T$ and the Hessian $\nabla^2 f(0, 0)$ has both positive eigenvalue $\lambda_1 = 2$ and negative eigenvalue $\lambda_2 = -2$. Let us compute the corresponding eigenvalues. For $\lambda_1 = 2$, we have

$$(\nabla^2 f(x, y) - \lambda_1 I_2) \begin{pmatrix} v_1^1 \\ v_2^1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & -4 \end{pmatrix} \begin{pmatrix} v_1^1 \\ v_2^1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

which leads to the eigenvector

$$v^1 = \begin{pmatrix} v_1^1 \\ 0 \end{pmatrix}, \quad v_1^1 \neq 0.$$

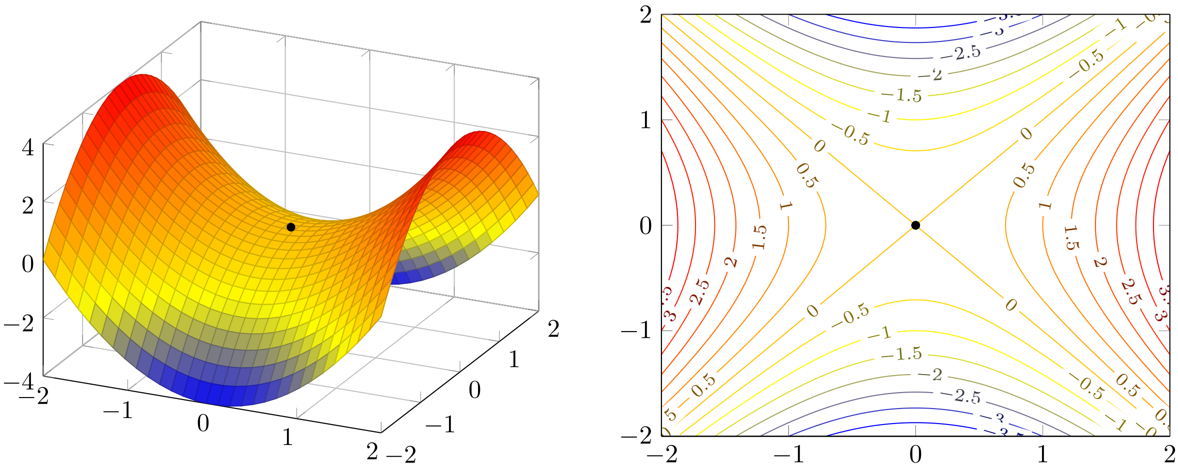
Similarly, for $\lambda_2 = -2$, we have

$$(\nabla^2 f(x, y) - \lambda_2 I_2) \begin{pmatrix} v_1^2 \\ v_2^2 \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} v_1^2 \\ v_2^2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

which leads to the eigenvector

$$v^2 = \begin{pmatrix} 0 \\ v_2^2 \end{pmatrix}, \quad v_2^2 \neq 0.$$

It is easy to see that f is increasing in the direction of v^1 and is decreasing in the direction of v^2, which implies that $(0,0)$ is a saddle point; see Figure 12.2.



(A) smooth function. (B) nonsmooth function.

FIGURE 12.2. Subfigure (a) stands for the plot of the function $f(x) = x^2 - y^2$, and Subfigure (b) illustrates its contour plot.

THEOREM 141 (Second-order necessary optimality conditions). *Let $f : \mathbb{R}^n \to \mathbb{R}$ be a twice continuously differentiable function on an open neighborhood $B(x^*, r)$ ($f \in \mathcal{C}^2(B(x^*, r))$), for a local minimizer x^* and $r > 0$. Then,*

$$\nabla f(x^*) = 0, \quad \nabla^2 f(x^*) \succcurlyeq 0. \tag{12.6}$$

THEOREM 142 (Second-order sufficient optimality conditions). *Let $f : \mathbb{R}^n \to \mathbb{R}$ be a twice continuously differentiable function on an open neighborhood $B(x^*, r)$ ($f \in \mathcal{C}^2(B(x^*, r))$) for a point x^* and $r > 0$, and let $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ be positive definite ($\nabla^2 f(x^*) \succ 0$). Then, x^* is a local minimizer of f.*

THEOREM 143 ((Strict) global optimizer). *Let $f : \mathbb{R}^n \to \mathbb{R}$ be a $\mathcal{C}^2$ function (twice differentiable with continuous derivatives), and let x^* be its critical point. Then, setting $p = x - x^*$ for arbitrary $x \in \mathbb{R}^n$,*

(a) the point x^* is a global minimizer of f if $\nabla^2 f(x^* + tp)$ is positive semidefinite;
(b) the point x^* is a strict global minimizer of f if $\nabla^2 f(x^* + tp)$ is positive definite;
(c) the point x^* is a global maximizer of f if $\nabla^2 f(x^* + tp)$ is negative semidefinite;
(d) the point x^* is a strict global maximizer of f if $\nabla^2 f(x^* + tp)$ is negative definite.

THEOREM 144 (Saddle point). *Let $f : \mathbb{R}^n \to \mathbb{R}$ be a $\mathcal{C}^2$ function (twice differentiable with continuous derivatives), and let x^* be its critical point. If $\nabla^2 f(x^*)$ is indefinite, then x^* is a saddle point of f.*

Example 145. *Let us consider functions $f : \mathbb{R}^2 \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ given by*

$$f(x, y) = \cos(x)\cos(y).$$

We begin with the first-order optimality condition

$$\nabla f(x, y) = \begin{bmatrix} -\sin(x)\cos(y) \\ -\cos(x)\sin(y) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

It is clear that the set of critical points of f is given by

$$\Omega(f) = \left\{(x, y) \in \mathbb{R}^2 \mid x = k_1\pi \text{ or } y = k_2\pi + \tfrac{\pi}{2} \text{ and } x = k_3\pi + \tfrac{\pi}{2} \text{ or } y = k_4\pi\right\}.$$

Then,

$$\nabla^2 f(x, y) = \begin{bmatrix} -\cos(x)\cos(y) & \sin(x)\sin(y) \\ \sin(x)\sin(y) & -\cos(x)\cos(y) \end{bmatrix}$$

For $x = k_1\pi$ and $y = k_4\pi$ such that both of them are even or both of them are odd, we have

$$\nabla^2 f(x,y) = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix},$$

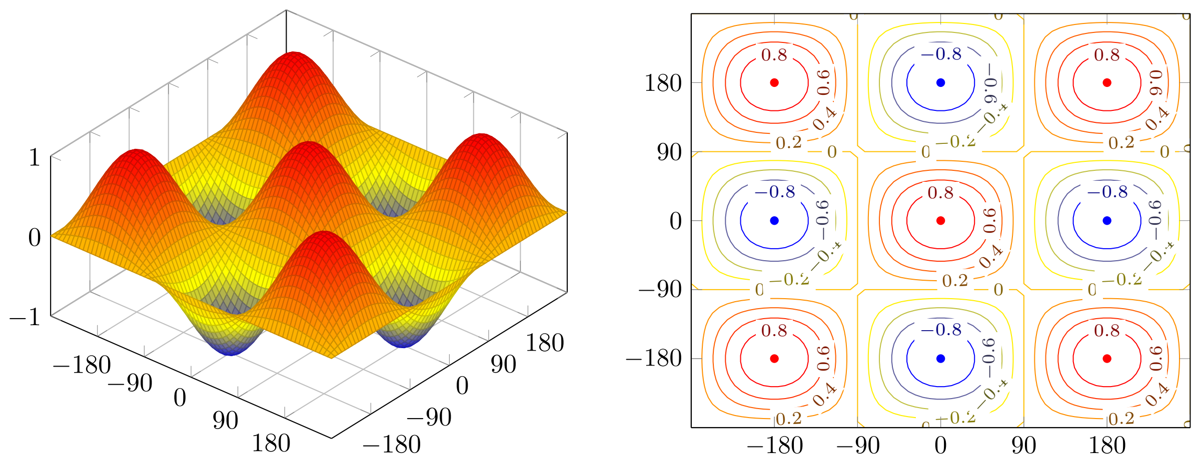
which has eigenvalue $\lambda_1 = \lambda_2 = -1$, i.e., $\nabla^2 f(x,y)$ is negative definite and so a local maximum. For $x = k_1\pi$ and $y = k_4\pi$ with one of k_1 and k_4 is odd, we have

$$\nabla^2 f(x,y) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

which is positive definite (the identity matrix), i.e., such points satisfy the second-order necessary conditions (blue points in Subfigure (B) of Figure 12.3). For $x = k_2\pi + \frac{\pi}{2}$ and $y = k_3\pi + \frac{\pi}{2}$ for arbitrary k_2 and k_3, we have

$$\nabla^2 f(x,y) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{or} \quad \nabla^2 f(x,y) = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix},$$

that are indefinite matrices with eigenvalues $\lambda_1 = -1$ and $\lambda_2 = 1$. Therefore, these points are saddle points.



(A) 3D plot of f. (B) contour plot of f.

FIGURE 12.3. Subfigure (A) stands for the plot of $f(x,y) = \cos(x)\cos(y)$, and Subfigure (B) illustrates its contour plot, where blue points are candidates for local minimizers.

12.2.1. Newton's method. Let $f : \mathbb{R}^n \to \mathbb{R}$ be a twice continuously differentiable function ($f \in \mathcal{C}^2$), and let $x^k \in \mathbb{R}^n$ and $\nabla^2 f(x^k)$ be positive definite. Then, we consider the following quadratic approximation

$$f(x) = f(x^k) + \left\langle \nabla f(x^k), x - x^k \right\rangle + \tfrac{1}{2} \left\langle x - x^k, \nabla^2 f(x^k)(x - x^k) \right\rangle. \tag{12.7}$$

Setting $d = x - x^k$, we have

$$q(d) = f(x^k + d) = f(x^k) + \left\langle \nabla f(x^k), d \right\rangle + \tfrac{1}{2} \left\langle d, \nabla^2 f(x^k) d \right\rangle. \tag{12.8}$$

To minimize this quadratic function, we write the first-order optimality conditions as

$$\nabla f(x^k) + \nabla^2 f(x^k) d = 0,$$

which is a linear system of equations. Since $\nabla^2 f(x^k)$ is positive definite, it is invertible, i.e.,

$$d = -[\nabla^2 f(x^k)]^{-1} \nabla f(x^k). \tag{12.9}$$

The positive definiteness of $\nabla^2 f(x^k)$ implies that

$$\left\langle \nabla f(x^k), d \right\rangle = -\left\langle \nabla f(x^k), [\nabla^2 f(x^k)]^{-1} \nabla f(x^k) \right\rangle < 0, \tag{12.10}$$

which means that d is a descent direction that we can use to minimize the original function f. For a given point x^k, setting $d^k = d = -[\nabla^2 f(x^k)]^{-1} \nabla f(x^k)$, we define the new iteration as

$$x^{k+1} = x^k + d^k, \tag{12.11}$$

which named as **Newton's method**. We describe this method in the following algorithm.

Algorithm 2: Newton Algorithm

Input: $x^0 \in \mathbb{R}^n$, $\varepsilon > 0$, $k = 0$;

begin

  repeat

    Compute d^k by solving the linear system of equations

$$\nabla^2 f(x^k) d^k = -\nabla f(x^k); \tag{12.12}$$

    Set $x^{k+1} = x^k + d^k$ and $k := k + 1$;

  until $\|\nabla f(x^k)\| \leq \varepsilon$

  $x^{\text{best}} = x^k$;

end

Output: x^{best}

It is clear that the most costly part of Newton's method is finding a solution of the linear system of equations $eq : NewtonLinSys$, which should be solved once in each iteration. Therefore, the effective solution of this linear system of equations is a crucial task for the efficient implementation of Newton's method.

Bibliography

[1] Gilbert Strang. *Introduction to Linear Algebra*, volume 3. Wellesley-Cambridge Press Wellesley, MA, 1993.